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RECOMBINANT YEAST CELL

Abstract

A recombinant yeast cell functionally expressing: —a nucleic acid sequence encoding a native protein having transketolase activity (EC 2.2.1.1); and —a nucleic acid sequence encoding a heterologous protein having transketolase activity (EC 2.2.1.1).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a U.S. National Stage Application under 35 U.S.C. § 371 of International Application No. PCT/EP2022/068917, filed Jul. 7, 2022, which claims priority to U.S. Application No. 63/220,896, filed Jul. 12, 2021, and European Application No. 21185148.0, filed Jul. 12, 2021, all of which are hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

[0002] The invention relates to a recombinant yeast cell and to a process for the production of ethanol wherein said recombinant yeast cell is used.

INCORPORATION BY REFERENCE OF THE SEQUENCE LISTING

[0003] The present application is being filed with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled IFF34056-US-PCT_SEQ_Listing.xml, created on Jul. 31, 2024, which is 188,359 bytes in size. The information in the electronic format of the Sequence Listing is incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

[0004] Microbial fermentation processes are applied to industrial production of a broad and rapidly expanding range of chemical compounds from renewable carbohydrate feedstocks. Especially in anaerobic fermentation processes, redox balancing of the cofactor couple NADH/NAD^{sup.+} can cause important constraints on product yields. This challenge is exemplified by the formation of glycerol as major by-product in the industrial production of—for instance—fuel ethanol by *Saccharomyces cerevisiae*, a direct consequence of the need to re-oxidize NADH formed in biosynthetic reactions.

[0005] Ethanol production by *Saccharomyces cerevisiae* is currently, by volume, the single largest fermentation process in industrial biotechnology. Various approaches have been proposed to improve the fermentative properties of organisms used in industrial biotechnology by genetic modification. A major challenge relating to the stoichiometry of yeast-based production of ethanol, is that substantial amounts of NADH-dependent side-products such as glycerol are generally formed as a by-product, especially under anaerobic and oxygen-limited conditions or under conditions where respiration is otherwise constrained or absent. It has been estimated that, in typical industrial ethanol processes, up to about 4 wt. % of the sugar feedstock is converted into glycerol (Nissen et al., “Anaerobic and aerobic batch cultivations of *Saccharomyces cerevisiae* mutants impaired in glycerol Synthesis”, (2000), Yeast, vol. 16, pages 463-474). Under conditions

that are ideal for anaerobic growth, the conversion into glycerol may even be higher, up to about 10%.

[0006] Glycerol production under anaerobic conditions is primarily linked to redox metabolism. During anaerobic growth of *S. cerevisiae*, sugar dissimilation occurs via alcoholic fermentation. In this process, the NADH formed in the glycolytic glyceraldehyde-3-phosphate dehydrogenase reaction is reoxidized by converting acetaldehyde, formed by decarboxylation of pyruvate to ethanol via NAD^{sup.+}-dependent alcohol dehydrogenase. The fixed stoichiometry of this redox-neutral dissimilatory pathway causes problems when a net reduction of NAD^{sup.+} to NADH occurs elsewhere in metabolism. Under anaerobic conditions, NADH reoxidation in *S. cerevisiae* is strictly dependent on reduction of sugar to glycerol. Glycerol formation is initiated by reduction of the glycolytic intermediate dihydroxyacetone phosphate (DHAP) to glycerol 3-phosphate (glycerol-3P), a reaction catalyzed by NAD^{sup.+}-dependent glycerol 3-phosphate dehydrogenase. Subsequently, the glycerol 3-phosphate formed in this reaction is hydrolysed by glycerol-3-phosphatase to yield glycerol and inorganic phosphate. Consequently, glycerol is a major by-product during anaerobic production of ethanol by *S. cerevisiae*, which is undesired as it reduces overall conversion of sugar to ethanol. Further, the presence of glycerol in effluents of ethanol production plants may impose costs for waste-water treatment.

[0007] In literature, several different approaches have been reported that could help to reduce the amount of the byproduct glycerol and divert carbon to ethanol resulting in an increased yield of ethanol per gram of fermented carbohydrate.

[0008] In WO2011/010923, WO2014/081803, WO2014/129898, WO2015/107496, WO2015/148272, WO2018/172328 and WO2018/228836, several alternative reduction pathways to reoxidize the NADH produced, also referred to as alternative “redox sinks”, have been suggested and implemented into (recombinant) yeast cells.

[0009] For example, in WO 2011/010923, the NADH-related side-product (glycerol) formation in a process for the production of ethanol from a carbohydrate containing feedstock is addressed by providing a recombinant yeast cell comprising one or more recombinant nucleic acid sequences encoding an NAD⁺ dependent acetylating acetaldehyde dehydrogenase (EC 1.2.1.10) activity. The cell may for example lack enzymatic activity needed for the NADH dependent glycerol synthesis or the cell may have a reduced enzymatic activity with respect to the NADH dependent glycerol synthesis compared to its corresponding wild-type yeast cell.

[0010] WO2014/129898 describes a recombinant cell functionally expressing heterologous nucleic acid sequences encoding for ribulose-1,5-phosphate carboxylase/oxygenase (EC 4.1.1.39; herein abbreviated as “Rubisco”), and optionally molecular chaperones for Rubisco, and phosphoribulokinase (EC 2.7.1.19; herein abbreviated as “PRK”). In addition, reference is made to the use of carbon dioxide as an electron acceptor in a recombinant chemotrophic micro-organism.

[0011] Although the described processes and yeast cells are advantageous, there is a continuing need for improvement. In an industrial environment the reduction in glycerol production by the above recombinant yeast cells can potentially affect their osmotolerance and their stress response to the external environment. Especially under challenging process conditions, for example when applying a fermentation medium having a high dry solids content and/or a high fermentation temperature, this may lead to a decline of the cell population and/or cell activity at the end of the fermentation period. It would be an advancement in the art to provide a process, and yeast cells for use therein, wherein the yeast cells have an improved robustness under high dry solids/high dry matter conditions and/or high temperatures. In addition, it would be an advancement in the art to provide yeast cells that have a reduced accumulation of glucose and/or total sugar content within the yeast cell. That is, it would be an advancement in the art to achieve a continued performance of the yeast cell and/or a low concentration of remaining glucose at the end of the fermentation, even where a high concentration of glucose is present at the start and/or throughout the fermentation.

SUMMARY OF THE INVENTION

[0012] The inventors have now surprising found that the processes and yeast cells of the prior art can be improved by using a specific combination of transketolase proteins.

[0013] Accordingly the invention provides a recombinant yeast cell functionally expressing: [0014] a nucleic acid sequence encoding a native protein having transketolase activity; and [0015] a nucleic acid sequence encoding a heterologous protein having transketolase activity.

In addition, the invention provides a process for the production of ethanol, comprising converting a carbon source, such as a carbohydrate or another organic carbon source, using the above recombinant yeast cell, thereby suitably forming ethanol.

[0016] Advantageously, use of the above recombinant yeast cell and/or the above process results in an improved robustness. Such is especially advantageous when a medium having a high dry solids content is applied and/or if a high fermentation temperature is applied.

[0017] A process for the production of ethanol from a carbon source, such as a carbohydrate, can advantageously be carried out in the presence of a saccharolytic enzyme, such as glucoamylase, to convert polysaccharides and/or oligosaccharides into glucose. When the process is carried out in a medium with a high dry matter content, for example after starting the process with a high concentration of corn mash, the concentration of glucose in the medium can become very high. Without wishing to be bound by any kind of theory, it is believed that a high concentration of glucose can cause osmotic stress for the yeast cell, causing the yeast cell to stop performing and even die.

[0018] Without wishing to be bound by any kind of theory it is believed that, compared to a yeast cell not comprising the claimed combination of transketolase proteins, the above recombinant yeast cell allows for reduced accumulation of glucose and/or other sugars within the yeast cell, thereby suitably allowing for an improved robustness.

[0019] The advantages are illustrated by the examples. In the examples fermentation is carried out at a high dry matter content of 36% w/w. As illustrated by the examples the recombinant yeast cell according to the invention, and the process according to the invention, allow for a continued performance of the yeast cell and/or continued conversion of the glucose. Even in a medium comprising a concentration of glucose as high as 36% w/w and/or temperatures as high as 3200, the recombinant yeast cell is still alive and converting carbohydrates into ethanol after 66 hours. As a result a low concentration of remaining glucose can be obtained at the end of the fermentation, even where a high concentration of glucose is present at the start and/or throughout the fermentation.

BRIEF DESCRIPTION OF THE SEQUENCE LISTING

[0020] This application contains a Sequence Listing in computer readable form, which is incorporated herein by reference. An overview is provided by Table 1 below.

TABLE-US-00001 TABLE 1 Overview of sequence listings: SEQ ID No: Enzyme Name Origin

Type	SEQ ID No:	Enzyme Name	Origin
1	Transketolase	<i>Saccharomyces cerevisiae</i> protein	SEQ ID No: 1
2	Transketolase	<i>Saccharomyces cerevisiae</i> DNA	SEQ ID No: 2
3	Transketolase (DAS1)	<i>Komagataella phaffii</i> protein	SEQ ID No: 3
4	Transketolase (DAS2)	<i>Komagataella phaffii</i> protein	SEQ ID No: 4
5	Transketolase	<i>Candida tenuis</i> protein	SEQ ID NO: 5
6	Transketolase	<i>Debaryomyces</i> protein	SEQ ID NO: 6
7	Transketolase	<i>Kuraishia capsulata</i> protein	SEQ ID NO: 7
8	Transketolase	<i>Candida arabinofementans</i> protein	SEQ ID NO: 8
9	Transketolase	<i>Metschnikowia pulcherrima</i> protein	SEQ ID NO: 9
10	Transketolase	<i>Meyerozyma</i> protein	SEQ ID NO: 10
11	Transketolase	<i>Diutina rugosa</i> protein	SEQ ID NO: 11
12	Transketolase	<i>Pachysolen tannophilus</i> protein	SEQ ID NO: 12
13	Transketolase	<i>Rasamsonia emersonii</i> protein	SEQ ID NO: 13
14	Transketolase	<i>Kluyveromyces marxianus</i> protein (strain DMKU3-1042/BCC 29191/NBRC 104275)	SEQ ID NO: 14
15	Transketolase	<i>Ogataea parapolyomorpha</i> protein	SEQ ID NO: 15
16	Transketolase	Artificial sequence; based on protein Transketolase protein from <i>Komagataella phaffii</i>	SEQ ID NO: 16
17	Transketolase	<i>Komagataella phaffii</i> protein (<i>Pichia pastoris</i>)	SEQ ID NO: 17
18	Transketolase	<i>Komagataella phaffii</i> DNA (<i>Pichia pastoris</i>)	SEQ ID NO: 18

Transketolase *Kluyveromyces lactis* protein SEQ ID NO: 20 Transketolase *Kluyveromyces lactis* DNA SEQ ID NO: 21 Promoter Motif Artificial sequence DNA SEQ ID NO: 22 Promoter Motif Artificial sequence DNA SEQ ID NO: 23 Promoter ANB1 *Saccharomyces cerevisiae* DNA SEQ ID NO: 24 Promoter DAN1 *Saccharomyces cerevisiae* DNA SEQ ID NO: 25 Ribulose-1,5-biphosphate *Thiobacillus denitrificans* protein carboxylase oxygenase (RuBisCO) as previously described in WO2014/129898 SEQ ID NO: 26 Ribulose-1,5-biphosphate Artificial sequence; based on DNA carboxylase oxygenase cbbM gene from *Thiobacillus* (RuBisCO) as previously described in WO2014/129898 SEQ ID NO: 27 Phosphoribulokinase (PRK) as *Spinacea oleracea* protein previously described in WO2014/129898 SEQ ID NO: 28 Phosphoribulokinase (PRK) as Artificial sequence; based on DNA previously described in prk gene from *Spinacea* WO2014/129898 *oleracea*, codon optimized SEQ ID NO: 29 (long) chaperone groEL as Artificial sequence; Translated protein previously described in protein sequence based on WO2014/129898 GroEL of *Escherichia coli* SEQ ID NO: 30 (long) chaperone groEL as Artificial sequence; based on DNA previously described in GroEL gene from *Escherichia* WO2014/129898 *coli*, codon optimized SEQ ID NO: 31 (short) chaperone groES as Artificial sequence; Translated protein previously described in protein sequence based on WO2014/129898 GroES of *Escherichia coli* SEQ ID NO: 32 (short) chaperone groES as Artificial sequence; based on DNA previously described in GroES gene from *Escherichia* WO2014/129898 *coli*, codon optimized SEQ ID NO: 33 Phosphoketolase *Bifidobacterium animalis* protein SEQ ID NO: 34 Phosphoketolase *Bifidobacterium adolescentis* protein SEQ ID NO: 35 Phosphoketolase *Bifidobacterium lactis* protein SEQ ID NO: 36 Phosphoketolase *Leuconostoc mesenteroides* protein SEQ ID NO: 37 Phosphotransacetylase *Bacillus subtilis* protein SEQ ID NO: 38 Phosphotransacetylase *Lactobacillus plantarum* protein SEQ ID NO: 39 Phosphotransacetylase *Bifidobacterium adoloscentis* protein SEQ ID NO: 40 Phosphotransacetylase *Methosarcina thermophila* protein SEQ ID NO: 41 Acetate kinase *Bifidobacterium adolescentis* protein SEQ ID NO: 42 Acetate kinase *Aspergillus nidulans* protein SEQ ID NO: 43 Bifunctional NAD.sup.+ dependent *Escherichia coli* protein acetylating acetaldehyde/alcohol dehydrogenase (adhE) SEQ ID NO: 44 Ethanolamine utilizing protein *Escherichia coli* protein (eutE) SEQ ID NO: 45 acetaldehyde dehydrogenase *Lactobacillus plantarum* protein (acdH) SEQ ID NO: 46 acetaldehyde dehydrogenase *Listeria innocua* protein (acdH) SEQ ID NO: 47 acetaldehyde/alcohol *Staphylococcus aureus* protein dehydrogenase (adhE) SEQ ID NO: 48 acetylating acetaldehyde *Escherichia coli* protein dehydrogenase (mhpF) SEQ ID NO: 49 Acetyl-coenzyme A *Saccharomyces cerevisiae* protein synthetase 1 (ACS1) SEQ ID NO: 50 Acetyl-coenzyme A *Saccharomyces cerevisiae* protein synthetase 2 (ACS2) SEQ ID NO: 51 Alcohol dehydrogenase 1 *Saccharomyces cerevisiae* protein (ADH1) SEQ ID NO: 52 Alcohol dehydrogenase 2 *Saccharomyces cerevisiae* protein (ADH2) SEQ ID NO: 53 Alcohol dehydrogenase 3 *Saccharomyces cerevisiae* protein (ADH3) SEQ ID NO: 54 Alcohol dehydrogenase 4 *Saccharomyces cerevisiae* protein (ADH4) SEQ ID NO: 55 Alcohol dehydrogenase 5 *Saccharomyces cerevisiae* protein (ADH5) SEQ ID NO: 56 Glucoamylase (GA) *Punctularia strigosoazonata* protein SEQ ID NO: 57 Glucoamylase (GA) *Punctularia strigosoazonata* protein SEQ ID NO: 58 Glucoamylase (GA) Artificial sequence, based on protein *Punctularia strigosoazonata* SEQ ID NO: 59 Glycerol dehydrogenase *Klebsiella pneumoniae* protein (Kpne_gldA) SEQ ID NO: 60 Glycerol dehydrogenase *Enterococcus aerogenes* protein (Eaer_gldA) SEQ ID NO: 61 Glycerol dehydrogenase *Yersinia aldovae* protein (Ya_gldA) SEQ ID NO: 62 Glycerol dehydrogenase *Escherichia coli* protein (Ec_gldA) SEQ ID NO: 63 Glycerol dehydrogenase Artificial sequence; based on DNA (Ec_gldA) gldA gene from *Escherichia coli* - codon optimized SEQ ID NO: 64 Dihydroxyacetone kinase *Saccharomyces cerevisiae* protein (DAK1) SEQ ID NO: 65 Dihydroxyacetone kinase *Saccharomyces cerevisiae* protein (DAK2) SEQ ID NO: 66 Dihydroxyacetone kinase *Klebsiella pneumoniae* protein (Kpne_dhaK) SEQ ID NO: 67 Dihydroxyacetone kinase *Yarrowia lipolytica* protein (Ylip_DAK1) SEQ ID NO: 68

Dihydroxyacetone kinase *Schizosaccharomyces pombe* protein (Spom_DAK1) SEQ ID NO: 69
 Dihydroxyacetone kinase *Saccharomyces cerevisiae* DNA (DAK1) SEQ ID NO: 70
 Dihydroxyacetone kinase *Saccharomyces cerevisiae* DNA (DAK2) SEQ ID NO: 71 Glycerol
 transporter *Danio rerio* protein Aquaporin 9 (Drer_T3) SEQ ID NO: 72 Glycerol transporter
Zygosaccharomyces rouxii protein ZYRO0E01210p (Zrou_T5) SEQ ID NO: 73 “fragmentA”
 Artificial Sequence DNA SEQ ID NO: 74 “fragmentB” Artificial Sequence DNA SEQ ID NO: 75
 “fragmentC” Artificial Sequence DNA SEQ ID NO: 76 “fragmentD” Artificial Sequence DNA
 SEQ ID NO: 77 “fragmentE” Artificial Sequence DNA SEQ ID NO: 78 repair DNA fragment
 Artificial Sequence DNA SEQ ID NO: 79 Terminator EBO1 (Sc_EBO1t) *Saccharomyces*
cerevisiae DNA SEQ ID NO: 80 Promoter ENO2 (Sc_ENO2p) *Saccharomyces cerevisiae* DNA
 SEQ ID NO: 81 *S. cerevisiae* YOX1 terminator *Saccharomyces cerevisiae* DNA (Sc_YOX1t) SEQ
 ID NO: 82 INT14.02 protospacer Artificial Sequence DNA SEQ ID NO: 83
 Sc_INT14.02_FLANK5 *Saccharomyces cerevisiae* DNA SEQ ID NO: 84 Sc_INT14.02_FLANK3
Saccharomyces cerevisiae DNA SEQ ID NO: 85 INT95 protospacer Artificial Sequence DNA SEQ
 ID NO: 86 Sc_INT95B_FLANK5 *Saccharomyces cerevisiae* DNA SEQ ID NO: 87
 Sc_INT95B_FLANK3 *Saccharomyces cerevisiae* DNA SEQ ID NO: 88 GPD2 protospacer
 Artificial Sequence DNA SEQ ID No: 89 NR1 nitrate reductase *Zea mays* Protein [Uniprot
 accession No: (amino Q9XGW5] acid sequenc) SEQ ID No: 90 NIA Nitrate reductase *Spinacia*
oleracea Protein [Uniprot accession No: P23312] SEQ ID No: 91 nirD Nitrite reductase
Escherichia coli Protein (small subunit) SEQ ID No: 92 nirB Nitrite reductase *Escherichia coli*
 Protein (large subunit) SEQ ID No: 93 niiA Nitrite reductase *Emericella nidulans* Protein
 (*Aspergillus nidulans*) SEQ ID No: 94 YNT1 Nitrate/Nitrite *Pichia angusta* Protein transporter
 (*Hansenula polymorpha*)
 [0021] In the context of this patent application, each of the above protein/amino acid sequences is
 preferably encoded by a DNA/nucleic acid sequence that is codon-pair optimized for expression in
 a yeast, more preferably for expression in a *Saccharomyces cerevisiae* yeast.

Description

DETAILED DESCRIPTION OF THE INVENTION

Definitions

[0022] Unless defined otherwise or clearly indicated by context, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art.

[0023] Throughout the present specification and the accompanying claims, the words “comprise” and “include” and variations such as “comprises”, “comprising”, “includes” and “including” are to be interpreted inclusively. That is, these words are intended to convey the possible inclusion of other elements or integers not specifically recited, where the context allows.

[0024] The articles “a” and “an” are used herein to refer to one or to more than one (i.e. to one or at least one) of the grammatical object of the article. By way of example, “an element” may mean one element or more than one element. When referring to a noun (e.g. a compound, an additive, etc.) in the singular, the plural is meant to be included. Thus, when referring to a specific moiety, e.g. “gene”, this means “at least one” of that gene, e.g. “at least one gene”, unless specified otherwise.

[0025] When referring to a compound of which several isomers exist (e.g. a D and an L enantiomer), the compound in principle includes all enantiomers, diastereomers and cis/trans isomers of that compound that may be used in the particular aspect of the invention; in particular when referring to such as compound, it includes the natural isomer(s).

[0026] Unless explicitly indicated otherwise, the various embodiments of the invention described herein can be cross-combined.

[0027] The term “carbon source” refers to a source of carbon, preferably a compound or molecule

comprising carbon. Preferably the carbon source is a carbohydrate. A carbohydrate is understood herein to be an organic compound made of carbon, oxygen and hydrogen. Suitably the carbon source may be selected from the group consisting of mono-, di- and/or polysaccharides, acids and acid salts. More preferably the carbon source is a compound selected from the group consisting of glucose, arabinose, xylose, galactose, mannose, rhamnose, fructose, glycerol, and acetic acid or a salt thereof.

[0028] The terms “dry matter” and “dry solids”, abbreviated respectively as “DM” and “DS”, are used interchangeably herein and refer to material remaining after removal of water. Dry matter content can be determined by any method known to the person skilled in the art therefore.

[0029] The term “ferment”, and variations thereof such as “fermenting”, “fermentation” and/or “fermentative”, is used herein in a classical sense, i.e. to indicate that a process is or has been carried out under anaerobic conditions. An anaerobic fermentation is herein defined to be a fermentation carried out under anaerobic conditions. Anaerobic conditions are herein defined as conditions without any oxygen or in which essentially no oxygen is consumed by the yeast cell. Conditions in which essentially no oxygen is consumed suitably corresponds to an oxygen consumption of less than 5 mmol/l.Math.h.sup.-1, in particular to an oxygen consumption of less than 2.5 mmol/l.Math.h.sup.-1, or less than 1 mmol/l.Math.h.sup.-1. More preferably 0 mmol/L/h is consumed (i.e. oxygen consumption is not detectable). This suitably corresponds to a dissolved oxygen concentration in a culture broth of less than 5% of air saturation, more suitably to a dissolved oxygen concentration of less than 1% of air saturation, or less than 0.2% of air saturation.

[0030] The term “fermentation process” refers to a process for the preparation or production of a fermentation product.

[0031] The term “cell” refers to a eukaryotic or prokaryotic organism, preferably occurring as a single cell. In the present invention the cell is a recombinant yeast cell. That is, the recombinant cell is selected from the group of genera consisting of yeast.

[0032] The terms “yeast” and “yeast cell” are used herein interchangeably and refer to a phylogenetically diverse group of single-celled fungi, most of which are in the division of Ascomycota and Basidiomycota. The budding yeasts (“true yeasts”) are classified in the order Saccharomycetales. The yeast cell according to the invention is preferably a yeast cell derived from the genus of *Saccharomyces*. More preferably the yeast cell is a yeast cell of the species *Saccharomyces cerevisiae*.

[0033] The term “recombinant”, for example referring to a “recombinant yeast”, a “recombinant cell”, “recombinant micro-organism” and/or “recombinant strain” as used herein, refers to a yeast, cell, micro-organism or strain, respectively, containing nucleic acid which is the result of one or more genetic modifications. Simply put the yeast, cell, micro-organism or strain contains a different combination of nucleic acid from (either of) its parent(s). To construe a recombinant yeast, cell, micro-organism or strain, recombinant DNA technique(s) and/or another mutagenic technique(s) can be used. For example a recombinant yeast and/or a recombinant yeast cell may comprise nucleic acid not present in the corresponding wild-type yeast and/or cell, which nucleic acid has been introduced into that yeast and/or yeast cell using recombinant DNA techniques (i.e. a transgenic yeast and/or cell), or which nucleic acid not present in said wild-type yeast and/or cell is the result of one or more mutations—for example using recombinant DNA techniques or another mutagenesis technique such as UV-irradiation—in a nucleic acid sequence present in said wild-type yeast and/or yeast cell (such as a gene encoding a wild-type polypeptide) or wherein the nucleic acid sequence of a gene has been modified to target the polypeptide product (encoding it) towards another cellular compartment. Further, the term “recombinant” may suitably relate to a yeast, cell, micro-organism or strain from which nucleic acid sequences have been removed, for example using recombinant DNA techniques.

[0034] By a recombinant yeast comprising or having a certain activity is herein understood that the recombinant yeast may comprise one or more nucleic acid sequences encoding for a protein having

such activity. Hence allowing the recombinant yeast to functionally express such a protein or enzyme.

[0035] The term “functionally expressing” means that there is a functioning transcription of the relevant nucleic acid sequence, allowing the nucleic acid sequence to actually be transcribed, for example resulting in the synthesis of a protein.

[0036] The term “transgenic” as used herein, for example referring to a “transgenic yeast” and/or a “transgenic cell”, refers to a yeast and/or cell, respectively, containing nucleic acid not naturally occurring in that yeast and/or cell and which has been introduced into that yeast and/or cell using for example recombinant DNA techniques, such as a recombinant yeast and/or cell.

[0037] The term “mutated” as used herein regarding proteins or polypeptides means that, as compared to the wild-type or naturally occurring protein or polypeptide sequence, at least one amino acid has been replaced with a different amino acid, inserted into, or deleted from the amino acid sequence. The replacement, insertion or deletion of the amino acid can for example be achieved via mutagenesis of nucleic acids encoding these amino acids. Mutagenesis is a well-known method in the art, and includes, for example, site-directed mutagenesis by means of PCR or via oligonucleotide-mediated mutagenesis as described in Sambrook et al., *Molecular Cloning-A Laboratory Manual*, 2nd ed., Vol. 1-3 (1989), published by Cold Spring Harbor Publishing).

[0038] The term “mutated” as used herein regarding genes means that, as compared to the wild-type or naturally occurring nucleic acid sequence, at least one nucleotide in the nucleic acid sequence of a gene or a regulatory sequence thereof, has been replaced with a different nucleotide, inserted into, or deleted from the nucleic acid sequence. The replacement, insertion or deletion of the amino acid can for example be achieved via mutagenesis, resulting for example in the transcription of a protein sequence with a qualitatively or quantitatively altered function or the knock-out of that gene. In the context of this invention an “altered gene” has the same meaning as a mutated gene.

[0039] The term “gen” or “gene”, as used herein, refers to a nucleic acid sequence that can be transcribed into mRNAs that are then translated into protein. A gene encoding for a certain protein refers to the one or more nucleic acid sequence(s) encoding for such a protein.

[0040] The term “nucleic acid” or “nucleotide” as used herein, refers to a monomer unit in a deoxyribonucleotide or ribonucleotide polymer, i.e. a polynucleotide, in either single or double-stranded form, and unless otherwise limited, encompasses known analogues having the essential nature of natural nucleotides in that they hybridize to single-stranded nucleic acids in a manner similar to naturally occurring nucleotides (e. g., peptide nucleic acids). For example, a certain enzyme that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to the reference nucleotide sequence encoding the enzyme. A polynucleotide can be full-length or a subsequence of a native or heterologous structural or regulatory gene. Unless otherwise indicated, the term includes reference to the specified sequence as well as the complementary sequence thereof. Thus, DNAs or RNAs with backbones modified for stability or for other reasons are “polynucleotides” as that term is intended herein. Moreover, DNAs or RNAs comprising unusual bases, such as inosine, or modified bases, such as tritylated bases, to name just two examples, are polynucleotides as the term is used herein. It will be appreciated that a great variety of modifications have been made to DNA and RNA that serve many useful purposes known to those of skill in the art. The term polynucleotide as it is employed herein embraces such chemically, enzymatically or metabolically modified forms of polynucleotides, as well as the chemical forms of DNA and RNA characteristic of viruses and cells, including among other things, simple and complex cells.

[0041] The terms “nucleotide sequence” and “nucleic acid sequence” are used interchangeably herein. An example of a nucleic acid sequence is a DNA sequence.

[0042] The terms “polypeptide”, “peptide” and “protein” are used interchangeably herein to refer to a polymer of amino acid residues, for example illustrated by an amino acid sequence. The terms

apply to amino acid polymers in which one or more amino acid residue is an artificial chemical analogue of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers. The essential nature of such analogues of naturally occurring amino acids is that, when incorporated into a protein, that protein is specifically reactive to antibodies elicited to the same protein but consisting entirely of naturally occurring amino acids. The terms “polypeptide”, “peptide” and “protein” are also inclusive of modifications including, but not limited to, glycosylation, lipid attachment, sulphation, gamma-carboxylation of glutamic acid residues, hydroxylation and ADP-ribosylation.

[0043] The term “enzyme” refers herein to a protein having a catalytic function. Where a protein catalyzes a certain biological reaction, the terms “protein” and “enzyme” may be used interchangeable herein. When an enzyme is mentioned with reference to an enzyme class (EC), the enzyme class is a class wherein the enzyme is classified or may be classified, on the basis of the Enzyme Nomenclature provided by the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (NC-IUBMB), which nomenclature may be found at <http://www.chem.qmul.ac.uk/iubmb/enzyme/>. Other suitable enzymes that have not (yet) been classified in a specified class but may be classified as such, are meant to be included.

[0044] If referred herein to a protein or a nucleic acid sequence, such as a gene, by reference to a accession number, this number in particular is used to refer to a protein or nucleic acid sequence (gene) having a sequence as can be found via www.ncbi.nlm.nih.gov/, (as available on 1 Oct. 2020) unless specified otherwise.

[0045] Every nucleic acid sequence herein that encodes a polypeptide also includes any conservatively modified variants thereof. This includes that, by reference to the genetic code, it describes every possible silent variation of the nucleic acid. The term “conservatively modified variants” applies to both amino acid and nucleic acid sequences. With respect to particular nucleic acid sequences, conservatively modified variants refers to those nucleic acids which encode identical or conservatively modified variants of the amino acid sequences due to the degeneracy of the genetic code. The term “degeneracy of the genetic code” refers to the fact that a large number of functionally identical nucleic acids encode any given protein. For instance, the codons GCA, GCC, GCG and GCU all encode the amino acid alanine. Thus, at every position where an alanine is specified by a codon, the codon can be altered to any of the corresponding codons described without altering the encoded polypeptide. Such nucleic acid variations are “silent variations” and represent one species of conservatively modified variation.

[0046] The term “functional homologue” (or in short “homologue”) of a polypeptide and/or amino acid sequence having a specific sequence (e.g. “SEQ ID NO: X”), as used herein, refers to a polypeptide and/or amino acid sequence comprising said specific sequence with the proviso that one or more amino acids are mutated, substituted, deleted, added, and/or inserted, and which polypeptide has (qualitatively) the same enzymatic functionality for substrate conversion.

[0047] The term “functional homologue” (or in short “homologue”) of a polynucleotide and/or nucleic acid sequence having a specific sequence (e.g. “SEQ ID NO: X”), as used herein, refers to a polynucleotide and/or nucleic acid sequence comprising said specific sequence with the proviso that one or more nucleic acids are mutated, substituted, deleted, added, and/or inserted, and which polynucleotide encodes for a polypeptide sequence that has (qualitatively) the same enzymatic functionality for substrate conversion. With respect to nucleic acid sequences, the term functional homologue is meant to include nucleic acid sequences which differ from another nucleic acid sequence due to the degeneracy of the genetic code and encode the same polypeptide sequence.

[0048] Sequence identity is herein defined as a relationship between two or more amino acid (polypeptide or protein) sequences or two or more nucleic acid (polynucleotide) sequences, as determined by comparing the sequences. Usually, sequence identities or similarities are compared over the whole length of the sequences compared. In the art, “identity” also means the degree of sequence relatedness between amino acid or nucleic acid sequences, as the case may be, as

determined by the match between strings of such sequences.

[0049] Amino acid or nucleotide sequences are said to be homologous when exhibiting a certain level of similarity. Two sequences being homologous indicate a common evolutionary origin. Whether two homologous sequences are closely related or more distantly related is indicated by “percent identity” or “percent similarity”, which is high or low respectively. Although disputed, to indicate “percent identity” or “percent similarity”, “level of homology” or “percent homology” are frequently used interchangeably. A comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. The skilled person will be aware of the fact that several different computer programs are available to align two sequences and determine the homology between two sequences (Kruskal et al., “*An overview of sequence comparison: Time warps, string edits, and macromolecules*”, (1983), Society for Industrial and Applied Mathematics (SIAM), Vol 25, No. 2, pages 201-237 and D. and the handbook edited by Sankoff and J. B. Kruskal, (ed.), “*Time warps, string edits and macromolecules: the theory and practice of sequence comparison*”, (1983), pp. 1-44, published by Addison-Wesley Publishing Company, Massachusetts USA).

[0050] The percent identity between two amino acid sequences can be determined using the Needleman and Wunsch algorithm for the alignment of two sequences. (Needleman et al “*A General Method Applicable to the Search for Similarities in the Amino Acid Sequence of Two Proteins*” (1970) J. Mol. Biol. Vol. 48, pages 443-453). The algorithm aligns amino acid sequences as well as nucleotide sequences. The Needleman-Wunsch algorithm has been implemented in the computer program NEEDLE. For the purpose of this invention the NEEDLE program from the EMBOSS package is used (version 2.8.0 or higher, see Rice et al, “*EMBOSS: The European Molecular Biology Open Software Suite*” (2000), Trends in Genetics vol. 16, (6) pages 276-277, <http://emboss.bioinformatics.nl/>). For protein sequences, EBLOSUM62 is used for the substitution matrix. For nucleotide sequences, EDNAFULL is used. Other matrices can be specified. The optional parameters used for alignment of amino acid sequences are a gap-open penalty of 10 and a gap extension penalty of 0.5. The skilled person will appreciate that all these different parameters will yield slightly different results but that the overall percentage identity of two sequences is not significantly altered when using different algorithms.

[0051] The homology or identity is the percentage of identical matches between the two full sequences over the total aligned region including any gaps or extensions. The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment including the gaps. The identity defined as herein can be obtained from NEEDLE and is labelled in the output of the program as “IDENTITY”.

[0052] The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment after subtraction of the total number of gaps in the alignment. The identity defined as herein can be obtained from NEEDLE by using the NOBRIEF option and is labelled in the output of the program as “longest-identity”.

[0053] A variant of a nucleotide or amino acid sequence disclosed herein may also be defined as a nucleotide or amino acid sequence having one or more substitutions, insertions and/or deletions as compared to the nucleotide or amino acid sequence specifically disclosed herein (e.g. in the sequence listing).

[0054] Optionally, in determining the degree of amino acid similarity, the skilled person may also take into account so-called “conservative” amino acid substitutions, as will be clear to the skilled person. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and threonine; a group of amino acids having amide-containing side chains is

asparagine and glutamine; a group of amino acids having aromatic side chains is phenylalanine, tyrosine, and tryptophan; a group of amino acids having basic side chains is lysine, arginine, and histidine; and a group of amino acids having sulphur-containing side chains is cysteine and methionine. In an embodiment, conservative amino acids substitution groups are: valine-leucine-isoleucine, phenylalanine-tyrosine, lysine-arginine, alanine-valine, and asparagine-glutamine. Substitutional variants of the amino acid sequence disclosed herein are those in which at least one residue in the disclosed sequences has been removed and a different residue inserted in its place. Preferably, the amino acid change is conservative. In an embodiment, conservative substitutions for each of the naturally occurring amino acids are as follows: Ala to Ser; Arg to Lys; Asn to Gln or His; Asp to Glu; Cys to Ser or Ala; Gln to Asn; Glu to Asp; Gly to Pro; His to Asn or Gln; Ile to Leu or Val; Leu to Ile or Val; Lys to Arg; Gln or Glu; Met to Leu or Ile; Phe to Met, Leu or Tyr; Ser to Thr; Thr to Ser; Trp to Tyr; Tyr to Trp or Phe; and, Val to Ile or Leu.

[0055] Nucleotide sequences of the invention may also be defined by their capability to hybridise with parts of specific nucleotide sequences disclosed herein, respectively, under moderate, or preferably under stringent hybridisation conditions. Stringent hybridisation conditions are herein defined as conditions that allow a nucleic acid sequence of at least about 25, preferably about 50 nucleotides, 75 or 100 and most preferably of about 200 or more nucleotides, to hybridise at a temperature of about 65° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at 65° C. in a solution comprising about 0.1 M salt, or less, preferably 0.2×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours and preferably washing is performed for at least one hour with at least two changes of the washing solution. These conditions will usually allow the specific hybridisation of sequences having about 90% or more sequence identity. Moderate conditions are herein defined as conditions that allow a nucleic acid sequences of at least 50 nucleotides, preferably of about 200 or more nucleotides, to hybridise at a temperature of about 45° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at room temperature in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours, and preferably washing is performed for at least one hour with at least two changes of the washing solution. These conditions will usually allow the specific hybridisation of sequences having up to 50% sequence identity. The person skilled in the art will be able to modify these hybridisation conditions in order to specifically identify sequences varying in identity between 50% and 90%.

[0056] “Expression” refers to the transcription of a gene into structural RNA (rRNA, tRNA) or messenger RNA (mRNA) with subsequent translation into a protein.

[0057] “Overexpression” refers to expression of a gene, respectively a nucleic acid sequence, by a recombinant cell in excess to its expression in a corresponding wild-type cell. Such overexpression can for example be arranged for by: increasing the frequency of transcription of one or more nucleic acid sequences, for example by operational linking of the nucleic acid sequence to a promoter functional within the recombinant cell; and/or by increasing the number of copies of a certain nucleic acid sequence.

[0058] The terms “upregulate”, “upregulated” and “upregulation” refer to a process by which a cell increases the quantity of a cellular component, such as RNA or protein. Such an upregulation may be in response to or caused by a genetic modification.

[0059] By the term “pathway” or “metabolic pathway” is herein understood a series of chemical reactions in a cell that build and breakdown molecules.

[0060] Nucleic acid sequences (i.e. polynucleotides) or proteins (i.e. polypeptides) may be native or heterologous to the genome of the host cell.

[0061] “Native”, “homologous” or “endogenous” with respect to a host cell, means that the nucleic acid sequence does naturally occur in the genome of the host cell or that the protein is naturally

produced by that cell. The terms “native”, “homologous” and “endogenous” are used interchangeably herein.

[0062] As used herein, “heterologous” may refer to a nucleic acid sequence or a protein. For example, “heterologous”, with respect to the host cell, may refer to a polynucleotide that does not naturally occur in that way in the genome of the host cell or that a polypeptide or protein is not naturally produced in that manner by that cell. A heterologous nucleic acid sequence is a nucleic acid that originates from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention. For example, a promoter operably linked to a native structural gene is from a species different from that from which the structural gene is derived, or, if from the same species, one or both are substantially modified from their original form. A heterologous protein may originate from a foreign species or, if from the same species, is substantially modified from its original form by deliberate human intervention. That is, heterologous protein expression involves expression of a protein that is not naturally expressed in that way in the host cell. The term “heterologous expression” refers to the expression of heterologous nucleic acids in a host cell. The expression of heterologous proteins in eukaryotic host cell systems such as yeast are well known to those of skill in the art. A polynucleotide comprising a nucleic acid sequence of a gene encoding a certain protein or enzyme with a specific activity can be expressed in such a eukaryotic system. In some embodiments, transformed/transfected cells may be employed as expression systems for the expression of the enzymes. Expression of heterologous proteins in yeast is well known. Sherman, F., et al., *Methods in Yeast Genetics*, (1986), published by Cold Spring Harbor Laboratory, is a well-recognized work describing the various methods available to express proteins in yeast. Two widely utilized yeasts are *Saccharomyces cerevisiae* and *Pichia pastoris*. Vectors, strains, and protocols for expression in *Saccharomyces* and *Pichia* are known in the art and available from commercial suppliers (e.g., Invitrogen). Suitable vectors usually have expression control sequences, such as promoters, including 3-phosphoglycerate kinase or alcohol oxidase, and an origin of replication, termination sequences and the like as desired.

[0063] As used herein “promoter” is a DNA sequence that directs the transcription of a (structural) gene or other (part of) nucleic acid sequence. Suitably, a promoter is located in the 5'-region of a gene, proximal to the transcriptional start site of a (structural) gene. Promoter sequences may be constitutive, inducible or repressible. In an embodiment there is no (external) inducer needed.

[0064] The term “vector” as used herein, includes reference to an autosomal expression vector and to an integration vector used for integration into the chromosome.

[0065] The term “expression vector” refers to a DNA molecule, linear or circular, that comprises a segment encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional segments may include promoter and terminator sequences, and may optionally include one or more origins of replication, one or more selectable markers, an enhancer, a polyadenylation signal, and the like. Expression vectors are generally derived from plasmid or viral DNA, or may contain elements of both. In particular an expression vector comprises a nucleic acid sequence that comprises in the 5' to 3' direction and operably linked: (a) a yeast-recognized transcription and translation initiation region, (b) a coding sequence for a polypeptide of interest, and (c) a yeast-recognized transcription and translation termination region.

[0066] “Plasmid” refers to autonomously replicating extrachromosomal DNA which is not integrated into a microorganism's genome and is usually circular in nature.

[0067] An “integration vector” refers to a DNA molecule, linear or circular, that can be incorporated in a microorganism's genome and provides for stable inheritance of a gene encoding a polypeptide of interest. The integration vector generally comprises one or more segments comprising a gene sequence encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional

segments may include promoter and terminator sequences, and one or more segments that drive the incorporation of the gene of interest into the genome of the target cell, usually by the process of homologous recombination. Typically, the integration vector will be one which can be transferred into the target cell, but which has a replicon which is nonfunctional in that organism. Integration of the segment comprising the gene of interest may be selected if an appropriate marker is included within that segment.

[0068] By “host cell” is herein understood a cell, such as a yeast cell, that is to be transformed with one or more nucleic acid sequences encoding for one or more heterologous proteins, to construe a transformed cell, also referred to as a recombinant cell. For example, the transformed cell may contain a vector and may support the replication and/or expression of the vector.

[0069] “Transformation” and “transforming”, as used herein, refers to the insertion of an exogenous polynucleotide into a host cell, irrespective of the method used for the insertion, for example, direct uptake, transduction, f-mating or electroporation. The exogenous polynucleotide may be maintained as a non-integrated vector, for example, a plasmid, or alternatively, may be integrated into the host cell genome. “Transformation” and “transforming”, as used herein, refers to the insertion of an exogenous polynucleotide (i.e. an exogenous nucleic acid sequence) into a host cell, irrespective of the method used for the insertion, for example, direct uptake, transduction, f-mating or electroporation. The exogenous polynucleotide may be maintained as a non-integrated vector, for example, a plasmid, or alternatively, may be integrated into the host cell genome.

[0070] By “constitutive expression” and “constitutively expressing” is herein understood that there is a continuous transcription of a nucleic acid sequence. That is, the nucleic acid sequence is transcribed in an ongoing manner. Constitutively expressed genes are always “on”.

[0071] By “anaerobic constitutive expression” is herein understood that nucleic acid sequence is constitutively expressed in an organism under anaerobic conditions. That is, under anaerobic conditions the nucleic acid sequence is transcribed in an ongoing manner, i.e. under such anaerobic conditions the genes are always “on”.

[0072] By “disruption” is herein understood any disruption of activity, including, but not limited to, deletion, mutation and reduction of the affinity of the disrupted gene and expression of RNA complementary to such disrupted gene. It includes all nucleic acid modifications such as nucleotide deletions or substitutions, gene knock-outs, and other actions which affect the translation or transcription of the corresponding polypeptide and/or which affect the enzymatic (specific) activity, its substrate specificity, and/or or stability. It also includes modifications that may be targeted on the coding sequence or on the promotor of the gene. A gene disruptant is a cell that has one or more disruptions of the respective gene. Native to yeast herein is understood as that the gene is present in the yeast cell before the disruption.

[0073] The term “encoding” has the same meaning as “coding for”. Thus, by way of example, “one or more genes encoding a protein having activity X” has the same meaning as “one or more genes coding for a protein having activity X”.

[0074] As far as genes or nucleic acid sequences encoding a protein or an enzyme are concerned, the phrase “a nucleic acid sequence encoding a X”, respectively “one or more nucleic acid sequences encoding a X”, wherein X denotes a certain protein or (enzymatic) activity, has the same meaning as “a nucleic acid sequence encoding a protein having X activity”, respectively “one or more nucleic acid sequences encoding a protein having X activity”. Thus, by way of example, “one or more nucleic acid sequences encoding a transketolase” has the same meaning as “one or more nucleic acid sequences encoding a protein having transketolase activity”. As indicated above, the article “a” refers to “one or more”.

[0075] By a “redox sink” is herein understood a metabolic pathway that, overall, consumes or oxidizes NADH into NAD⁺ and/or prevents or reduces the consumption or reduction of NAD⁺ into NADH. A non-native metabolic pathway is a metabolic pathway that does not occur in the corresponding wild-type cell. Hence, a non-native metabolic pathway forming a redox sink is

preferably a non-native metabolic pathway that, as compared to a corresponding wild-type yeast cell, increases NADH consumption and/or decreases NAD⁺ consumption. By increasing NADH consumption and/or decreasing NAD⁺ consumption advantageously an (additional) non-native redox sink can be created within the cell.

[0076] The abbreviation “NADH” refers to reduced, hydrogenated form of nicotinamide adenine dinucleotide. The abbreviation “NAD⁺” refers to the oxidized form of nicotinamide adenine dinucleotide. Nicotinamide adenine dinucleotide may act as a so-called cofactor, assisting in biochemical reactions and/or transformations in a cell.

[0077] “NADH dependent” or “NAD⁺ dependent” is herein equivalent to NADH specific and “NADH dependency” or “NAD⁺ dependency” is herein equivalent to NADH specificity.

[0078] By a “NADH dependent” or “NAD⁺ dependent” enzyme is herein understood an enzyme that is exclusively depended on NADH/NAD⁺ as a co-factor or that is predominantly dependent on NADH/NAD⁺ as a cofactor, i.e. as contrasted to other types of co-factor. By an “exclusive NADH/NAD⁺ dependent” enzyme is herein understood an enzyme that has an absolute requirement for NADH/NAD⁺ over NADPH/NADP⁺. That is, it is only active when NADH/NAD⁺ is applied as cofactor. By a “predominantly NADH/NDA⁺-dependent” enzyme is herein understood an enzyme that has a higher specificity and/or a higher catalytic efficiency for NADH/NAD⁺ as a cofactor than for NADPH/NADP⁺ as a cofactor.

[0079] The enzyme's specificity characteristics can be described by the formula:

$$1 < K_{\text{sub.m}} \text{NADP}^{\text{sup.}+} / K_{\text{sub.m}} \text{NAD}^{\text{sup.}+} < \infty (\text{infinity})$$

wherein $K_{\text{sub.m}}$ is the so-called Michaelis constant.

[0080] For a predominantly NADH-dependent enzyme, preferably

$K_{\text{sub.m}} \text{NADP}^{\text{sup.}+} / K_{\text{sub.m}} \text{NAD}^{\text{sup.}+}$ is between 1 and 1000, between 1 and 500, between 1 and 200, between 1 and 100, between 1 and 50, between 1 and 10, between 5 and 100, between 5 and 50, between 5 and 20 or between 5 and 10.

[0081] The $K_{\text{sub.m}}$'s for the enzymes herein can be determined as enzyme specific, for NAD^{sup.+} and NADP^{sup.+} respectively, using know analysis techniques, calculations and protocols. These are described for instance in Lodish et al., Molecular Cell Biology 6^{sup.th} Edition, Ed. Freeman, pages 80 and 81, e.g. Figure 3-22. For an predominantly NADH-dependent enzyme, preferably the ratio of the catalytic efficiency for NADPH/NADP⁺ as a cofactor ($k_{\text{sub.cat}} / K_{\text{sub.m}}$)^{sup.}NADP⁺ to NADH/NAD⁺ as cofactor ($k_{\text{sub.cat}} / K_{\text{sub.m}}$)^{sup.}NAD⁺, i.e. the catalytic efficiency ratio ($k_{\text{sub.cat}} / K_{\text{sub.m}}$)^{sup.}NADP⁺ : ($k_{\text{sub.cat}} / K_{\text{sub.m}}$)^{sup.}NAD⁺, is more than 1:1, more preferably equal to or more than 2:1, still more preferably equal to or more than 5:1, even more preferably equal to or more than 10:1, yet even more preferably equal to or more than 20:1, even still more preferably equal to or more than 100:1, and most preferably equal to or more than 1000:1. There is no upper limit, but for practical reasons the predominantly NADH-dependent enzyme may have a catalytic efficiency ratio ($k_{\text{sub.cat}} / K_{\text{sub.m}}$)^{sup.}NADP⁺ : ($k_{\text{sub.cat}} / K_{\text{sub.m}}$)^{sup.}NAD⁺ of equal to or less than 1.000.000.000:1 (i.e. 1.10^{sup.9:1}).

The Yeast Cell

[0082] The recombinant yeast cell is preferably a yeast cell, or derived from a yeast cell, from the genus of Saccharomycetaceae or the genus of Schizosaccharomycetaceae. That is, preferably the host cell from which the recombinant yeast cell is derived is a yeast cell from the genus of Saccharomycetaceae or the genus of Schizosaccharomycetaceae.

[0083] Examples of suitable yeast cells include *Saccharomyces*, such as *Saccharomyces cerevisiae*, *Saccharomyces eubayanus*, *Saccharomyces jurei*, *Saccharomyces pastorianus*, *Saccharomyces beticus*, *Saccharomyces fermentati*, *Saccharomyces paradoxus*, *Saccharomyces uvarum* and *Saccharomyces bayanus*.

[0084] Examples of suitable yeast cells further include *Schizosaccharomyces*, such as *Schizosaccharomyces pombe*, *Schizosaccharomyces japonicus*, *Schizosaccharomyces octosporus*

and *Schizosaccharomyces cryophilus*.

[0085] Other exemplary yeasts include *Torulaspora* such as *Torulaspora delbrueckii*; *Kluyveromyces* such as *Kluyveromyces marxianus*; *Pichia* such as *Pichia stipitis*, *Pichia pastoris* or *Pichia angusta*; *Zygosaccharomyces* such as *Zygosaccharomyces bailii*; *Brettanomyces* such as *Brettanomyces intermedius*; *Brettanomyces bruxellensis*, *Brettanomyces anomalus*, *Brettanomyces custersianus*, *Brettanomyces naardenensis*, *Brettanomyces nanus*, *Dekkera bruxellensis* and *Dekkera anomala*; *Metschnikowia*, *Issatchenkia*, such as *Issatchenkia orientalis*, *Kloeckera* such as *Kloeckera apiculata*; and *Aureobasidium* such as *Aureobasidium pullulans*.

[0086] The yeast cell is preferably a yeast cell of the genus *Schizosaccharomyces*, herein also referred to as a *Schizosaccharomyces* yeast cell, or a yeast cell of the genus *Saccharomyces*, herein also referred to as a *Saccharomyces* yeast cell. More preferably the yeast cell is a yeast cell derived from a yeast cell of the species *Saccharomyces cerevisiae*, herein also referred to as a *Saccharomyces cerevisiae* yeast cell. That is, preferably the host cell from which the recombinant yeast cell is derived is a yeast cell from the species *Saccharomyces cerevisiae*.

[0087] Preferably the yeast cell is an industrial yeast cell. The living environments of yeast cells in industrial processes are significantly different from that in the laboratory. Industrial yeast cells must be able to perform well under multiple environmental conditions which may vary during the process. Such variations include changes in nutrient sources, pH, ethanol concentration, temperature, oxygen concentration, etc., which together have potential impact on the cellular growth and ethanol production of the yeast cell. An industrial yeast cell can be understood to refer to a yeast cell that, when compared to a laboratory counterpart, has a more robust performance. That is, when compared to a laboratory counterpart, the industrial yeast cell shows less variation in performance when one or more environmental conditions selected from the group of nutrient sources, pH, ethanol concentration, temperature, oxygen concentration, are varied during fermentation. Preferably, the yeast cell is constructed on the basis of an industrial yeast cell as a host, wherein the construction is conducted as described hereinafter. Examples of industrial yeast cells are Ethanol Red® (Fermentis) Fermiol® (DSM) and Thermosacc® (Lallemand).

[0088] The recombinant yeast cell described herein may be derived from any host cell capable of producing a fermentation product. Preferably the host cell is a yeast cell, more preferably an industrial yeast cell as described herein above. Preferably the yeast cell described herein is derived from a host cell having the ability to produce ethanol.

[0089] The yeast cell described herein may be derived from the host cell through any technique known by one skilled in the art to be suitable therefore. Such techniques may include any one or more of mutagenesis, recombinant DNA technology (including, but not limited to, CRISPR-CAS techniques), selective and/or adaptive evolution, mating, cell fusion, and/or cytoduction between yeast strains. Suitably the one or more desired genes are incorporated in the yeast cell by a combination of one or more of the above techniques.

[0090] The recombinant yeast cells according to the invention are preferably inhibitor tolerant, i.e. they can withstand common inhibitors at the level that they typically have with common pretreatment and hydrolysis conditions, so that the recombinant yeast cells can find broad application, i.e. it has high applicability for different feedstock, different pretreatment methods and different hydrolysis conditions. In an embodiment the recombinant yeast cell is inhibitor tolerant. Inhibitor tolerance is resistance to inhibiting compounds. The presence and level of inhibitory compounds in lignocellulose may vary widely with variation of feedstock, pretreatment method hydrolysis process. Examples of categories of inhibitors are carboxylic acids, furans and/or phenolic compounds. Examples of carboxylic acids are lactic acid, acetic acid or formic acid. Examples of furans are furfural and hydroxy-methylfurfural. Examples of phenolic compounds are vanillin, syringic acid, ferulic acid and coumaric acid. The typical amounts of inhibitors are for carboxylic acids: several grams per liter, up to 20 grams per liter or more, depending on the feedstock, the pretreatment and the hydrolysis conditions. For furans: several hundreds of

milligrams per liter up to several grams per liter, depending on the feedstock, the pretreatment and the hydrolysis conditions. For phenolics: several tens of milligrams per liter, up to a gram per liter, depending on the feedstock, the pretreatment and the hydrolysis conditions.

[0091] In an embodiment, the recombinant yeast cell is a cell that is naturally capable of alcoholic fermentation, preferably, anaerobic alcoholic fermentation. A recombinant yeast cell preferably has a high tolerance to ethanol, a high tolerance to low pH (i.e. capable of growth at a pH lower than about 5, about 4, about 3, or about 2.5) and towards organic and/or a high tolerance to elevated temperatures.

Native Transketolase

[0092] The recombinant yeast cell is suitably functionally expressing: [0093] a nucleic acid sequence encoding a native protein having transketolase activity; and [0094] a nucleic acid sequence encoding a heterologous protein having transketolase activity.

[0095] A protein having transketolase activity is herein also referred to as “transketolase protein”, “transketolase enzyme” or simply “transketolase”. The “transketolase” is herein abbreviated as “TKL”.

[0096] In analogy, a native protein having transketolase activity is herein also referred to as “native transketolase protein”, “native transketolase enzyme” or simply “native transketolase”.

[0097] Transketolase is an enzyme that is active within the pentose phosphate pathway of a yeast cell. The genes encoding for this pentose phosphate pathway are herein also referred to as the “PPP” genes. Preferably references in this specification to the pentose phosphate pathway are to be understood as references to the non-oxidative part of the pentose phosphate pathway. The enzymes active within the pentose phosphate pathway include the enzymes ribulose-5-phosphate isomerase (RKI), ribulose-5-phosphate epimerase (RPE), transketolase (TKL) and transaldolase (TAL).

[0098] The enzyme “transketolase” (belonging to enzyme class EC 2.2.1.1) is herein defined as an enzyme that catalyses the reaction: D-ribose 5-phosphate+D-xylulose 5-phosphate $\rightleftharpoons$ Math.sedoheptulose 7-phosphate+D-glyceraldehyde 3-phosphate; and vice versa.

[0099] The enzyme is also known as glycolaldehydetransferase or sedoheptulose-7-phosphate:D-glyceraldehyde-3-phosphate glycolaldehydetransferase. A certain transketolase can be further defined by its amino acid sequence. Likewise a transketolase can be further defined by a nucleotide sequence encoding the transketolase. As explained in detail above under definitions, a certain transketolase that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the transketolase.

[0100] The nucleic acid sequences encoding the native protein having transketolase activity can in itself be homologous nucleic acid sequences, heterologous nucleic acid sequences or a mixture of homologous and heterologous nucleic acid sequences, provided that the protein that is encoded by such nucleic acid sequences is a transketolase protein that is native, i.e. endogenous, to the host cell. *Saccharomyces cerevisiae* is a preferred host cell. Hence, preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell, functionally expressing a nucleic acid sequence encoding a protein having transketolase activity that is native to the *Saccharomyces cerevisiae* yeast cell.

[0101] More preferably the nucleic acid sequence encoding the native protein having transketolase activity is native to the host cell.

[0102] The recombinant yeast cell is therefore preferably functionally expressing: [0103] a native nucleic acid sequence encoding a native protein having transketolase activity; and [0104] a heterologous nucleic acid sequence encoding a heterologous protein having transketolase activity.

[0105] Wild-type yeasts may comprise one or two native transketolase genes. In addition to a first transketolase gene “TKL1”, some yeasts, such as for example *Saccharomyces cerevisiae*, comprises the paralog “TKL2”, a second transketolase gene.

[0106] Suitably the recombinant yeast cells according to the invention may comprise a native

TKL1 gene and/or a native TKL2 gene.

[0107] That is, suitably the recombinant yeast cell may comprise: [0108] a nucleic acid sequence encoding for native TKL1 (e.g. a gene “TKL1”); or [0109] a nucleic acid sequence encoding for native TKL2 (e.g. a gene “TKL2”); or [0110] both a nucleic acid sequence encoding for native TKL1 (e.g. a gene “TKL1”) and a nucleic acid sequence encoding for native TKL2 (e.g. a gene “TKL2”).

[0111] Preferably the recombinant yeast cell comprises a native nucleotide sequence encoding for native transketolase TKL1. That is, preferably the recombinant yeast cell comprises a native TKL1 gene.

[0112] The recombinant yeast cell may comprise one or more copies, suitably in the range from equal to or more than 1 to equal to or less than 30 copies, preferably in the range equal to or more than 1 to equal to or less than 20 copies, of a native gene encoding a transketolase. More preferably the recombinant yeast cell comprises one, two, three, four, five, six, seven, eight, nine, ten, eleven or twelve copies of a native gene encoding a transketolase.

[0113] The recombinant yeast cell can be a recombinant yeast cell, wherein the native nucleic acid sequence encoding for the native protein having transketolase activity is under control of a TKL promoter, as described in detail herein below.

[0114] As indicated above, host cells from the species *Saccharomyces cerevisiae* are preferred. The amino acid sequence of native transketolase 1 of *Saccharomyces cerevisiae* is illustrated by SEQ ID NO: 1. The native nucleic acid sequence encoding transketolase 1 in *Saccharomyces cerevisiae* is illustrated by SEQ ID NO: 2.

[0115] Preferably the native protein having transketolase activity comprises or consists of: the amino acid sequence of SEQ ID NO: 1; or a functional homologue of SEQ ID NO: 1, suitably native to the host cell, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 1.

[0116] Preferably the, preferably native, nucleic acid sequence encoding the native transketolase protein therefore comprises or consists of: [0117] the nucleic acid sequence of SEQ ID NO: 2; or [0118] a functional homologue of SEQ ID NO: 2, encoding for a native transketolase, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 2.

[0119] More preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell comprising, respectively functionally expressing, a native protein having transketolase activity, which protein comprises or consists of the amino acid sequence of SEQ ID NO: 1. Most preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell comprising, respectively functionally expressing, a nucleic acid sequence comprising or consisting of SEQ ID NO: 2.

Heterologous Transketolase

[0120] As indicated above, the recombinant yeast cell is suitably functionally expressing: [0121] a nucleic acid sequence encoding a native protein having transketolase activity; and [0122] a nucleic acid sequence encoding a heterologous protein having transketolase activity.

[0123] A heterologous protein having transketolase activity is herein also referred to as “heterologous transketolase protein”, “heterologous transketolase enzyme” or simply “heterologous transketolase”.

[0124] The recombinant yeast cell is suitably functionally expressing a heterologous nucleic acid sequence encoding the heterologous protein having transketolase activity.

[0125] Such heterologous nucleic acid sequence encoding for the heterologous protein having transketolase activity, respectively the heterologous transketolase, is suitably present in addition to the, preferably native, nucleic acid sequence encoding for the native protein having transketolase

activity, respectively the native transketolase.

[0126] Preferably the recombinant yeast cell comprises a heterologous nucleic acid sequence encoding for a heterologous transketolase in addition to a native nucleic acid sequence encoding for a native transketolase.

[0127] Such heterologous nucleic acid sequence encoding for the transketolase is preferably under control of a TKL promoter as detailed herein below.

[0128] The recombinant yeast cell can comprise, respectively functionally express, one or more heterologous nucleic acid sequence(s) encoding for one or more heterologous transketolase protein(s). That is the recombinant yeast cell can comprise one or more heterologous transketolase (s).

[0129] Preferably the one or more heterologous transketolase(s) comprise(s) or consist(s) of:

[0130] the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19; or [0131] a functional homologue of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19; or [0132] a functional homologue of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19, comprising an amino acid sequence having one or more mutations, substitutions, insertions and/or deletions when compared with the amino sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19, where more preferably the amino acid sequence of any such functional homologue has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions as compared to the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19.

[0133] Preferably the recombinant yeast cell comprises, respectively functionally expresses: [0134] one or more heterologous nucleic acid sequences encoding for one or more amino acid sequence(s) chosen from the group consisting of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19; and/or [0135] functional homologues thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with any of those; and/or [0136] functional homologues thereof comprising a nucleic acid sequence having one or more mutations, substitutions, insertions and/or deletions when compared therewith.

[0137] More preferably the nucleic acid sequence of any such functional homologues has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than

75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions as compared to such nucleic acid sequences.

[0138] More preferably the heterologous transketolase is derived from a *Komagataella phaffii*, a yeast species also referred to as "*Pichia pastoris*". For example, preferably the heterologous transketolase comprises or consists of an amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 16 or SEQ ID NO: 17; or a functional homologues thereof comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 16 or SEQ ID NO: 17; or a functional homologue thereof comprising an amino acid sequence having one or more mutations, substitutions, insertions and/or deletions when compared with the amino sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 16 or SEQ ID NO: 17, where more preferably the amino acid sequence of any such functional homologue has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions as compared to the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 16 or SEQ ID NO: 17.

[0139] In order to allow for a good expression of the heterologous transketolase in the host cell, it can be advantageous to use a heterologous transketolase that may have an amino acid sequence having equal to or more than 30%, equal to or more than 35%, equal to or more than 40%, equal to or more than 45%, equal to or more than 50%, equal to or more than 55%, equal to or more than 60%, equal to or more than 65%, equal to or more than 70%, equal to or more than 75%, equal to or more than 80%, equal to or more than 85%, equal to or more than 90% equal to or more than 95%, equal to or more than 98% or equal to or more than 99% sequence identity with the amino acid sequence of the native transketolase of the host cell.

[0140] However, it may also be preferred for the heterologous transketolase to be a heterologous transketolase that is not regulated by native (i.e. endogenous) regulators of the host cell. That is, preferably the heterologous transketolase is a transketolase enzyme of which the activity cannot be increased or decreased by molecules that are natively produced by the host cell. In order to avoid native regulators, it can be advantageous to use a heterologous transketolase in the host cell that may have an amino acid sequence having equal to or less than 99%, equal to or less than 98%, equal to or less than 95%, equal to or less than 90%, equal to or less than 85%, equal to or less than 80%, equal to or less than 75%, equal to or less than 70%, or equal to or less than 65% sequence identity with the amino acid sequence of the native transketolase of the host cell.

[0141] Therefore, more preferably the heterologous transketolase comprises or consists of an amino acid sequence having a percentage identity with the amino acid sequence of the native transketolase of the host cell in the range of equal to or more than 30% to equal to or less than 80%, more preferably in the range of equal to or more than 30%, or equal to or more than 35%, to equal to or less than 75%, and most preferably in the range of equal to or more than 35% to equal to or less than 70% or even equal to or less than 65%.

[0142] More preferably any heterologous nucleic acid sequence encoding for the heterologous transketolase comprises or consists of a nucleic acid sequence having a percentage identity with the nucleic acid sequence encoding for the native transketolase of the host cell in the range of equal to or more than 30% to equal to or less than 80%, more preferably in the range of equal to or more than 30%, or equal to or more than 35%, to equal to or less than 75%, and most preferably in the range of equal to or more than 35% to equal to or less than 70% or even equal to or less than 65%.

[0143] Host cells from the species *Saccharomyces cerevisiae* are preferred. As indicated above, the amino acid sequence of native transketolase 1 of *Saccharomyces cerevisiae* is illustrated by SEQ ID NO: 1, the native nucleic acid sequence encoding transketolase 1 in *Saccharomyces cerevisiae*

is illustrated by SEQ ID NO: 2.

[0144] The recombinant yeast cell is therefore preferably a recombinant *Saccharomyces cerevisiae* yeast cell, functionally expressing: [0145] a nucleic acid sequence encoding a native transketolase, wherein the native transketolase preferably comprises or consists of an amino acid sequence of SEQ ID NO: 1; and [0146] a nucleic acid sequence encoding a heterologous transketolase, wherein the heterologous transketolase preferably comprises or consists of an amino acid sequence having in the range of equal to or more than 30% to equal to or less than 80%, more preferably in the range of equal to or more than 30%, or equal to or more than 35%, to equal to or less than 75%, and most preferably in the range of equal to or more than 35% to equal to or less than 70% or even equal to or less than 65%, sequence identity with the amino acid sequence of SEQ ID NO: 1.

[0147] Similarly, the recombinant yeast cell is preferably a recombinant *Saccharomyces cerevisiae* yeast cell, functionally expressing: [0148] a native nucleic acid sequence encoding a native transketolase, wherein the native nucleic acid preferably comprises or consists of the nucleic acid sequence of SEQ ID NO: 2; and [0149] a heterologous nucleic acid sequence encoding a heterologous transketolase, wherein the heterologous nucleic acid sequence preferably comprises or consists of a nucleic acid sequence having in the range of equal to or more than 30% to equal to or less than 80%, more preferably in the range of equal to or more than 35%, or equal to or more than 35%, to equal to or less than 75%, and most preferably in the range of equal to or more than 35% to equal to or less than 70% or even equal to or less than 65%, sequence identity with the nucleic acid sequence of SEQ ID NO: 2.

[0150] Most preferably the recombinant yeast cell comprises a, suitably heterologous, nucleic acid sequence encoding the heterologous protein having transketolase activity, wherein the nucleic acid sequence comprises or consists of: [0151] an nucleic acid sequence of SEQ ID NO: 18 or SEQ ID NO: 20; or [0152] a functional homologue of SEQ ID NO: 18 or SEQ ID NO: 20, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 18 or SEQ ID NO: 20; or [0153] a functional homologue of SEQ ID NO: 18 or SEQ ID NO: 20, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 20 or SEQ ID NO: 22, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 18 or SEQ ID NO: 20.

[0154] The recombinant yeast cell may comprise one, two, or more copies of a heterologous nucleic acid sequence (e.g. a heterologous gene) encoding for the heterologous transketolase and/or one, two, or more copies of a native nucleic acid sequence (e.g. a native gene) encoding for the native transketolase.

[0155] Most preferably the recombinant yeast cell may comprise one, two, three, four, five, six, seven, eight, nine, ten, eleven or twelve copies of a heterologous nucleic acid sequence (e.g. a heterologous gene) encoding for the heterologous transketolase and/or one, two, three, four, five, six, seven, eight, nine, ten, eleven or twelve copies of a native nucleic acid sequence (e.g. a native gene) encoding for the native transketolase.

[0156] Preferably the recombinant yeast cell is a recombinant yeast cell comprising, respectively functionally expressing: [0157] one, two or more copies of a heterologous nucleic acid sequence comprises or consists of a nucleic acid sequence having in the range of equal to or more than 30% to equal to or less than 80%, more preferably in the range of equal to or more than 35% to equal to or less than 75%, and most preferably in the range of equal to or more than 35% to equal to or less than 70% or even equal to or less than 65%, sequence identity with the nucleic acid sequence encoding for the native transketolase; and/or [0158] one, two or more copies of a nucleic acid

sequence encoding for any of the above mentioned heterologous transketolases; and/or [0159] one, two or more copies of a nucleic acid sequence of SEQ ID NO: 18 and/or SEQ ID NO: 20; [0160] and/or [0161] one, two or more copies of a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 18 and/or SEQ ID NO: 20; and/or [0162] one, two or more copies of a nucleic acid sequence having one or more mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of respectively SEQ ID NO: 18 and/or SEQ ID NO: 20, wherein more preferably this nucleic acid sequence has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of respectively SEQ ID NO: 18 and/or SEQ ID NO: 20.

[0163] Most preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell comprising, respectively functionally expressing: [0164] i) one, two or more copies of the nucleic acid sequence of SEQ ID NO: 10; and [0165] ii) one, two or more copies of a nucleic acid sequence of SEQ ID NO: 18 and/or SEQ ID NO: 20; and/or one, two or more copies of a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 18 and/or SEQ ID NO: 20; and/or one, two or more copies of a nucleic acid sequence having one or more mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of respectively SEQ ID NO: 18 and/or SEQ ID NO: 20, wherein more preferably this nucleic acid sequence has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of respectively SEQ ID NO: 18 and/or SEQ ID NO: 20.

Optional Overexpression of One or More Other Enzymes of the PPP Pathway

[0166] The recombinant yeast cell may further optionally comprise one or more genetic modifications in the other PPP-genes, i.e. RKI, RPE and TAL, that increase the flux of the pentose phosphate pathway. Advantageously, such genetic modification(s) may lead to a further increased flux through the non-oxidative part of the pentose phosphate pathway.

[0167] The recombinant yeast cell may thus optionally comprise one or more additional genetic modifications to overexpress one or more other enzymes of the (non-oxidative part of) the pentose phosphate pathway. For example, the recombinant yeast cell may comprise one or more nucleic acid sequences to overexpress one or more of the enzymes selected from the group consisting of ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase and transaldolase.

[0168] The enzyme “ribulose 5-phosphate epimerase” (EC 5.1.3.1) is herein defined as an enzyme that catalyses the epimerisation of D-xylulose 5-phosphate into D-ribulose 5-phosphate and vice versa. The enzyme is also known as phosphoribulose epimerase; erythrose-4-phosphate isomerase; phosphoketopentose 3-epimerase; xylulose phosphate 3-epimerase; phosphoketopentose epimerase; ribulose 5-phosphate 3-epimerase; D-ribulose phosphate-3-epimerase; D-ribulose 5-phosphate epimerase; D-ribulose-5-P 3-epimerase; D-xylulose-5-phosphate 3-epimerase; pentose-5-phosphate 3-epimerase; or D-ribulose-5-phosphate 3-epimerase. A ribulose 5-phosphate epimerase may be further defined by its amino acid sequence. Likewise a ribulose 5-phosphate epimerase may be defined by a nucleotide sequence encoding the enzyme as well as by a nucleotide sequence hybridising to a reference nucleotide sequence encoding a ribulose 5-phosphate epimerase. The nucleotide sequence encoding for ribulose 5-phosphate epimerase is herein designated as RPE or RPE1.

[0169] The enzyme “ribulose 5-phosphate isomerase” (EC 5.3.1.6) is herein defined as an enzyme

that catalyses direct isomerisation of D-ribose 5-phosphate into D-ribulose 5-phosphate and vice versa. The enzyme is also known as phosphopentosisomerase; phosphoriboisomerase; ribose phosphate isomerase; 5-phosphoribose isomerase; D-ribose 5-phosphate isomerase; D-ribose-5-phosphate ketol-isomerase; or D-ribose-5-phosphate aldose-ketose-isomerase. A ribulose 5-phosphate isomerase may be further defined by its amino acid sequence. Likewise a ribulose 5-phosphate isomerase may be defined by a nucleotide sequence encoding the enzyme as well as by a nucleotide sequence hybridising to a reference nucleotide sequence encoding a ribulose 5-phosphate isomerase. The nucleotide sequence encoding for ribulose 5-phosphate isomerase is herein designated RKI or RKI1.

[0170] The enzyme “transaldolase” (EC 2.2.1.2) is herein defined as an enzyme that catalyses the reaction: sedoheptulose 7-phosphate+D-glyceraldehyde 3-phosphate $\leftrightarrow$ D-erythrose 4-phosphate+D-fructose 6-phosphate and vice versa. The enzyme is also known as dihydroxyacetone transferase; dihydroxyacetone synthase; formaldehyde transketolase; or sedoheptulose-7-phosphate:D-glyceraldehyde-3-phosphate glyceronetransferase. A transaldolase may be further defined by its amino acid sequence. Likewise a transaldolase may be defined by a nucleotide sequence encoding the enzyme as well as by a nucleotide sequence hybridising to a reference nucleotide sequence encoding a transaldolase. The nucleotide sequence encoding for transketolase from is herein designated TAL or TAL1.

TKL Promoter

[0171] The recombinant yeast cell is preferably a recombinant yeast cell, wherein the nucleic acid sequence encoding for the native protein having transketolase activity and/or the nucleic acid sequence encoding for the heterologous protein having transketolase activity is/are under control of a promoter (the “TKL promoter”), which TKL promoter has an anaerobic/aerobic expression ratio for the transketolase of 2 or more. Herewith is suitably meant that the expression of the heterologous and/or native transketolase (“TKL”) is at least a factor 2 higher under anaerobic conditions than under aerobic conditions. The above can alternatively be phrased as the recombinant yeast cell functionally expressing a nucleic acid sequence encoding the native transketolase, respectively the heterologous transketolase, wherein this native transketolase, respectively this heterologous transketolase, is under control of a promoter (the “TKL promoter”) which has a TKL expression ratio .sub.anaerobic/aerobic of 2 or more.

[0172] The TKL promoter can suitably be operably linked to the nucleic acid sequence encoding the protein having transketolase activity. Preferably, the TKL promoter is located in the 5'-region of a TKL gene, more preferably it is located proximal to the transcriptional start site of a TKL gene.

[0173] Preferably the TKL promoter is ROX1 repressed. ROX1 is herein Heme-dependent repressor of hypoxic gene(s); that mediates aerobic transcriptional repression of hypoxia induced genes such as COX5b and CYC7; the repressor function is regulated through decreased promoter occupancy in response to oxidative stress; and contains an HMG domain that is responsible for DNA bending activity; involved in the hyperosmotic stress resistance. ROX1 is regulated by oxygen.

[0174] Without wishing to be limited by any kind of theory it is believed that the regulation of ROX1 may function as follows: According to Kwast et al., “*Genomic Analysis of Anaerobically induced genes in Saccharomyces cerevisiae: Functional roles of ROX1 and other factors in mediating the anoxic response*”, (2002), Journal of bacteriology vol 184, no 1 pages 250-265, herein incorporated by reference,; “Although Rox1 functions in an O₂-independent manner, its expression is oxygen (heme) dependent, activated by the heme-dependent transcription factor Hap1 [19]. Thus, as oxygen levels fall to those that limit heme biosynthesis [20], ROX1 is no longer transcribed [21], its protein levels fall [22], and the genes it regulates are derepressed”. Further details and suitable motifs are provided by Keng, T. (1992), “*HAP1 and ROX1 form a regulatory pathway in the repression of HEM13 transcription in Saccharomyces cerevisiae*”, Mol. Cell. Biol. 12: pages 2616-2623, and Ter Kinde and de Steensma, “*A microarray-assisted screen for potential*

Hap1 and Rox1 target genes in Saccharomyces cerevisiae", (2002), Yeast 19: pages 825-840, incorporated herein by reference.

[0175] Preferably, the TKL promoter comprises a ROX1 binding motif. The TKL promoter may suitably comprise one or more ROX1 binding motif(s).

[0176] More preferably the TKL promoter can comprise in its nucleic acid sequence one or more copies of the motif NNNATTGTTNNN. Herein "N" represents a nucleic acid chosen from the group consisting of Adenine (A), Guanine (G), Cytosine (C) and Thymine (T). Such motif is illustrated by SEQ ID NO: 21.

[0177] More preferably, the TKL promoter comprises or consists of a nucleic acid sequence that is identical to the nucleic acid sequence of a, preferably native, promoter of a gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5, HEM13, YNR014W, YAR028W, FUN 57, COX5B, OYE2, SUR2, FRDS1, PIS1, LAC1, YGR035C, YAL028W, EUG1, HEM14, ISU2, ERG26, YMR252C and SML1, more preferably FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5 and HEM13, or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith. The reference to a native promoter is herein to the promoter that is native to the host cell.

[0178] Preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell and preferably the TKL promoter is a native promoter of a *Saccharomyces cerevisiae* gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5, HEM13, YNR014W, YAR028W, FUN 57, COX5B, OYE2, SUR2, FRDS1, PIS1, LAC1, YGR035C, YAL028W, EUG1, HEM14, ISU2, ERG26, YMR252C and SML1.

[0179] In addition or in the alternative, the TKL promoter preferably comprises in its nucleic acid sequence one or more copies of the motifs: TCGTTYAG and/or AAAAATTGTTGA. Herein "Y" represents C or T. The AAAAATTGTTGA motif is illustrated by SEQ ID NO: 22.

[0180] The TKL promoter can also comprise or consist of a nucleic acid sequence that is identical to the nucleic acid sequence of a, preferably native, promoter of a DAN, TIR or PAU gene. For example, the TKL promoter can suitably comprise or consist of a nucleic acid sequence of a, preferably native, promoter of a gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, YLL025W, YOR394W, YHL046C, YMR325W, YAL068C, YPL282C, PAU2, and PAU4 or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith. The reference to a native promoter is herein to the promoter that is native to the host cell.

[0181] Preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell and preferably the TKL promoter is a native promoter of a *Saccharomyces cerevisiae* gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, YLL025W, YOR394W, YHL046C, YMR325W, YAL068C, YPL282C, PAU2, and PAU4.

[0182] More preferably, the TKL promoter can comprise or consist of a sequence that is identical to the nucleic acid sequence of a, preferably native, promoter of a gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041 W, YKL224C, PAU3, and YLL025W or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith.

[0183] The nucleic acid sequence of the *S. cerevisiae* ANB1 promoter is illustrated in SEQ ID NO: 23. The nucleic acid sequence of the *S. cerevisiae* DAN1 promoter is illustrated in SEQ ID NO: 24. [0184] Preferred TKL promoters can thus comprise or consist of: [0185] a nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24; or [0186] a functional homologue of the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24; or [0187] a functional homologue of the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24, having one or more mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24, wherein more preferably the nucleic acid sequence has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24.

[0188] The TKL promoter can also be a synthetic oligonucleotide. That is, the TKL promoter may be a product of artificial oligonucleotide synthesis. Artificial oligonucleotide synthesis is a method in synthetic biology that is used to create artificial oligonucleotides, such as genes, in the laboratory. Commercial gene synthesis services are now available from numerous companies worldwide, some of which have built their business model around this task. Current gene synthesis approaches are most often based on a combination of organic chemistry and molecular biological techniques and entire genes may be synthesized “de novo”, without the need for precursor template DNA.

[0189] The TKL promoter has a TKL expression ratio .sub.anaerobic/aerobic of 2 or more, preferably of 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more. By a TKL expression ratio .sub.anaerobic/aerobic of 2 or more is suitably meant that the expression of the enzyme transketolase (“TKL”) is, under further identical expression conditions, at least a factor 2 higher under anaerobic conditions than under aerobic conditions.

[0190] There is no upper limit, and the TKL promoter can be a TKL promoter that allows the promoted transketolase gene to be expressed only at anaerobic conditions and not at aerobic conditions.

[0191] For practical reasons a TKL expression ratio .sub.anaerobic/aerobic in the range from equal to or more than 2 to equal to or less than 10×10 (i.e. 10×10) or to or less than 10×4 (i.e. 104) can be considered.

[0192] As indicated above, “Expression” herein refers to the transcription of a gene into structural RNA (rRNA, tRNA) or messenger RNA (mRNA) with subsequent translation into a protein.

[0193] The TKL expression ratio can for example be determined by measuring the amount of Transketolase (TKL) protein of cells grown under aerobic and anaerobic conditions. The amount of TKL protein can be determined by proteomics or any other method known to quantify protein amounts.

[0194] It is also possible to determine the level or transketolase (TKL) expression ratio by measuring the transketolase (TKL) activity of cells grown under aerobic and anaerobic conditions, e.g. in a cell-free extract.

[0195] In addition or in the alternative to the above, the level or TKL expression ratio can be determined by measuring the transcription level (e.g. as amount of mRNA) of the TKL gene of cells grown under aerobic and anaerobic conditions. The skilled person knows how to determine translation levels using methods commonly known in the art, e.g. Q-PCR, real-time PCR, northern blot, RNA-seq.

[0196] The TKL promoter advantageously enables higher expression of transketolase during

anaerobic conditions than under aerobic conditions. In the process according to the invention, the recombinant yeast cell preferably expresses transketolase, where the amount of transketolase expressed under anaerobic conditions is a multiplication factor higher than the amount of transketolase expressed under aerobic conditions and wherein this multiplication factor is preferably 2 or more, more preferably 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more.

Increased Flux

[0197] Preferably the genetic modification(s) made in respect of the PPP-genes, i.e. with respect to TKL1 and optionally RKI, RPE and TAL, cause an increased flux of the non-oxidative part of the pentose phosphate pathway is herein understood to mean a modification that increases the flux by at least a factor of about 1.1, about 1.2, about 1.5, about 2, about 5, about 10 or about 20 as compared to the flux in a strain which is genetically identical except for the genetic modification causing the increased flux. The flux of the non-oxidative part of the pentose phosphate pathway may be measured by growing the modified host on xylose as sole carbon source, determining the specific xylose consumption rate and subtracting the specific xylitol production rate from the specific xylose consumption rate, if any xylitol is produced. However, the flux of the non-oxidative part of the pentose phosphate pathway is proportional with the growth rate on xylose as sole carbon source, preferably with the anaerobic growth rate on xylose as sole carbon source. There is a linear relation between the growth rate on xylose as sole carbon source ($\mu_{\text{sub.max}}$) and the flux of the non-oxidative part of the pentose phosphate pathway. The specific xylose consumption rate ($Q_{\text{sub.s}}$) is equal to the growth rate (μ) divided by the yield of biomass on sugar ($Y_{\text{sub.xs}}$) because the yield of biomass on sugar is constant (under a given set of conditions: anaerobic, growth medium, pH, genetic background of the strain, etc.; i.e. $Q_{\text{sub.s}} = \mu / Y_{\text{sub.xs}}$). Therefore the increased flux of the non-oxidative part of the pentose phosphate pathway may be deduced from the increase in maximum growth rate under these conditions unless transport (uptake is limiting).

[0198] One or more genetic modifications that increase the flux of the pentose phosphate pathway may be introduced in the host cell in various ways. These including e.g. achieving higher steady state activity levels of xylulose kinase and/or one or more of the enzymes of the non-oxidative part pentose phosphate pathway and/or a reduced steady state level of unspecific aldose reductase activity. These changes in steady state activity levels may be effected by selection of mutants (spontaneous or induced by chemicals or radiation) and/or by recombinant DNA technology e.g. by overexpression or inactivation, respectively, of genes encoding the enzymes or factors regulating these genes.

[0199] In a preferred host cell, the genetic modification comprises overexpression of at least one enzyme of the (non-oxidative part) pentose phosphate pathway. Preferably the enzyme is selected from the group consisting of the enzymes encoding for ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase, transketolase and transaldolase. Various combinations of enzymes of the (non-oxidative part) pentose phosphate pathway may be overexpressed. E.g. the enzymes that are overexpressed may be at least the enzymes ribulose-5-phosphate isomerase and ribulose-5-phosphate epimerase; or at least the enzymes ribulose-5-phosphate isomerase and transketolase; or at least the enzymes ribulose-5-phosphate isomerase and transaldolase; or at least the enzymes ribulose-5-phosphate epimerase and transketolase; or at least the enzymes ribulose-5-phosphate epimerase and transaldolase; or at least the enzymes transketolase and transaldolase; or at least the enzymes ribulose-5-phosphate epimerase, transketolase and transaldolase; or at least the enzymes ribulose-5-phosphate isomerase, transketolase and transaldolase; or at least the enzymes ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase, and transaldolase; or at least the enzymes ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase, and transketolase. In one embodiment of the invention each of the enzymes ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase, transketolase and transaldolase are overexpressed in the host cell. More preferred is a host cell in which the genetic modification comprises at least overexpression of both

the enzymes transketolase and transaldolase as such a host cell is already capable of anaerobic growth on xylose. In fact, under some conditions host cells overexpressing only the transketolase and the transaldolase already have the same anaerobic growth rate on xylose as do host cells that overexpress all four of the enzymes, i.e. the ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase, transketolase and transaldolase. Moreover, host cells overexpressing both of the enzymes ribulose-5-phosphate isomerase and ribulose-5-phosphate epimerase are preferred over host cells overexpressing only the isomerase or only the epimerase as overexpression of only one of these enzymes may produce metabolic imbalances.

Redox Sink

[0200] Preferably the recombinant yeast cell can further comprise one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink.

[0201] For example, these one or more genetic modifications can be one or more genetic modifications for the functional expression of one or more, optionally heterologous, nucleic acid sequences encoding for one or more NAD⁺/NADH dependent proteins that function in a metabolic pathway to convert NADH to NAD⁺. Several examples of such metabolic pathways exist, as illustrated further below.

[0202] For example, the “one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink” can be chosen from the group consisting of: [0203] a) one or more genetic modifications comprising or consisting of: [0204] a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase activity (EC 4.1.2.9 or EC 4.1.2.22, PKL); and/or [0205] a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8); and/or [0206] a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0207] and/or [0208] b) one or more genetic modifications comprising or consisting of: [0209] a, preferably heterologous, nucleic acid sequence encoding for a protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or [0210] a, preferably heterologous, nucleic acid sequences encoding for a protein having phosphoribulokinase (PRK) activity; and [0211] optionally, a, preferably heterologous, nucleic acid sequence encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

and/or [0212] c) one or more genetic modifications comprising or consisting of: [0213] a, preferably heterologous, nucleic acid sequence encoding a protein comprising NADH dependent acetylating acetaldehyde dehydrogenase activity.

[0214] For example, WO2014/081803 describes a recombinant microorganism expressing a heterologous phosphoketolase, phosphotransacetylase or acetate kinase and bifunctional acetaldehyde-alcohol dehydrogenase, incorporated herein by reference; and WO2015/148272 describes a recombinant *S. cerevisiae* strain expressing a heterologous phosphoketolase, phosphotransacetylase and acetylating acetaldehyde dehydrogenase, incorporated herein by reference. Further WO2018172328A1 describes a recombinant cell that may comprise one or more (heterologous) genes coding for an enzyme having phosphoketolase activity. The phosphoketolase (PKL) routes described in WO2014/081803, WO2015/148272 and WO2018172328A1, all incorporated herein by reference, provide preferred metabolic pathways to convert NADH to NAD⁺ and the NADH dependent phosphoketolase described therein is a preferred NADH dependent protein for application in the current invention.

Rubisco

[0215] As indicated above, the recombinant yeast cell may advantageously functionally express one or more, preferably heterologous, nucleic acid sequences encoding for ribulose-1,5-phosphate carboxylase/oxygenase (EC4.1.1.39; Rubisco), and optionally one or more molecular chaperones

for Rubisco.

[0216] More preferably the recombinant yeast cell functionally expresses: [0217] a heterologous nucleic acid sequence encoding a protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or [0218] a heterologous nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity; and/or [0219] optionally one or more heterologous nucleic acid sequence encoding one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0220] The protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity is herein also referred to as “ribulose-1,5-biphosphate carboxylase oxygenase”, “ribulose-1,5-biphosphate carboxylase oxygenase protein”, “ribulose-1,5-biphosphate carboxylase oxygenase enzyme”, “Rubisco enzyme”, “Rubisco protein” or simply “Rubisco”. A ribulose-1,5-biphosphate carboxylase oxygenase may be further defined by its amino acid sequence. Likewise a ribulose-1,5-biphosphate carboxylase oxygenase may be further defined by a nucleotide sequence encoding the ribulose-1,5-biphosphate carboxylase oxygenase. As explained in detail above under definitions, a certain ribulose-1,5-biphosphate carboxylase oxygenase that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the ribulose-1,5-biphosphate carboxylase oxygenase.

Preferences for the Rubisco protein and the nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0221] The Rubisco protein may suitably be selected from the group of eukaryotic and prokaryotic Rubisco proteins. The Rubisco protein is preferably from a non-phototrophic organism. For example, the Rubisco protein may be from a chemolithoautotrophic microorganism. Good results have been achieved with a bacterial Rubisco protein. Preferably, the Rubisco protein originates from a *Thiobacillus*, in particular, *Thiobacillus denitrificans*, which is chemolithoautotrophic.

[0222] The Rubisco protein may be a single-subunit Rubisco protein or a Rubisco protein having more than one subunit. Preferably the Rubisco protein is a single-subunit Rubisco protein. Good results have been obtained with a Rubisco protein that is a so-called form-II Rubisco protein.

[0223] Especially good results were achieved with a Rubisco protein encoded by a cbbM gene, also referred to as CbbM.

[0224] A preferred Rubisco protein is the Rubisco protein encoded by the cbbM gene from *Thiobacillus denitrificans*. SEQ ID NO: 25 shows the amino acid sequence of a suitable Rubisco protein, encoded by the cbbM gene from *Thiobacillus denitrificans*. SEQ ID NO: 26 illustrates the nucleic acid sequence of the cbbM gene from *Thiobacillus denitrificans*, codon optimized for *S. cerevisiae*.

[0225] Preferably the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity thus comprises or consists of: [0226] an amino acid sequence of SEQ ID NO: 25; or [0227] a functional homologue of SEQ ID NO: 25, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 25; or [0228] a functional homologue of SEQ ID NO: 25, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 25, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 25.

[0229] Preferable the nucleic acid sequence encoding the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity comprises or consists of: [0230] a nucleic acid sequence of SEQ ID NO: 26; or [0231] a functional homologue of SEQ ID NO: 26, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least

80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 26; or [0232] a functional homologue of SEQ ID NO: 26, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 26, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 26.

[0233] Examples of other suitable Rubisco polypeptides and their origin are given in Table 1 of WO2014/129898, incorporated herein by reference, and in Table 2 below, with reference to the sequence identity with the amino acid sequence of SEQ ID NO:25.

[0234] The nucleic acid sequence (e.g. the gene) encoding for the ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) protein may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898 and by the article of Guadalupe-Medina et al., “Carbon dioxide fixation by Calvin-Cycle enzymes improves ethanol yield in yeast”, published in Biotechnol, Biofuels, 2013, vol. 6, p. 125, both herein incorporated by reference.

TABLE-US-00002 TABLE 2 Natural Rubisco polypeptides suitable for expression MAX Source
Accession no. ID (%) *Thiobacillus denitrificans* AAA99178.2 100 *Sideroxydans lithotrophicus* ES-
1 YP_003522651.1 94 *Thiothrix nivea* DSM 5205 ZP_10101642.1 91 *Halothiobacillus*
neapolitanus c2 YP_003262978.1 90 *Acidithiobacillus ferrooxidans* YP_002220242.1 88 ATCC
53993 *Rhodospirillum rubrum* T118 YP_522655.1 86 *Thiorhodococcus dreschii* AZ1
ZP_08824342.1 85 uncultured prokaryote AGE14067.1 82

[0235] As indicated above, the Rubisco protein is suitably functionally expressed in the recombinant yeast cell, at least during use in a fermentation process.

[0236] The nucleic acid sequence encoding for the Rubisco protein can be present in one, two or more copies with the recombinant yeast cell. Without wishing to be bound by any kind of theory it is believed that the robustness of the recombinant yeast cell is best served when the nucleic acid sequence (e.g. the gene) encoding for the Rubisco protein is present in the recombinant yeast cell in less than 12 copies, more preferably less than 8 copies. Preferably the recombinant yeast cell therefore comprises in the range from equal to or more than 1 copy, more preferably equal to or more than 2 copies, to equal to or less than 7 copies, more preferably equal to or less than 6 copies of a nucleic acid sequence (e.g. a gene) encoding for a Rubisco protein. The recombinant yeast cell may for example comprise one, two, three, four, five, six or seven copies of a nucleic acid sequence encoding for ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco).

[0237] To increase the likelihood that the Rubisco protein is expressed at sufficient levels and in active form in the transformed (recombinant) host cells of the invention, the nucleic acid sequence encoding the Rubisco protein and other proteins as described herein (see below), are preferably adapted to optimise their codon usage to that of the host cell in question. The adaptiveness of a nucleic acid sequence encoding an enzyme to the codon usage of a host cell may be expressed as codon adaptation index (CAI). The codon adaptation index is herein defined as a measurement of the relative adaptiveness of the codon usage of a gene towards the codon usage of highly expressed genes in a particular host cell or organism. The relative adaptiveness (w) of each codon is the ratio of the usage of each codon, to that of the most abundant codon for the same amino acid. The CAI index is defined as the geometric mean of these relative adaptiveness values. Non-synonymous codons and termination codons (dependent on genetic code) are excluded. CAI values range from 0 to 1, with higher values indicating a higher proportion of the most abundant codons (see Sharp and Li, “The codon adaptation index—a measure of directional synonymous codon usage bias, and its potential applications”, (1987), published in Nucleic Acids Research vol. 15, pages 1281-1295; also see: Jansen et al., “Revisiting the codon adaptation index from a whole-genome perspective:

analyzing the relationship between gene expression and codon occurrence in yeast using a variety of models”, (2003), Nucleic Acids Res. Vol. 31(8), pages 2242-51). An adapted nucleic acid sequence preferably has a CAI of at least 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8 or 0.9. Preferably, the sequences have been codon optimized for expression in the fungal host cell in question, such as for example *Saccharomyces cerevisiae* cells.

[0238] Preferably the functionally expressed Rubisco protein has an activity, defined by the rate of ribulose-1,5-bisphosphate-dependent ^{14}C -bicarbonate incorporation by cell extracts of at least $1 \text{ nmol} \cdot \text{Math} \cdot \text{min} \cdot \text{sup} \cdot -1 \cdot \text{Math} \cdot (\text{mg protein}) \cdot \text{sup} \cdot -1$, in particular an activity of at least $2 \text{ nmol} \cdot \text{Math} \cdot \text{min} \cdot \text{sup} \cdot -1 \cdot \text{Math} \cdot (\text{mg protein}) \cdot \text{sup} \cdot -1$, more in particular an activity of at least $4 \text{ nmol} \cdot \text{Math} \cdot \text{min} \cdot \text{sup} \cdot -1 \cdot \text{Math} \cdot (\text{mg protein}) \cdot \text{sup} \cdot -1$. The upper limit for the activity is not critical. In practice, the activity may be about $200 \text{ nmol} \cdot \text{Math} \cdot \text{min} \cdot \text{sup} \cdot -1 \cdot \text{Math} \cdot (\text{mg protein}) \cdot \text{sup} \cdot -1$ or less, in particular $25 \text{ nmol} \cdot \text{Math} \cdot \text{min} \cdot \text{sup} \cdot -1 \cdot \text{Math} \cdot (\text{mg protein}) \cdot \text{sup} \cdot -1$, more in particular $15 \text{ nmol} \cdot \text{Math} \cdot \text{min} \cdot \text{sup} \cdot -1 \cdot \text{Math} \cdot (\text{mg protein}) \cdot \text{sup} \cdot -1$ or less, e.g. about $10 \text{ nmol} \cdot \text{Math} \cdot \text{min} \cdot \text{sup} \cdot -1 \cdot \text{Math} \cdot (\text{mg protein}) \cdot \text{sup} \cdot -1$ or less. The conditions for an assay for determining this Rubisco activity are as found in the Examples (e.g. Example 4) of WO2014/129898, incorporated herein by reference.

Phosphoribulokinase

[0239] Preferably recombinant yeast cell is also functionally expressing a heterologous nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity (EC2.7.1.19; PRK).

[0240] The protein having phosphoribulokinase (PRK) activity is herein also referred to as “phosphoribulokinase protein”, “phosphoribulokinase enzyme”, “phosphoribulokinase”, “PRK enzyme”, “PRK protein” or simply “PRK”. Preferences for the PRK protein and the nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0241] A functionally expressed phosphoribulokinase (PRK, (EC 2.7.1.19)) according to the invention is capable of catalyzing the chemical reaction

$\text{ATP} + \text{D-ribulose 5-phosphate} \rightarrow \text{ADP} + \text{D-ribulose 1,5-bisphosphate}$

Thus, the two substrates of this enzyme are ATP and D-ribulose 5-phosphate; its two products are ADP and D-ribulose 1,5-bisphosphate.

[0242] The PRK protein belongs to the family of transferases, specifically those transferring phosphorus-containing groups (phosphotransferases) with an alcohol group as acceptor. The systematic name of this enzyme class is ATP:D-ribulose-5-phosphate 1-phosphotransferase. Other names in common use include phosphopentokinase, ribulose-5-phosphate kinase, phosphopentokinase, phosphoribulokinase (phosphorylating), 5-phosphoribulose kinase, ribulose phosphate kinase, PKK, PRuK, and PRK. The PRK enzyme participates in carbon fixation. A phosphoribulokinase (PRK) protein may be further defined by its amino acid sequence. Likewise a phosphoribulokinase (PRK) protein may be further defined by a nucleotide sequence encoding the phosphoribulokinase (PRK). As explained in detail above under definitions, a certain phosphoribulokinase (PRK) that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the phosphoribulokinase (PRK).

[0243] The PRK can be from a prokaryote or a eukaryote. Good results have been achieved with a PRK originating from a eukaryote. Preferably the PRK protein originates from a plant selected from Caryophyllales, in particular from Amaranthaceae, more in particular from *Spinacia*.

[0244] A preferred PRK protein is the PRK protein from *Spinacia*. SEQ ID NO: 27 shows the amino acid sequence of such PRK protein from *Spinacia*. SEQ ID NO: 28 illustrates the nucleic acid sequence of the *prk* gene from *Spinacia oleracea*—codon optimized for *S. cerevisiae*.

[0245] Preferably the protein having phosphoribulokinase (PRK) activity thus comprises or consists of: [0246] an amino acid sequence of SEQ ID NO: 27; or [0247] a functional homologue

of SEQ ID NO: 27, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 27; or [0248] a functional homologue of SEQ ID NO: 27, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 27, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 27.

[0249] Preferable the nucleic acid sequence encoding the protein having phosphoribulokinase (PRK) activity comprises or consists of: [0250] a nucleic acid sequence of SEQ ID NO: 28; or [0251] a functional homologue of SEQ ID NO: 28, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 28; or [0252] a functional homologue of SEQ ID NO: 28, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 28, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 28.

[0253] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphoribulokinase (PRK) activity may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898, herein incorporated by reference.

[0254] Examples of suitable PRK polypeptides and their origin are given in Table 2 of WO2014/129898, incorporated herein by reference, and in Table 3 below, with reference to the sequence identity with the amino acid sequence of SEQ ID NO:27.

TABLE-US-00003 TABLE 3 Natural PRK polypeptides suitable for expression with identity to PRK from *Spinacia* MAX Source Accession no. ID (%) *Spinacia oleracea* P09559.1 100 *Medicago truncatula* XP_003612664.1 88 *Arabidopsis thaliana* NP_174486.1 87 *Vitis vinifera* XP_002263724.1 86 *Closterium peracerosum* BAL03266.1 82 *Zea mays* NP_001148258.1 78

[0255] The nucleic acid sequences encoding for the PRK protein may be under the control of a promoter (the “PRK promoter”) that enables higher expression under anaerobic conditions than under aerobic conditions. Examples of such promoters are described in WO2017/216136A1 and WO2018/228836, both herein incorporated by reference. More preferably such promoter has a PRK expression ratio .sub.anaerobic/aerobic of 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more. Further preferences are as described in WO2018/228836, incorporated herein by reference.

Rubisco Chaperones

[0256] Optionally, the recombinant yeast cell further comprises one or more, preferably heterologous, nucleic acid sequences encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0257] Suitably such molecular chaperones are also referred herein as “chaperone protein”, “chaperonin” or simply “chaperone”. Preferences for the chaperones and the nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0258] Preferably the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0259] Chaperonins are proteins that provide favorable conditions for the correct folding of other

proteins, thus preventing aggregation. Newly made proteins usually must fold from a linear chain of amino acids into a three-dimensional form. Chaperonins belong to a large class of molecules that assist protein folding, called molecular chaperones. The energy to fold proteins is supplied by adenosine triphosphate (ATP). A review article about chaperones that is useful herein is written by Yébenes et al., “*Chaperonins: two rings for folding*” (2011), Trends in Biochemical Sciences, Vol. 36, No. 8, pages 424-432, incorporated herein by reference.

[0260] The chaperone or chaperones may be prokaryotic chaperones or eukaryotic chaperones. In addition, the chaperones may be homologous or heterologous. For example, the recombinant yeast cell may comprises one or more nucleic acid sequence encoding one or more homologous or heterologous, prokaryotic or eukaryotic, molecular chaperones, which—when expressed—are capable of functionally interacting with an enzyme in the recombinant yeast cell, in particular with at least one of Rubisco and PRK.

[0261] Suitably the chaperone or chaperones are derived from a bacterium, more preferably from *Escherichia*, in particular *E. coli*. Preferred chaperones are GroEL and GroEs from *E. coli*. Other preferred chaperones are chaperones from *Saccharomyces*, in particular *Saccharomyces cerevisiae* Hsp10 and Hsp60.

[0262] If the chaperones are naturally expressed in an organelle such as a mitochondrion (examples are Hsp60 and Hsp10 of *Saccharomyces cerevisiae*) relocation to the cytosol can be achieved e.g. by modifying the native signal sequence of the chaperonins. In eukaryotes the proteins Hsp60 and Hsp10 are structurally and functionally nearly identical to GroEL and GroES, respectively. Thus, it is contemplated that Hsp60 and Hsp10 from any recombinant yeast cell may serve as a chaperone for the Rubisco. This is described for example by Zeilstra-Ryalls et al., “*The universally conserved GroE (Hsp60) chaperonins*”, (1991), Annu Rev Microbiol. vol. 45, pages 301-325; and Horwich et al., “*Two Families of Chaperonin: Physiology and Mechanism*” (2007), Annu. Rev. Cell. Dev. Biol. Vol. 23, pages 115-145, both herewith incorporated by reference.

[0263] Good results have been achieved with a recombinant yeast cell comprising both the heterologous chaperones GroEL and GroES.

[0264] As an alternative to GroES a functional homologue of GroES may be present, in particular a functional homologue comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of GroES, respectively the amino sequence of SEQ ID NO: 31.

[0265] SEQ ID NO:31 provides a preferred translated protein sequence, based on GroES of *Escherichia coli*. SEQ ID NO: 32 provides a synthetic nucleic acid sequence, based on GroES from *Escherichia coli*, codon optimized for expression in *Saccharomyces cerevisiae*.

[0266] Examples of suitable natural chaperones polypeptide homologous to GroES are given in Table 4.

TABLE-US-00004 TABLE 4 Natural chaperones homologous to GroES polypeptides suitable for expression >gi|115388105|ref|XP_001211558.1|:2-101 10 kDa heat shock protein, mitochondrial [*Aspergillus terreus* NIH2624] >gi|116196854|ref|XP_001224239.1|:1-102 conserved hypothetical protein [*Chaetomium globosum* CBS 148.51] >gi|119175741|ref|XP_001240050.1|:3-102 hypothetical protein CIMG_09671 [*Coccidioides immitis* RS] >gi|119471607|ref|XP_001258195.1|:12-111 chaperonin, putative [*Neosartorya fischeri* NRRL181] >gi|121699818|ref|XP_001268174.1|:8-106 chaperonin, putative [*Aspergillus clavatus* NRRL 1] >gi|126274604|ref|XP_001387607.1|:2-102 predicted protein [*Scheffersomyces stipitis* CBS 6054] >gi|146417701|ref|XP_001484818.1|:5-106 conserved hypothetical protein [*Meyerozyma guilliermondii* ATCC 6260] >gi|154303611|ref|XP_001552212.1|:1-102 10 kDa heat shock protein, mitochondrial [*Botryotinia fuckeliana* B05.10] >gi|156049571|ref|XP_001590752.1|:1-102 hypothetical protein SS1G_08492 [*Sclerotinia sclerotiorum* 1980] >gi|156840987|ref|XP_001643870.1|:1-103 hypothetical protein Kpol_495p10 [*Vanderwaltozyma*

polyspora DSM 70924] >gi|169608295|ref|XP_001797567.1|:1-101 hypothetical protein
 SNOG_07218 [*Phaeosphaeria nodorum* SN15] >gi|171688384|ref|XP_001909132.1|:1-102
 hypothetical protein [*Podospira anserina* S mat+] >gi|189189366|ref|XP_001931022.1|:71-168 10
 kDa chaperonin [*Pyrenophora tritici-repentis* Pt-1C-BFP] >gi|19075598|ref|NP_588098.1|:1-102
 mitochondrial heat shock protein Hsp10 (predicted) [*Schizosaccharomyces pombe* 972h-]
 >gi|212530240|ref|XP_002145277.1|:3-100 chaperonin, putative [*Talaromyces marneffe* ATCC
 18224] >gi|212530242|ref|XP_002145278.1|:3-95 chaperonin, putative [*Talaromyces marneffe*
 ATCC 18224] >gi|213404320|ref|XP_002172932.1|:1-102 mitochondrial heat shock protein Hsp10
 [*Schizosaccharomyces japonicus* yFS275] >gi|225557301|gb|EEH05587.1|:381-478 pre-mRNA
 polyadenylation factor fip1 [*Ajellomyces capsulatus* G186AR] >gi|225684092|gb|EEH22376.1|:3-
 100 heat shock protein [*Paracoccidioides brasiliensis* Pb03
 >gi|238490530|ref|XP_002376502.1|:2-104 chaperonin, putative [*Aspergillus flavus* NRRL3357
 >gi|238878220|gb|EEQ41858.1|:1-106 10 kDa heat shock protein, mitochondrial [*Candida*
albicans WO-1] >gi|240280207|gb|EER43711.1|:426-523 pre-mRNA polyadenylation factor fip1
 [*Ajellomyces capsulatus* H143] >gi|241950445|ref|XP_002417945.1|:1-103 10 kDa chaperonin,
 putative; 10 kDa heat shock protein mitochondrial (hsp10), putative [*Candida dubliniensis* CD36]
 >gi|242819222|ref|XP_002487273.1|:90-182 chaperonin, putative [*Talaromyces stipitatus* ATC
 >gi|254566327|ref|XP_002490274.1|:1-102 Putative protein of unknown function [*Komagataella*
pastoris GS115] >gi|254577241|ref|XP_002494607.1|:1-103 ZYRO0A05434p
 [*Zygosaccharomyces rouxii*] >gi|255717999|ref|XP_002555280.1|:1-103 KLTH0G05588p
 [*Lachancea thermotolerans*] >gi|255956581|ref|XP_002569043.1|:2-101 Pc21g20560 [*Penicillium*
chrysogenum Wisconsin 54-1255] >gi|258572664|ref|XP_002545094.1|:16-108 chaperonin GroS
 [*Uncinocarpus reesii* 1704] >gi|261190594|ref|XP_002621706.1|:3-100 chaperonin [*Ajellomyces*
dermatitidis SLH14081] >gi|295664909|ref|XP_002793006.1|:3-100 10 kDa heat shock protein,
 mitochondrial [*Paracoccidioides* sp. 'lutzii' Pb01] >gi|296412657|ref|XP_002836039.1|:76-177
 hypothetical protein [*Tuber melanosporum* Mel28] >gi|302307854|ref|NP_984626.2|:2-102
 AEL235Wp [*Ashbya gossypii* ATCC 10895] >gi|302894117|ref|XP_003045939.1|:1-102 predicted
 protein [*Nectria haematococca* mpVI 77- 13-4] >gi|303318351|ref|XP_003069175.1|:3-100 10 kDa
 heat shock protein, mitochondrial, putative [*Coccidioides posadasii* C735 delta SOWgp]
 >gi|310795300|gb|EFQ30761.1|:1-102 chaperonin 10 kDa subunit [*Glomerella graminicola*
 M1.001] >gi|315053085|ref|XP_003175916.1|:12-109 chaperonin GroS [*Arthroderma gypseum*
 CBS 118893] >gi|317032114|ref|XP_001394060.2|:334-433 heat shock protein [*Aspergillus niger*
 CBS 513.88] >gi|317032116|ref|XP_001394059.2|:2-101 heat shock protein [*Aspergillus niger*
 CBS 513.88] >gi|320583288|gb|EFW97503.1|:6-106 chaperonin, putative heat shock protein,
 putative [*Ogataea parapolymorpha* DL-1] >gi|320591507|gb|EFX03946.1|:1-102 heat shock
 protein [*Grosmannia clavigera* kw1407] >gi|322700925|gb|EFY92677.1|:1-102 chaperonin
 [*Metarhizium acridum* CQMa 102] >gi|325096696|gb|EGC50006.1|:409-506 pre-mRNA
 polyadenylation factor fip1 [*Ajellomyces capsulatus* H88] >gi|326471604|gb|EGD95613.1|:14-111
 chaperonin 10 Kd subunit [*Trichophyton tonsurans* CBS112818]
 >gi|327293056|ref|XP_003231225.1|:3-100 chaperonin [*Trichophyton rubrum* CBS 118892]
 >gi|330942654|ref|XP_003306155.1|:37-136 hypothetical protein PTT_19211 [*Pyrenophora teres*
f. teres 0-1] >gi|336268042|ref|XP_003348786.1|:47-147 hypothetical protein SMAC_01809
 [*Sordaria macrospora* khell] >gi|340519582|gb|EGR49820.1|:1-109 predicted protein
 [*Trichoderma reesei* QM6a] >gi|340960105|gb|EGS21286.1|:3-103 putative mitochondrial 10 kDa
 heat shock protein [*Chaetomium thermophilum* var. *thermophilum* DSM 1495]
 >gi|342883802|gb|EGU84224.1|:1-102 hypothetical protein FOXB_05181 [*Fusarium oxysporum*
 Fo5176] >gi|344302342|gb|EGW32647.1|:2-102 hypothetical protein SPAPADRAFT_61712
 [*Spathaspora passalidarum* NRRL Y-27907] >gi|345570750|gb|EGX53571.1|:1-102 hypothetical
 protein AOL_s00006g437 [*Arthrobotrys oligospora* ATCC 24927]
 >gi|346321154|gb|EGX90754.1|:1-102 chaperonin [*Cordyceps militaris* CM01]

>gi|346970393|gb|EGY13845.1|:1-102 heat shock protein [*Verticillium dahliae* VdLs.17]
 >gi|354548296|emb|CCE45032.1|:1-106 hypothetical protein CPAR2_700360 [*Candida parapsilosis*] >gi|358385052|gb|EHK22649.1|:1-102 hypothetical protein TRIVIDRAFT_230640 [*Trichoderma virens* Gv 29-8] >gi|358393422|gb|EHK42823.1|:1-101 hypothetical protein TRIATDRAFT_258186 [*Trichoderma atroviride* IMI 206040] >gi|361126733|gb|EHK98722.1|:1-97 putative 10 kDa heat shock protein, mitochondrial [*Glare lozoyensis* 74030]
 >gi|363753862|ref|XP_003647147.1|:2-102 hypothetical protein Ecym_5593 [*Eremothecium cymbalariae* DBVPG#7215] >gi|365758401|gb|EHN00244.1|:1-106 Hsp10p [*Saccharomyces cerevisiae* × *Saccharomyces kudriavzevii* VIN7] >gi|365987664|ref|XP_003670663.1|:1-103 hypothetical protein NDAI_0F01010 [*Naumovozyma dairenensis* CBS 421]
 >gi|366995125|ref|XP_003677326.1|:1-103 hypothetical protein NCAS_0G00860 [*Naumovozyma castellii* CBS 4309] >gi|366999797|ref|XP_003684634.1|:1-103 hypothetical protein TPHA_0C00430 [*Tetrapisispora phaffii* CBS 4417] >gi|367009030|ref|XP_003679016.1|:1-103 hypothetical protein TDEL_0A04730 [*Torulaspora delbruekii*]
 >gi|367023138|ref|XP_003660854.1|:1-104 hypothetical protein MYCTH_59302 [*Myceliophthora thermophila* ATCC 42464] >gi|367046344|ref|XP_003653552.1|:1-102 hypothetical protein THITE_2116070 [*Thielavia terrestris* NRRL8126] >gi|378726440|gb|EHY52899.1|:9-109 chaperonin GroES [*Exophiala dermatitidis* NIH/UT8656] >gi|380493977|emb|CCF33483.1|:1-102 chaperonin 10 kDa subunit [*Colletotrichum higginsianu*] >gi|385305728|gb|EIF49680.1|:1-102 10 kda heat shock mitochondrial [*Dekkera bruxellensis* AWRI1499]
 >gi|389628546|ref|XP_003711926.1|:1-102 hsp10-like protein [*Magnaporthe oryzae* 70-15]
 >gi|396462608|ref|XP_003835915.1|:1-101 similar to 10 kDa heat shock protein [*Leptosphaeria maculans* JN3] >gi|398392541|ref|XP_003849730.1|:1-102 hypothetical protein MYCGRDRAFT_105721 [*Zymoseptoria tritici* IPO323] >gi|400597723|gb|EJP65453.1|:24-124 chaperonin 10 kDa subunit [*Beauveria bassiana* ARSEF 2860] >gi|401623646|gb|EJS41738.1|:1-106 hsp10p [*Saccharomyces arboricola* H-6] >gi|401842164|gb|EJT44422.1|:1-92 HSP10-like protein [*Saccharomyces kudriavzevii* IFO 1802] >gi|402084027|gb|EJT79045.1|:1-102 hsp10-like protein [*Gaeumannomyces graminis* var. *tritici*] >gi|403215209|emb|CCK69709.1|:1-104 hypothetical protein KNAG_0C06130 [*Kazachstania naganishii* CBS 8797]
 >gi|406604629|emb|CCH43969.1|:4-100 hypothetical protein BN7_3524 [*Wickerhamomyces ciferrii*] >gi|406867021|gb|EKD20060.1|:56-156 hypothetical protein MBM_02012 [*Marssonina brunnea* f. sp. 'multigermtubi' MB_m1] >gi|407926227|gb|EKG19196.1|:74-174 GroES-like protein [*Macrophomina phaseolina* MS6] >gi|408398157|gb|EKJ77291.1|:11-111 hypothetical protein FPSE_02566 [*Fusarium pseudograminearum* CS3096]
 >gi|410082063|ref|XP_003958610.1|:1-103 hypothetical protein KAFR_0H00660 [*Kazachstania africana* CBS2517] >gi|425777664|gb|EKV15823.1|:58-157 Chaperonin, putative [*Penicillium digitatum* Pd1] >gi|440639680|gb|ELR09599.1|:1-102 chaperonin GroES [*Geomyces destructans* 20631-21] >gi|444323906|ref|XP_004182593.1|:1-105 hypothetical protein TBLA_0J00760 [*Tetrapisisporablattae* CBS 6284] >gi|448083208|ref|XP_004195335.1|:2-101 Piso0_005888 [*Millerozyma farinosa* CBS 7064] >gi|448087837|ref|XP_004196425.1|:2-102 Piso0_005888 [*Millerozyma farinosa* CBS 7064] >gi|448534948|ref|XP_003870866.1|:1-106 Hsp10 protein [*Candida orthopsilosis* Co 90-125] >gi|449295977|gb|EMC91998.1|:1-102 hypothetical protein BAUCODRAFT_39148 [*Baudoinia compn*] >gi|46123659|ref|XP_386383.1|:3-103 hypothetical protein FG06207.1 [*Gibberella zeae* PH-1] >gi|50289455|ref|XP_447159.1|:1-103 hypothetical protein [*Candida glabrata* CBS 138] >gi|50308731|ref|XP_454370.1|:1-103 hypothetical protein [*Kluyveromyces lactis* NRRL Y- 1140] >gi|50411066|ref|XP_457014.1|:1-106 DEHA2B01122p [*Debaryomyces hansenii* CBS767] >gi|50545998|ref|XP_500536.1|:1-102 YALI0B05610p [*Yarrowia lipolytica*] >gi|51013895|gb|AAT93241.1|:1-106 YOR020C [*Saccharomyces cerevisiae*]
 >gi|6324594|ref|NP_014663.1|:1-106 Hsp10p [*Saccharomyces cerevisiae* S288c]
 >gi|67523953|ref|XP_660036.1|:2-101 hypothetical protein AN2432.2 [*Aspergillus nidulans* FGSC

A4] >gi|70992219|ref|XP_750958.1|:12-106 chaperonin [*Aspergillus fumigatus* Af293]
>gi|85079266|ref|XP_956315.1|:1-104 hypothetical protein NCU04334 [*Neurospora crassa*
OR74A]

[0267] Table 4: Natural chaperones homologous to GroES polypeptides suitable for expression

[0268] As an alternative to GroEL a functional homologue of GroEL may be present, in particular a functional homologue comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of GroEL, respectively the amino sequence of SEQ ID NO: 29.

[0269] SEQ ID NO:29 provides a preferred translated protein sequence, based on GroEL of *Escherichia coli*. SEQ ID NO: 30 provides a synthetic nucleic acid sequence, based on GroEL from *Escherichia coli*, codon optimized for expression in *Saccharomyces cerevisiae*.

[0270] Suitable natural chaperones polypeptides homologous to GroEL are given in Table 5.

TABLE-US-00005 TABLE 5 Natural chaperones homologous to GroEL polypeptides suitable for expression >gi|115443330|ref|XP_001218472.1| heat shock protein 60, mitochondrial precursor

[*Aspergillus terreus* NIH2624] >gi|114188341|gb|EAU30041.1| heat shock protein 60, mitochondrial precursor [*Aspergillus terreus* NIH2624] >gi|119480793|ref|XP_001260425.1| antigenic mitochondrial protein HSP60, putative [*Neosartorya fischeri* NRRL 181]

>gi|119408579|gb|EAW18528.1| antigenic mitochondrial protein HSP60, putative [*Neosartorya fischeri* NRRL 181] >gi|126138730|ref|XP_001385888.1| hypothetical protein PICST_90190 [*Scheffersomyces stipitis* CBS 6054] >gi|126093166|gb|ABN67859.1| mitochondrial groEL-type heat shock protein [*Scheffersomyces stipitis* CBS 6054] >gi|145246630|ref|XP_001395564.1| heat shock protein 60 [*Aspergillus niger* CBS 513.88] >gi|134080285|emb|CAK46207.1| unnamed protein product [*Aspergillus niger*] >gi|350636909|gb|EHA25267.1| hypothetical protein ASPNIDRAFT_54001 [*Aspergillus niger* ATCC 1015] >gi|146413148|ref|XP_001482545.1| heat shock protein 60, mitochondrial precursor [*Meyerozyma guilliermondii* ATCC 6260]

>gi|154277022|ref|XP_001539356.1| heat shock protein 60, mitochondrial precursor [*Ajellomyces capsulatus* NAM1] >gi|150414429|gb|EDN09794.1| heat shock protein 60, mitochondrial precursor [*Ajellomyces capsulatus* NAM1] >gi|154303540|ref|XP_001552177.1| heat shock protein 60 [*Botryotinia fuckeliana* B05.10] >gi|347840915|emb|CCD55487.1| similar to heat shock protein 60 [*Botryotinia fuckeliana*] >gi|156063938|ref|XP_001597891.1| heat shock protein 60, mitochondrial precursor [*Sclerotinia sclerotiorum* 1980] >gi|154697421|gb|EDN97159.1| heat shock protein 60, mitochondrial precursor [*Sclerotinia sclerotiorum* 1980 UF-70]

>gi|156844469|ref|XP_001645297.1| hypothetical protein Kpol_1037p35 [*Vanderwaltozyma polyspora* DSM 70294] >gi|156115957|gb|EDO17439.1| hypothetical protein Kpol_1037p35 [*Vanderwaltozyma polyspora* DSM 70294] >gi|16416029|emb|CAB91379.2| probable heat-shock protein hsp60 [*Neurospora crassa*] >gi|350289516|gb|EGZ70741.1| putative heat-shock protein hsp60 [*Neurospora tetrasperma* FGSC 2509] >gi|169626377|ref|XP_001806589.1| hypothetical protein SNOG_16475 [*Phaeosphaeria nodorum* SN15] >gi|111055053|gb|EAT76173.1| hypothetical protein SNOG_16475 [*Phaeosphaeria nodorum* SN15]

>gi|169783766|ref|XP_001826345.1| heat shock protein 60 [*Aspergillus oryzae* RIB40] >gi|238493601|ref|XP_002378037.1| antigenic mitochondrial protein HSP60, putative [*Aspergillus flavus* NRRL3357] >gi|83775089|dbj|BAE65212.1| unnamed protein product [*Aspergillus oryzae* RIB40] >gi|220696531|gb|EED52873.1| antigenic mitochondrial protein HSP60, putative [*Aspergillus flavus* NRRL3357] >gi|391869413|gb|EIT78611.1| chaperonin, Cpn60/Hsp60p [*Aspergillus oryzae* 3.042] >gi|189190432|ref|XP_001931555.1| heat shock protein 60, mitochondrial precursor [*Pyrenophora tritici-repentis* Pt-1C-BFP] >gi|187973161|gb|EDU40660.1| heat shock protein 60, mitochondrial precursor [*Pyrenophora tritici-repentis* Pt-1C-BFP]

>gi|190348913|gb|EDK41467.2| heat shock protein 60, mitochondrial precursor [*Meyerozyma guilliermondii* ATCC 6260] >gi|225554633|gb|EEH02929.1| hsp60-like protein [*Ajellomyces*

capsulatus G186AR] >gi|238880068|gb|EEQ43706.1| heat shock protein 60, mitochondrial precursor [*Candida albicans* WO-1] >gi|239613490|gb|EEQ90477.1| chaperonin GroL [*Ajellomyces dermatitidis* ER-3] >gi|240276977|gb|EER40487.1| hsp60-like protein [*Ajellomyces capsulatus* H143] >gi|241958890|ref|XP_002422164.1| heat shock protein 60, mitochondrial precursor, putative [*Candida dubliniensis* CD36] >gi|223645509|emb|CAX40168.1| heat shock protein 60, mitochondrial precursor, putative [*Candida dubliniensis* CD36] >gi|254572906|ref|XP_002493562.1| Tetradecameric mitochondrial chaperonin [*Komagataella pastoris* GS115] >gi|238033361|emb|CAY71383.1| Tetradecameric mitochondrial chaperonin [*Komagataella pastoris* GS115] >gi|254579947|ref|XP_002495959.1| ZYRO0C07106p [*Zygosaccharomyces rouxii*] >gi|238938850|emb|CAR27026.1| ZYRO0C07106p [*Zygosaccharomyces rouxii*] >gi|255712781|ref|XP_002552673.1| KLTH0C10428p [*Lachancea thermotolerans*] >gi|238934052|emb|CAR22235.1| KLTH0C10428p [*Lachancea thermotolerans* CBS 6340] >gi|255721795|ref|XP_002545832.1| heat shock protein 60, mitochondrial precursor [*Candida tropicalis* MYA-3404] >gi|240136321|gb|EER35874.1| heat shock protein 60, mitochondrial precursor [*Candida tropicalis* MYA-3404] >gi|255941288|ref|XP_002561413.1| Pc16g11070 [*Penicillium chrysogenum* Wisconsin 54- 1255] >gi|211586036|emb|CAP93777.1| Pc16g11070 [*Penicillium chrysogenum* Wisconsin 54- 1255] >gi|259148241|emb|CAY81488.1| Hsp60p [*Saccharomyces cerevisiae* EC1118] >gi|260950325|ref|XP_002619459.1| heat shock protein 60, mitochondrial precursor [*Clavispora lusitaniae* ATCC 42720] >gi|238847031|gb|EEQ36495.1| heat shock protein 60, mitochondrial precursor [*Clavispora lusitaniae* ATCC 42720] >gi|261194577|ref|XP_002623693.1| chaperonin GroL [*Ajellomyces dermatitidis* SLH14081] >gi|239588231|gb|EEQ70874.1| chaperonin GroL [*Ajellomyces dermatitidis* SLH14081] >gi|327355067|gb|EGE83924.1| chaperonin GroL [*Ajellomyces dermatitidis* ATCC 18188] >gi|296422271|ref|XP_002840685.1| hypothetical protein [*Tuber melanosporum* Mel28] >gi|295636906|emb|CAZ84876.1| unnamed protein product [*Tuber melanosporum*] >gi|296809035|ref|XP_002844856.1| heat shock protein 60 [*Arthroderma otae* CBS 113480] >gi|238844339|gb|EEQ34001.1| heat shock protein 60 [*Arthroderma otae* CBS 113480] >gi|302308696|ref|NP_985702.2| AFR155Wp [*Ashbya gossypii* ATCC 10895] >gi|299790751|gb|AAS53526.2| AFR155Wp [*Ashbya gossypii* ATCC 10895] >gi|374108933|gb|AEY97839.1| FAFR155Wp [*Ashbya gossypii* FDAG1] >gi|302412525|ref|XP_003004095.1| heat shock protein [*Verticillium albo-atrum* VaMs. 102] >gi|261356671|gb|EEY19099.1| heat shock protein [*Verticillium albo-atrum* VaMs. 102] >gi|302505585|ref|XP_003014499.1| hypothetical protein ARB_07061 [*Arthroderma benhamiae* CBS 112371] >gi|291178320|gb|EFE34110.1| hypothetical protein ARB_07061 [*Arthroderma benhamiae* CBS 112371] >gi|302656385|ref|XP_003019946.1| hypothetical protein TRV_05992 [*Trichophyton verrucosum* HKI 0517] >gi|291183723|gb|EFE39322.1| hypothetical protein TRV_05992 [*Trichophyton verrucosum* HKI 0517] >gi|302915513|ref|XP_003051567.1| predicted protein [*Nectria haematococca* mpVI 77-13-4] >gi|256732506|gb|EEU45854.1| predicted protein [*Nectria haematococca* mpVI 77-13-4] >gi|310794550|gb|EFQ30011.1| chaperonin GroL [*Glomerella graminicola* M1.001] >gi|315048491|ref|XP_003173620.1| chaperonin GroL [*Arthroderma gypseum* CBS 118893] >gi|311341587|gb|EFR00790.1| chaperonin GroL [*Arthroderma gypseum* CBS 118893] >gi|320580028|gb|EFW94251.1| Tetradecameric mitochondrial chaperonin [*Ogataea parapolymorpha* DL-1] >gi|320586014|gb|EFW98693.1| heat shock protein mitochondrial precursor [*Grosmannia clavigera* kw1407] >gi|322692465|gb|EFY84374.1| Heat shock protein 60 precursor (Antigen HIS-62) [*Metarhizium acridum* CQMa 102] >gi|322705285|gb|EFY96872.1| Heat shock protein 60 (Antigen HIS-62) [*Metarhizium anisopliae* ARSEF 23] >gi|323303806|gb|EGA57589.1| Hsp60p [*Saccharomyces cerevisiae* FostersB] >gi|323307999|gb|EGA61254.1| Hsp60p [*Saccharomyces cerevisiae* FostersO] >gi|323332364|gb|EGA73773.1| Hsp60p [*Saccharomyces cerevisiae* AWRI796] >gi|326468648|gb|EGD92657.1| heat shock protein 60 [*Trichophyton tonsurans* CBS 112818]

>gi|32647866|gb|EGE03876.1| chaperonin GroL [*Trichophyton equinum* CBS 127.97]
 >gi|330915493|ref|XP_003297052.1| hypothetical protein PTT_07333 [*Pyrenophora teres* f. *teres* 0-1]
 >gi|311330479|gb|EFQ94847.1| hypothetical protein PTT_07333 [*Pyrenophora teres* f. *teres* 0-1]
 >gi|336271815|ref|XP_003350665.1| hypothetical protein SMAC_02337 [*Sordaria macrospora* k-hell]
 >gi|380094827|emb|CCC07329.1| unnamed protein product [*Sordaria macrospora* k- hell]
 >gi|336468236|gb|EGO56399.1| hypothetical protein NEUTE1DRAFT_122948 [*Neurospora tetrasperma* FGSC 2508]
 >gi|340522598|gb|EGR52831.1| hsp60 mitochondrial precursor-like protein [*Trichoderma reesei* QM6a]
 >gi|341038907|gb|EGS23899.1| mitochondrial heat shock protein 60-like protein [*Chaetomium thermophilum* var. *thermophilum* DSM 1495]
 >gi|342886297|gb|EGU86166.1| hypothetical protein FOXB_03302 [*Fusarium oxysporum* Fo5176]
 >gi|344230084|gb|EGV61969.1| chaperonin GroL [*Candida tenuis* ATCC 10573]
 >gi|344303739|gb|EGW33988.1| hypothetical protein SPAPADRAFT_59397 [*Spathaspora passalidarum* NRRL Y-27907]
 >gi|345560428|gb|EGX43553.1| hypothetical protein AOL_s00215g289 [*Arthrobotrys oligospora* ATCC 24927]
 >gi|346323592|gb|EGX93190.1| heat shock protein 60 (Antigen HIS-62) [*Cordyceps militaris* CM01]
 >gi|346975286|gb|EGY18738.1| heat shock protein [*Verticillium dahliae* VdLs. 17]
 >gi|354545932|emb|CCE42661.1| hypothetical protein CPAR2_203040 [*Candida parapsilosis*]
 >gi|358369894|dbj|GAA86507.1| heat shock protein 60, mitochondrial precursor [*Aspergillus kawachii* IFO 4308]
 >gi|358386867|gb|EHK24462.1| hypothetical protein TRIVIDRAFT_79041 [*Trichoderma virens* Gv29-8]
 >gi|358399658|gb|EHK48995.1| hypothetical protein TRIATDRAFT_297734 [*Trichoderma atroviride* IMI 206040]
 >gi|363750488|ref|XP_003645461.1| hypothetical protein Ecym_3140 [*Eremothecium cymbalariae* DBVPG#7215]
 >gi|356889095|gb|AET38644.1| Hypothetical protein Ecym_3140 [*Eremothecium cymbalariae* DBVPG#7215]
 >gi|365759369|gb|EHN01160.1| Hsp60p [*Saccharomyces cerevisiae* × *Saccharomyces kudriavzevii* VIN7]
 >gi|365764091|gb|EHN05616.1| Hsp60p [*Saccharomyces cerevisiae* × *Saccharomyces kudriavzevii* VIN7]
 >gi|365985626|ref|XP_003669645.1| hypothetical protein NDAI_0D00880 [*Naumovozya dairenensis* CBS 421]
 >gi|343768414|emb|CCD24402.1| hypothetical protein NDAI_0D00880 [*Naumovozya dairenensis* CBS 421]
 >gi|366995970|ref|XP_003677748.1| hypothetical protein NCAS_0H00890 [*Naumovozya castellii* CBS 4309]
 >gi|342303618|emb|CCC71399.1| hypothetical protein NCAS_0H00890 [*Naumovozya castellii* CBS 4309]
 >gi|367005154|ref|XP_003687309.1| hypothetical protein TPHA_0J00520 [*Tetrapisispora phaffii* CBS 4417]
 >gi|357525613|emb|CCE64875.1| hypothetical protein TPHA_0J00520 [*Tetrapisispora phaffii* CBS 4417]
 >gi|367017005|ref|XP_003683001.1| hypothetical protein TDEL_0G04230 [*Torulaspora delbrueckii*]
 >gi|359750664|emb|CCE93790.1| hypothetical protein TDEL_0G04230 [*Torulaspora delbrueckii*]
 >gi|367035486|ref|XP_003667025.1| hypothetical protein MYCTH_2097570 [*Myceliophthora thermophila* ATCC 42464]
 >gi|347014298|gb|AEO61780.1| hypothetical protein MYCTH_2097570 [*Myceliophthora thermophila* ATCC 42464]
 >gi|367055018|ref|XP_003657887.1| hypothetical protein THITE_127923 [*Thielavia terrestris* NRRL 8126]
 >gi|347005153|gb|AEO71551.1| hypothetical protein THITE_127923 [*Thielavia terrestris* NRRL 8126]
 >gi|378728414|gb|EHY54873.1| heat shock protein 60 [*Exophiala dermatitidis* NIH/UT8656]
 >gi|380494593|emb|CCF33032.1| heat shock protein 60 [*Colletotrichum higginsianum*]
 >gi|385305893|gb|EIF49836.1| heat shock protein 60 [*Dekkera bruxellensis* AWRI1499]
 >gi|389638386|ref|XP_003716826.1| heat shock protein 60 [*Magnaporthe oryzae* 70-15]
 >gi|351642645|gb|EHA50507.1| heat shock protein 60 [*Magnaporthe oryzae* 70-15]
 >gi|440474658|gb|ELQ43388.1| heat shock protein 60 [*Magnaporthe oryzae* Y34]
 >gi|440480475|gb|ELQ61135.1| heat shock protein 60 [*Magnaporthe oryzae* P131]
 >gi|393243142|gb|EJD50658.1| chaperonin GroL [*Auricularia delicata* TFB-10046 SS5]
 >gi|396494741|ref|XP_003844378.1| similar to heat shock protein 60 [*Leptosphaeria maculans* JN3]
 >gi|312220958|emb|CBY00899.1| similar to heat shock protein 60 [*Leptosphaeria maculans*

JN3] >gi|398393428|ref|XP_003850173.1| chaperone ATPase HSP60 [*Zymoseptoria tritici* IPO323]
 >gi|339470051|gb|EGP85149.1| hypothetical protein MYCGRDRAFT_75170 [*Zymoseptoria tritici* IPO323] >gi|401624479|gb|EJS42535.1| hsp60p [*Saccharomyces arboricola* H-6]
 >gi|401842294|gb|EJT44530.1| HSP60-like protein [*Saccharomyces kudriavzevii* IFO 1802]
 >gi|402076594|gb|EJT72017.1| heat shock protein 60 [*Gaeumannomyces graminis* var. *tritici* R3-111a-1] >gi|403213867|emb|CCK68369.1| hypothetical protein KNAG_0A07160 [*Kazachstania naganishii* CBS 8797] >gi|406606041|emb|CCH42514.1| Heat shock protein 60, mitochondrial [*Wickerhamomyces ciferrii*] >gi|406863285|gb|EKD16333.1| heat shock protein 60 [*Marssonina brunnea* f. sp. 'multigermutubi' MB_m1] >gi|407922985|gb|EKG16075.1| Chaperonin Cpn60 [*Macrophomina phaseolina* MS6] >gi|408399723|gb|EKJ78816.1| hypothetical protein FPSE_00959 [*Fusarium pseudograminearum* CS3096] >gi|410083028|ref|XP_003959092.1| hypothetical protein KAFR_0I01760 [*Kazachstania africana* CBS 2517]
 >gi|372465682|emb|CCF59957.1| hypothetical protein KAFR_0I01760 [*Kazachstania africana* CBS 2517] >gi|444315528|ref|XP_004178421.1| hypothetical protein TBLA_0B00580 [*Tetrapisispora blattae* CBS 6284] >gi|387511461|emb|CCH58902.1| hypothetical protein TBLA_0B00580 [*Tetrapisispora blattae* CBS 6284] >gi|448090588|ref|XP_004197110.1| Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi|448095015|ref|XP_004198141.1| Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi|359378532|emb|CCE84791.1| Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi|359379563|emb|CCE83760.1| Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi|448526196|ref|XP_003869293.1| Hsp60 heat shock protein [*Candida orthopsilosis* Co 90- 125] >gi|380353646|emb|CCG23157.1| Hsp60 heat shock protein [*Candida orthopsilosis*] >gi|46123737|ref|XP_386422.1| HS60_AJECA Heat shock protein 60, mitochondrial precursor (Antigen HIS-62) [*Gibberella zeae* PH-1] >gi|50292099|ref|XP_448482.1| hypothetical protein [*Candida glabrata* CBS 138] >gi|49527794|emb|CAG61443.1| unnamed protein product [*Candida glabrata*] >gi|50310975| ref|XP_455510.1| hypothetical protein [*Kluyveromyces lactis* NRRL Y-1140] >gi|49644646| emb|CAG98218.1| KLLA0F09449p [*Kluyveromyces lactis*] >gi|50422027|ref|XP_459575.1| DEHA2E05808p [*Debaryomyces hansenii* CBS767] >gi|49655243|emb|CAG87802.1| DEHA2E05808p [*Debaryomyces hansenii* CBS767] >gi|50555023|ref|XP_504920.1| YALI0F02805p [*Yarrowia lipolytica*] >gi|49650790|emb|CAG77725.1| YALI0F02805p [*Yarrowia lipolytica* CLIB122] >gi|6323288|ref|NP_013360.1| Hsp60p [*Saccharomyces cerevisiae* S288c] >gi|123579|sp|P19882.1|HSP60_YEAST RecName: Full = Heat shock protein 60, mitochondrial; AltName: Full = CPN60; AltName: Full = P66; AltName: Full = Stimulator factor I 66 kDa component; Flags: Precursor >gi|171720|gb|AAA34690.1| heat shock protein 60 (HSP60) [*Saccharomyces cerevisiae*] >gi|577181|gb|AAB67380.1| Hsp60p: Heat shock protein 60 [*Saccharomyces cerevisiae*] >gi|151941093|gb|EDN59473.1| chaperonin [*Saccharomyces cerevisiae* YJM789] >gi|190405319|gb|EDV08586.1| chaperonin [*Saccharomyces cerevisiae* RM11-1a] >gi|207342889|gb|EDZ70518.1| YLR259Cp-like protein [*Saccharomyces cerevisiae* AWRI1631] >gi|256271752|gb|EEU06789.1| Hsp60p [*Saccharomyces cerevisiae* JAY291] >gi|285813676|tpg|DAA09572.1| TPA: chaperone ATPase HSP60 [*Saccharomyces cerevisiae* S288c] >gi|323353818|gb|EGA85673.1| Hsp60p [*Saccharomyces cerevisiae* VL3] >gi|349579966|dbj|GAA25127.1| K7_Hsp60p [*Saccharomyces cerevisiae* Kyokai no. 7] >gi|392297765|gb|EIW08864.1| Hsp60p [*Saccharomyces cerevisiae* CEN.PK113-7D] >gi|226279|prf|1504305A mitochondrial assembly factor >gi|68485963|ref|XP_713100.1| heat shock protein 60 [*Candida albicans* SC5314] >gi|68486010|ref|XP_713077.1| heat shock protein 60 [*Candida albicans* SC5314] >gi|6016258|sp|074261.1|HSP60_CANAL RecName: Full = Heat shock protein 60, mitochondrial; AltName: Full = 60 kDa chaperonin; AltName: Full = Protein Cpn60; Flags: Precursor >gi|3552009|gb|AAC34885.1| heat shock protein 60 [*Candida albicans*] >gi|46434552|gb|EAK93958.1| heat shock protein 60 [*Candida albicans* SC5314] >gi|46434577|gb|EAK93982.1| heat shock protein 60 [*Candida albicans* SC5314]

>gi|71001164|ref|XP_755263.1| antigenic mitochondrial protein HSP60 [*Aspergillus fumigatus* Af293] >gi|66852901|gb|EAL93225.1| antigenic mitochondrial protein HSP60, putative [*Aspergillus fumigatus* Af293] >gi|159129345|gb|EDP54459.1| antigenic mitochondrial protein HSP60, putative [*Aspergillus fumigatus* A1163] >gi|90970323|gb|ABE02805.1| heat shock protein 60 [*Rhizophagus intraradices*]

[0271] The recombinant yeast cell preferably comprises, respectively functionally expresses, a GroES chaperone and a GroEL chaperone. Preferably a 10 kDa chaperone (“GroES”) from Table 4 is combined with a matching 60 kDa chaperone (“GroEL”) from Table 5 of the same organism genus or species for expression in the recombinant yeast cell.

[0272] For instance: >gi|189189366|ref|XP_001931022.1|:71-168 10 kDa chaperonin [*Pyrenophora tritici-repentis*] expressed together with matching >gi|189190432|ref|XP_001931555.1| heat shock protein 60, mitochondrial precursor [*Pyrenophora tritici-repentis* Pt-1C-BFP]. All other combinations from Table 4 and 5 similarly made with same organism source are also available to the skilled person for expression. Furthermore, one may combine a chaperone from Table 4 from one organism with a chaperone from Table 5 from another organism, or one may combine GroES with a chaperone from Table 5, or one may combine GroEL with a chaperone from Table 4.

[0273] Preferably the molecular chaperone(s) thus comprise or consist of: [0274] an amino acid sequence of SEQ ID NO: 29 and/or SEQ ID NO: 31; or [0275] one or more functional homologue(s) of SEQ ID NO: 29 and/or SEQ ID NO: 31, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31; or [0276] one or more functional homologue(s) of SEQ ID NO: 29 and/or SEQ ID NO: 31, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31, more preferably one or more functional homologue(s) that has/have no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31.

[0277] Preferable the nucleic acid sequence(s) encoding the molecular chaperones comprise or consist of: [0278] a nucleic acid sequence of SEQ ID NO: 30 and/or SEQ ID NO: 32; or [0279] one or more functional homologue(s) of SEQ ID NO: 30 and/or SEQ ID NO: 32, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32; or [0280] one or more functional homologue(s) of SEQ ID NO: 30 and/or SEQ ID NO: 32, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32, more preferably one or more functional homologue(s) of that has/have no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32.

[0281] The nucleic acid sequence(s) encoding for the molecular chaperones may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898, herein incorporated by reference.

Phosphoketolase

[0282] As indicated above, the recombinant yeast cell can advantageously comprise a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase (PKL) activity

(EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0283] The recombinant cell may comprise one or more heterologous genes coding for a protein having phosphoketolase activity. Such a protein having phosphoketolase activity is herein also referred to as “phosphoketolase protein”, “phosphoketolase enzyme” or simply as “phosphoketolase”. Phosphoketolase is further herein abbreviated as “PKL” or “XFP”.

[0284] As used herein, a phosphoketolase catalyzes at least the conversion of D-xylulose 5-phosphate to D-glyceraldehyde 3-phosphate and acetyl phosphate. The phosphoketolase is involved in at least one of the following the reactions:

EC 4.1.2.9:

D-xylulose-5-phosphate+phosphatecustom-characteracetyl phosphate+D-glyceraldehyde 3-phosphate+H.sub.2O (IV)

D-ribulose-5-phosphate+phosphatecustom-characteracetyl phosphate+D-glyceraldehyde 3-phosphate+H.sub.2O (V)

EC 4.1.2.22:

D-fructose 6-phosphate+phosphatecustom-characteracetyl phosphate+D-erythrose 4-phosphate+H.sub.2O (VI)

[0285] A suitable enzymatic assay to measure phosphoketolase activity is described e.g. in Sonderegger et al., “*Metabolic Engineering of a Phosphoketolase Pathway for Pentose Catabolism in Saccharomyces cerevisiae*”, (2004), Applied & Environmental Microbiology, vol. 70(5), pages 2892-2897, incorporated herein by reference.

[0286] Preferably the protein having phosphoketolase (PKL) activity comprises or consists of:

[0287] an amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36; or [0288] a functional homologue of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36; or [0289] a functional homologue of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36.

[0290] Suitable nucleic acid sequences coding for an phosphoketolase protein may in be found in an organism selected from the group of *Aspergillus niger*, *Neurospora crassa*, *L. casei*, *L. plantarum*, *L. plantarum*, *B. adolescentis*, *B. bifidum*, *B. gallicum*, *B. animalis*, *B. lactis*, *L. pentosum*, *L. acidophilus*, *P. chrysogenum*, *A. nidulans*, *A. clavatus*, *L. mesenteroides*, and *O. oenii*.

[0291] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphoketolase (PKL) activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0292] The recombinant cell may comprise one or more (heterologous) genes coding for an enzyme

having phosphoketolase activity.

Phosphotransacetylase

[0293] As indicated above, the recombinant yeast cell can advantageously comprise a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase (PKL) activity (EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0294] As used herein, a phosphotransacetylase catalyzes at least the conversion of acetyl phosphate to acetyl-CoA.

[0295] The recombinant cell may comprise one or more heterologous genes coding for a protein having phosphotransacetylase activity. Such a protein having phosphotransacetylase activity is herein also referred to as “phosphotransacetylase protein”, “phosphotransacetylase enzyme” or simply as “phosphotransacetylase”. phosphotransacetylase is further herein abbreviated as “PTA”.

[0296] Preferably the protein having phosphotransacetylase (PTA) activity comprises or consists of: [0297] an amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40; or [0298] a functional homologue of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40; or [0299] a functional homologue of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40.

[0300] Suitable nucleic acid sequences coding for an enzyme having phosphotransacetylase may in be found in an organism selected from the group of *B. adolescentis*, *B. subtilis*, *C. cellulolyticum*, *C. phytofermentans*, *B. bifidum*, *B. animalis*, *L. mesenteroides*, *Lactobacillus plantarum*, *M. thermophila*, and *O. oeni*.

[0301] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphotransacetylase (PTA) activity may suitably be incorporated in the genome of the recombinant yeast cell.

Acetate Kinase

[0302] As indicated above, the recombinant yeast cell can comprise a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase (PKL) activity (EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0303] As used herein, an acetate kinase catalyzes at least the conversion of acetate to acetyl phosphate.

[0304] The recombinant cell may comprise one or more, preferably heterologous, genes coding for a protein having acetate kinase activity (EC 2.7.2.12). Such a protein having acetate kinase activity is herein also referred to as “acetate kinase protein”, “acetate kinase enzyme” or simply as “acetate kinase”. Acetate kinase is further herein abbreviated as “ACK”.

[0305] Preferably the protein having acetate kinase (ACK) activity comprises or consists of: [0306] an amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42; or [0307] a functional homologue

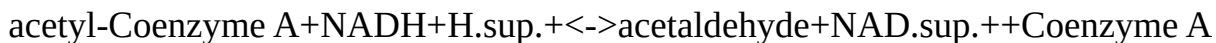
of SEQ ID NO: 41 or SEQ ID NO: 42, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42; or [0308] a functional homologue of SEQ ID NO: 41 or SEQ ID NO: 42, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42. [0309] The nucleic acid sequence (e.g. the gene) encoding for the protein having acetate kinase (ACK) activity may suitably be incorporated in the genome of the recombinant yeast cell.

Acetylating Acetaldehyde Dehydrogenase

[0310] As indicated above, the recombinant yeast cell can advantageously comprise and functionally express a, preferably heterologous, nucleic acid sequence encoding a protein comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10).

[0311] If an acetylating acetaldehyde dehydrogenase is present, more preferably, the recombinant yeast cell functionally expresses: [0312] a, preferably heterologous, nucleic acid sequence encoding a protein comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and [0313] a, suitably endogenous or heterologous, nucleic acid sequence encoding a protein having NAD^{sup.}+-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2); and [0314] a, suitably endogenous or heterologous, nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0315] Acetylating acetaldehyde dehydrogenase is an enzyme that catalyzes the conversion of acetyl-Coenzyme A to acetaldehyde (EC1.2.1.10). This conversion can be represented by the equilibrium reaction formula:



[0316] A protein having acetylating acetaldehyde dehydrogenase activity is herein also referred to as “acetylating acetaldehyde dehydrogenase protein”, “acetylating acetaldehyde dehydrogenase enzyme” or simply “acetylating acetaldehyde dehydrogenase”. Preferences for a acetylating acetaldehyde dehydrogenase and the nucleic sequences encoding for such are as described in WO2011/010923 and WO2019/063507, incorporated herein by reference.

[0317] The nucleic acid sequence encoding a protein having NAD^{sup.}+-dependent acetylating acetaldehyde dehydrogenase activity (EC1.2.1.10) is preferably a heterologous nucleic acid sequence. The encoded NAD^{sup.}+-dependent acetylating acetaldehyde dehydrogenase may therefore preferably be a heterologous NAD^{sup.}+-dependent acetylating acetaldehyde dehydrogenase.

[0318] It is possible for the protein having acetylating acetaldehyde dehydrogenase activity to be monofunctional or bifunctional.

[0319] The nucleic acid sequence encoding the NAD⁺ dependent acetylating acetaldehyde dehydrogenase may in principle originate from any organism comprising a nucleic acid sequence encoding said dehydrogenase. Known acetylating acetaldehyde dehydrogenases that can catalyse the NADH-dependent reduction of acetyl-Coenzyme A to acetaldehyde may in general be divided in three types of NAD⁺ dependent acetylating acetaldehyde dehydrogenase functional homologues:

[0320] 1) Bifunctional proteins that catalyse the reversible conversion of acetyl-CoA to acetaldehyde, and the subsequent reversible conversion of acetaldehyde to ethanol. These type of proteins advantageously have both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity. An example of this type of proteins is the AdhE protein in *E. coli* (Gen Bank No: NP_415757). AdhE appears to be the evolutionary product of a gene fusion. The

NH.sub.2-terminal region of the AdhE protein is highly homologous to aldehyde:NAD⁺ oxidoreductases, whereas the COOH-terminal region is homologous to a family of Fe.sup.2+ dependent ethanol:NAD⁺ oxidoreductases (see Membrillo-Hernandez et al., “*Evolution of the adhE Gene Product of Escherichia coli from a Functional Reductase to a Dehydrogenase*”, (2000) J. Biol. Chem. 275: pages 33869-33875, herein incorporated by reference). The *E. coli* AdhE is subject to metal-catalyzed oxidation and therefore oxygen-sensitive (see Tamarit et al.

“*Identification of the Major Oxidatively Damaged Proteins in Escherichia coli Cells Exposed to Oxidative Stress*” (1998) J. Biol. Chem. 273: pages 3027-3032, herein incorporated by reference). [0321] 2) Proteins that catalyse the reversible conversion of acetyl-Coenzyme A to acetaldehyde in strictly or facultative anaerobic micro-organisms but do not possess alcohol dehydrogenase activity. An example of this type of proteins has been reported in *Clostridium kluyveri* (see Smith et al.” *Purification, Properties, and Kinetic Mechanism of Coenzyme A-Linked Aldehyde Dehydrogenase from Clostridium kluyveri*” (1980) Arch. Biochem. Biophys. Vol. 203: pages 663-675, incorporated herein by reference). An acetylating acetaldehyde dehydrogenase has been annotated in the genome of *Clostridium kluyveri* DSM 555 (GenBank No: EDK33116). A homologous protein AcdH is identified in the genome of *Lactobacillus plantarum* (GenBank No: NP_784141). Another example of this type of proteins is the said gene product in *Clostridium beijerinckii* NRRL B593 (see Toth et al. “*The ald Gene, Encoding a Coenzyme A-Acylating Aldehyde Dehydrogenase, Distinguishes Clostridium beijerinckii and Two Other Solvent-Producing Clostridia from Clostridium acetobutylicum*”, (1999), Appl. Environ. Microbiol. Vol. 65: pages 4973-4980, GenBank No: AAD31841, incorporated herein by reference). [0322] 3) Proteins that are part of a bifunctional aldolase-dehydrogenase complex involved in 4-hydroxy-2-ketovalerate catabolism. Such bifunctional enzymes catalyze the final two steps of the meta-cleavage pathway for catechol, an intermediate in many bacterial species in the degradation of phenols, toluates, naphthalene, biphenyls and other aromatic compounds (Powlowski and Shingler “*Genetics and biochemistry of phenol degradation by Pseudomonas sp. CF600*” (1994) Biodegradation Vol. 5, pages 219-236, herein incorporated by reference). 4-Hydroxy-2-ketovalerate is first converted by 4-hydroxy-2-ketovalerate aldolase to pyruvate and acetaldehyde, subsequently acetaldehyde is converted by acetylating acetaldehyde dehydrogenase to acetyl-CoA. An example of this type of acetylating acetaldehyde dehydrogenase is the DmpF protein in *Pseudomonas* sp CF600 (GenBank No: CAA43226) (Shingler et al., “*Nucleotide Sequence and Functional Analysis of the Complete Phenol/3,4-Dimethylphenol Catabolic Pathway of Pseudomonas sp. Strain CF600*”, (1992), J. Bacteriol., Vol. 174, pages 711-724, incorporated herein by reference). The *E. coli* MphF protein (Ferrandez et al., “*Genetic Characterization and Expression in Heterologous Hosts of the 3-(3-Hydroxyphenyl) Propionate Catabolic Pathway of Escherichia coli K-12*” (1997) J. Bacteriol. 179: pages 2573-2581, GenBank No: NP_414885, incorporated herein by reference) is homologous to the DmpF protein in *Pseudomonas* sp. CF600.

[0323] In a preferred embodiment, the protein having acetylating acetaldehyde dehydrogenase activity is bifunctional and comprises both NAD⁺ dependent acetylating acetaldehyde dehydrogenase (EC 1.2.1.10) activity and NAD⁺ dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC 1.1.1.2).

[0324] A suitable nucleic acid sequence may in particular be found in an organism selected from the group of *Escherichia*, in particular *E. coli*; *Mycobacterium*, in particular *Mycobacterium marinum*, *Mycobacterium ulcerans*, *Mycobacterium tuberculosis*; *Carboxydotherrmus*, in particular *Carboxydotherrmus hydrogenoformans*; *Entamoeba*, in particular *Entamoeba histolytica*; *Shigella*, in particular *Shigella sonnei*; *Burkholderia*, in particular *Burkholderia pseudo mallei*, *Klebsiella*, in particular *Klebsiella pneumoniae*; *Azotobacter*, in particular *Azotobacter vinelandii*; *Azoarcus* sp; *Cupriavidus*, in particular *Cupriavidus taiwanensis*; *Pseudomonas*, in particular *Pseudomonas* sp. CF600; *Pelomaculum*, in particular *Pelotomaculum thermopropionicum*. Preferably, the nucleic acid sequence encoding the NAD.sup.+ dependent acetylating acetaldehyde dehydrogenase

originates from *Escherichia*, more preferably from *E. coli*.

[0325] Particularly suitable is an mhpF gene from *E. coli*, or a functional homologue thereof. This gene is described in Ferrandez et al., "Genetic Characterization and Expression in Heterologous Hosts of the 3-(3-Hydroxyphenyl) Propionate Catabolic Pathway of *Escherichia coli* K-12" (1997) J. Bacteriol. 179: pages 2573-2581. Good results have been obtained with *S. cerevisiae*, wherein an mhpF gene from *E. coli* has been incorporated. In a further advantageous embodiment the nucleic acid sequence encoding an (acetylating) acetaldehyde dehydrogenase is from *Pseudomonas*, in particular dmpF, e.g. from *Pseudomonas* sp. CF600.

[0326] Further, an acetylating acetaldehyde dehydrogenase (or nucleic acid sequence encoding such activity) may for instance be selected from the group of *Escherichia coli* adhE, *Entamoeba histolytica* adh2, *Staphylococcus aureus* adhE, *Piromyces* sp. E2 adhE, *Clostridium kluyveri* EDK33116, *Lactobacillus plantarum* acdH, *Escherichia coli* eutE, *Listeria innocua* acdH, and *Pseudomonas putida* YP 001268189.

[0327] Preferably the protein having NAD.sup.+ -dependent acetylating acetaldehyde dehydrogenase activity comprises or consists of: [0328] an amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48; or [0329] a functional homologue of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48 having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48; or [0330] a functional homologue of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48.

[0331] Most preferably the acetylating acetaldehyde dehydrogenase protein is a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity.

[0332] The nucleic acid sequence (e.g. the gene) encoding for the protein having acetylating acetaldehyde dehydrogenase activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0333] Examples of suitable enzymes are further illustrated below in tables 6(a) to 6(e) for BLAST of the listed enzymes, giving suitable alternative alcohol/acetaldehyde dehydrogenases.

TABLE-US-00006 TABLE 6(a) BLAST Query - adhE from *Escherichia coli* Identity Accession Description (%) number bifunctional acetaldehyde-CoA/alcohol 100 NP_309768.1 dehydrogenase [*Escherichia coli* O157:H7 str. Sakai] bifunctional acetaldehyde-CoA/alcohol 99 YP_540449.1 dehydrogenase [*Escherichia coli* UTI89] bifunctional acetaldehyde-CoA/alcohol 95 YP_001177024.1 dehydrogenase [*Enterobacter* sp. 638]

TABLE-US-00007 TABLE 6(b) BLAST Query - acdH from *Lactobacillus plantarum* Description Identity (%) Accession number acetaldehyde dehydrogenase [*Lactobacillus plantarum* 100 YP_004888365.1 WCFS1] acetaldehyde dehydrogenase [*Lactobacillus pentosus* 95 CCC16763.1 IG1] aldehyde-alcohol dehydrogenase [*Enterococcus* 58 WP_016251441.1 *cecorum*] aldehyde-alcohol dehydrogenase 2 [*Enterococcus* 57 WP_016623694.1 *faecalis*] bifunctional acetaldehyde-CoA/alcohol dehydrogenase 55 WP_010493695.1 [*Lactobacillus zae*] alcohol dehydrogenase [*Bacillus thuringiensis*] 54 WP_003280110.1 bifunctional acetaldehyde-CoA/alcohol

dehydrogenase, 53 WP_009931954.1 partial [*Listeria monocytogenes*]

TABLE-US-00008 TABLE 6(c) BLAST Query - eutE from *Escherichia coli* Description Identity (%) Accession number aldehyde oxidoreductase, ethanolamine utilization 100 NP_416950.1 protein [*Escherichia coli* str. K-12 substr. MG1655] ethanolamine utilization; acetaldehyde dehydrogenase 99 NP_289007.1 [*Escherichia coli* 0157:H7 str. EDL933] aldehyde dehydrogenase [*Escherichia albertii*] 99 WP_001075674.1

TABLE-US-00009 TABLE 6(d) BLAST Query - Lin1129 from *Listeria innocua* Description Identity (%) Accession number aldehyde dehydrogenase [*Listeria innocua*] > 100 NP_470466.1 emb|CAC96360.1| lin1129 [*Listeria innocua* Clip11262] ethanolamine utilization protein EutE [*Listeria innocua*] 99 WP_003761764.1 aldehyde dehydrogenase [*Listeria monocytogenes*] 95 AGR09081.1 hypothetical protein [*Enterococcus malodoratus*] 64 WP_010739890.1 aldehyde dehydrogenase [*Yersinia aldovae*] 59 WP_004699364.1 aldehyde dehydrogenase EutE [*Klebsiella pneumoniae*] 58 WP_004205473.1

TABLE-US-00010 TABLE 6(e) BLAST Query - adhE from *Staphylococcus aureus* Description Identity (%) Accession number bifunctional acetaldehyde-CoA/alcohol dehydrogenase 100 NP_370672.1 [*Staphylococcus aureus* subsp. aureus Mu50] aldehyde dehydrogenase family protein [*Staphylococcus* 99 YP_008127042.1 aureus CA-347] bifunctional acetaldehyde-CoA/alcohol dehydrogenase 85 WP_002495347.1 [*Staphylococcus epidermidis*] aldehyde-alcohol dehydrogenase 2 [*Enterococcus* 75 WP_016623694.1 faecalis]

Acetyl-Coenzyme A Synthetase

[0334] If the recombinant yeast cell functionally expresses a protein having acetylating acetaldehyde dehydrogenase activity, preferably the recombinant yeast cell is further functionally expressing: [0335] a nucleic acid sequence encoding a protein having NAD.sup.+ -dependent alcohol dehydrogenase activity (EC 1.1.1.1 or or EC1.1.1.2); and/or [0336] a nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0337] A protein having acetyl-Coenzyme A synthetase activity can herein also be referred to as “acetyl-Coenzyme A synthetase protein”, “acetyl-Coenzyme A synthetase enzyme” or simply “acetyl-Coenzyme A synthetase” or even “acetyl CoA synthetase”. The protein is further abbreviated herein as “ACS”.

[0338] The acetyl-Coenzyme A synthetase, also known as acetate-CoA ligase or acetyl-activating enzyme, catalyses the formation of acetyl-CoA from acetate, coenzyme A (CoA) and ATP as shown below:

ATP+acetate+CoA=AMP+diphosphate+acetyl-CoA

[0339] It is understood that the recombinant yeast cell may naturally comprise an endogenous gene encoding an acetyl-Coenzyme A synthetase protein. In the alternative, or in addition thereto, the recombinant yeast cell may comprise a heterologous nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0340] For example, the recombinant yeast cell according to the invention may comprise an acetyl-Coenzyme A synthetase, which may be present in the wild-type cell, as is for instance the case with *S. cerevisiae* which contains two acetyl-Coenzyme A synthetase isoenzymes encoded by the ACS1 (amino acid sequence illustrated as SEQ ID NO: 49) and ACS2 (amino acid sequence illustrated as SEQ ID NO: 50) genes (van den Berg et al (1996) J. Biol. Chem. 271:pages 28953-28959, incorporated herein by reference), or a host cell may be provided with one or more heterologous gene(s) encoding this activity, e.g. the ACS1 and/or ACS2 gene of *S. cerevisiae* or a functional homologue thereof may be incorporated into a cell lacking acetyl-Coenzyme A synthetase isoenzyme activity.

[0341] Preferably the protein having NAD.sup.+ -dependent acetyl-Coenzyme A synthetase activity comprises or consists of: [0342] an amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50; or [0343] a functional homologue of SEQ ID NO: 49 or SEQ ID NO: 50 having at least 40%, at least

45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50; or [0344] a functional homologue of SEQ ID NO: 49 or SEQ ID NO: 50 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50.

[0345] Preferably the recombinant yeast cell is a recombinant yeast cell wherein the, endogenous or heterologous, acetyl-Coenzyme A synthetase protein, is overexpressed, most preferably by using a suitable promoter as described for example in WO2011/010923, incorporated herein by reference. Any heterologous nucleic acid sequence (e.g. the gene) encoding for the protein having acetyl-Coenzyme A synthetase activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0346] Examples of suitable proteins having acetyl-Coenzyme A synthetase activity are listed in table 7. At the top of table 7 the ACS2 used in the examples and that is BLASTED is mentioned. TABLE-US-00011 TABLE 7 BLAST Query - ACS2 from *Saccharomyces cerevisiae* Description Identity (%) Accession number acetate -- CoA ligase ACS2 [*Saccharomyces cerevisiae* 100 NP_013254.1 S288c] acetyl CoA synthetase [*Saccharomyces cerevisiae* 99 EDN59693.1 YJM789] acetate -- CoA ligase [*Kluyveromyces lactis* NRRL Y- 85 XP_453827.1 1140] acetate -- CoA ligase [*Candida glabrata* CBS 138] 83 XP_445089.1 acetate -- CoA ligase [*Scheffersomyces stipitis* CBS 6054] 68 XP_001385819.1 acetyl-coenzyme A synthetase FacA [*Aspergillus* 63 EDP50475.1 *fumigatus* A1163] acetate -- CoA ligase facA-*Penicillium chrysogenum* 62 XP_002564696.1 [*Penicillium chrysogenum* Wisconsin 54-1255]

Alcohol Dehydrogenase

[0347] If the recombinant yeast cell functionally expresses a protein having acetylating acetaldehyde dehydrogenase activity, preferably the recombinant yeast cell is further functionally expressing: [0348] a nucleic acid sequence encoding a protein having NAD.sup.+ dependent alcohol dehydrogenase activity (EC 1.1.1.1 or or EC1.1.1.2); and/or [0349] a nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0350] A protein having alcohol dehydrogenase activity is herein also referred to as “alcohol dehydrogenase protein”, “alcohol dehydrogenase enzyme” or simply “alcohol dehydrogenase”. The protein is further abbreviated herein as “ADH”.

[0351] The alcohol dehydrogenase enzyme catalyses the conversion of acetaldehyde into ethanol.

[0352] It is understood that the recombinant yeast cell may naturally comprise an endogenous nucleic acid sequence encoding an alcohol dehydrogenase protein. In the alternative, or in addition thereto, the recombinant yeast cell may comprise a heterologous nucleic acid sequence encoding a protein having alcohol dehydrogenase activity

[0353] For example, the recombinant yeast cell may naturally comprise a gene encoding alcohol dehydrogenase, as is de case with *S. cerevisiae* (Amino acid sequences of the native *S. cerevisiae* alcohol dehydrogenases ADH1, ADH2, ADH3, ADH4 and ADH5 are illustrated respectively as SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 and SEQ ID NO: 55), see Lutstorf and Megnet, “Multiple Forms of Alcohol Dehydrogenase in *Saccharomyces Cerevisiae*”, (1968), Arch. Biochem. Biophys., vol. 126, pages 933-944, incorporated herein by reference, or Ciriacy, “Genetics of Alcohol Dehydrogenase in *Saccharomyces cerevisiae* I. Isolation and genetic analysis of *adh* mutants”, (1975), Mutat. Res. 29, pages 315-326, incorporated herein by reference).

[0354] Preferably, however, the recombinant yeast cell comprises alcohol dehydrogenase activity

within a, suitably heterologous, bifunctional enzyme having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity as described herein above. That is, most preferably the alcohol dehydrogenase protein is a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity. When the recombinant yeast cell comprises a heterologous nucleic acid sequence encoding a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity, any native nucleic acid sequences encoding for any native protein encoding alcohol dehydrogenase activity may or may not be disrupted and/or deleted.

[0355] The recombinant yeast cell may therefore advantageously be a recombinant yeast cell functionally expressing: [0356] one or more heterologous nucleic acid sequence(s) encoding a bifunctional protein having NAD.sup.+ -dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and NAD.sup.+ -dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2); and [0357] one or more, native or heterologous, nucleic acid sequence(s) encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1), wherein optionally one or more native nucleic acid sequence(s) encoding a protein having NAD.sup.+ -dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2) are disrupted or deleted.

[0358] Alternatively the recombinant yeast cell may advantageously be a recombinant yeast cell functionally expressing: [0359] one or more, native or heterologous, nucleic acid sequence(s) encoding a monofunctional protein having NAD.sup.+ -dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and [0360] one or more, native or heterologous, nucleic acid sequence(s) encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1); and [0361] one or more, native or heterologous, nucleic acid sequences(s) encoding a protein having NAD.sup.+ -dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2).

[0362] Preferences for the bifunctional protein are provided above and are as listed for the acetylating acetaldehyde dehydrogenase protein. If the protein is not bifunctional, the NAD.sup.+ -dependent alcohol dehydrogenase protein is preferably a protein having NAD.sup.+ -dependent alcohol dehydrogenase activity that comprises or consists of: [0363] an amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55; or [0364] a functional homologue of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55 having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55; or [0365] a functional homologue of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55.

[0366] Any heterologous nucleic acid sequence (e.g. the gene) encoding for the protein having NAD.sup.+ -dependent alcohol dehydrogenase activity may suitably be incorporated in the genome of the recombinant yeast cell.

Deletion or Disruption of Glycerol 3-Phosphate Phosphohydrolase and/or Glycerol 3-Phosphate Dehydrogenase

[0367] The recombinant yeast cell further may or may not comprise a deletion or disruption of one or more endogenous nucleotide sequence encoding a glycerol 3-phosphate phosphohydrolase gene and/or encoding a glycerol 3-phosphate dehydrogenase gene.

[0368] Preferably enzymatic activity needed for the NADH-dependent glycerol synthesis in the yeast cell is reduced or deleted. The reduction or deletion of the enzymatic activity of glycerol 3-phosphate phosphohydrolase and/or glycerol 3-phosphate dehydrogenase can be achieved by modifying one or more genes encoding a NAD-dependent glycerol 3-phosphate dehydrogenase (GPD) and/or one or more genes encoding a glycerol phosphate phosphatase (GPP), such that the enzyme is expressed considerably less than in the wild-type or such that the gene encodes a polypeptide with reduced activity. Such modifications can be carried out using commonly known biotechnological techniques, and may in particular include one or more knock-out mutations or site-directed mutagenesis of promoter regions or coding regions of the structural genes encoding GPD and/or GPP. Alternatively, yeast strains that are defective in glycerol production may be obtained by random mutagenesis followed by selection of strains with reduced or absent activity of GPD and/or GPP. *S. cerevisiae* GPD1, GPD2, GPP1 and GPP2 genes are shown in WO2011010923, and are disclosed in SEQ ID NO: 24-27 of that application.

[0369] Preferably the recombinant yeast is a recombinant yeast that further comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase (GPD) gene. The one or more of the glycerol phosphate phosphatase (GPP) genes may or may not be deleted or disrupted.

[0370] More preferably the recombinant yeast is a recombinant yeast that comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase 1 (GPD1) gene. The glycerol-3-phosphate dehydrogenase 2 (GPD2) gene may or may not be deleted or disrupted.

[0371] Most preferably the recombinant yeast is a recombinant yeast that comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase 1 (GPD1) gene, whilst the glycerol-3-phosphate dehydrogenase 2 (GPD2) gene remains active and/or intact. Preferably therefore, only one of the *S. cerevisiae* GPD1, GPD2, GPP1 and GPP2 genes is disrupted and deleted, whereas most preferably only GPD1 is chosen from the group consisting of GPD1, GPD2, GPP1 and GPP2 genes to be disrupted or deleted.

[0372] Without wishing to be bound to any kind of theory it is believed that a recombinant yeast according to the invention wherein the GPD1 gene, but not the GPD2 gene, is deleted or disrupted, can be advantageous when applied in a fermentation process where the glucose at the start of or during the fermentation, is preferably equal to or more than 80 g/L, more preferably equal to or more than 90 g/L, even more preferably equal to or more than 100 g/L, still more preferably equal to or more than 110 g/L, yet even more preferably equal to or more than 120 g/L, equal to or more than 130 g/L, equal to or more than 140 g/L, equal to or more than 150 g/L, equal to or more than 160 g/L, equal to or more than 170 g/L, or equal to or more than 180 g/L.

[0373] Preferably at least one gene encoding a GPD and/or at least one gene encoding a GPP is entirely deleted, or at least a part of the gene is deleted that encodes a part of the enzyme that is essential for its activity. Good results can be achieved with a *S. cerevisiae* cell, wherein the open reading frames of the GPD1 gene and/or of the GPD2 gene have been inactivated. Inactivation of a structural gene (target gene) can be accomplished by a person skilled in the art by synthetically synthesizing or otherwise constructing a DNA fragment consisting of a selectable marker gene flanked by DNA sequences that are identical to sequences that flank the region of the host cell's genome that is to be deleted. Suitably, good results can be obtained with the inactivation of the GPD1 and GPD2 genes in *Saccharomyces cerevisiae* by integration of the marker genes kanMX and hphMX4. Subsequently this DNA fragment is transformed into a host cell.

Transformed cells that express the dominant marker gene are checked for correct replacement of the region that was designed to be deleted, for example by a diagnostic polymerase chain reaction or Southern hybridization.

[0374] Thus, in the recombinant yeast cells of the invention, glycerol 3-phosphate phosphohydrolase activity in the cell and/or glycerol 3-phosphate dehydrogenase activity in the cell can be advantageously reduced.

Glucoamylase

[0375] Preferably, the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding for a glucoamylase (EC 3.2.1.20 or 3.2.1.3).

[0376] A protein having glucoamylase activity is herein also referred to as “glucoamylase enzyme”, “glucoamylase protein” or simply “glucoamylase”. Glucoamylase has herein been abbreviated as “GA”.

[0377] Glucoamylase, also referred to as amyloglucosidase, alpha-glucosidase, glucan 1,4-alpha glucosidase, maltase glucoamylase, and maltase-glucoamylase, catalyses at least the hydrolysis of terminal 1,4-linked alpha-D-glucose residues from non-reducing ends of amylose chains to release free D-glucose. A glucoamylase may be further defined by its amino acid sequence. Likewise a glucoamylase may be further defined by a nucleotide sequence encoding the glucoamylase. As explained in detail above under definitions, a certain glucoamylase that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the glucoamylase.

[0378] Preferably the protein having glucoamylase activity comprises or consists of: [0379] an amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58; or [0380] a functional homologue of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58; or [0381] a functional homologue of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58.

[0382] The polypeptide of SEQ ID NO: 56 encodes a “mature glucoamylase”, referring to the enzyme in its final form after translation and any post-translational modifications, such as N-terminal processing, C-terminal truncation, glycosylation, phosphorylation, etc.

[0383] In an embodiment the nucleotide sequence encodes a polypeptide having an amino acid sequence of SEQ ID NO: 57 or a variant thereof having an amino acid sequence identity of at least 50%, preferably at least 60%, 70%, 75%, 80%, 85%, 90%, 95, 98%, or 99% with the amino acid sequence of SEQ ID NO: 57. Amino acids 1-17 of the SEQ ID NO: 57 may encode for a native signal sequence.

[0384] In another embodiment the nucleotide sequence allowing the expression of a glucoamylase encodes a polypeptide having an amino acid sequence of SEQ ID NO: 58 or a variant thereof having an amino acid sequence identity of at least 50%, preferably at least 60%, 70%, 75%, 80%, 85%, 90%, 95, 98%, or 99% with the amino acid sequence of SEQ ID NO: 58. Amino acids 1-19 of the SEQ ID NO: 58 may encode for a signal sequence.

[0385] A signal sequence (also referred to as signal peptide, targeting signal, localization signal, localization sequence, transit peptide, leader sequence or leader peptide) can be present at the N-terminus of a polypeptide (here, the glucoamylase) where it signals that the polypeptide is to be excreted, for example outside the cell and into the media.

[0386] The nucleic acid sequence (e.g. the gene) encoding for the protein having glucoamylase activity may suitably be incorporated in the genome of the recombinant yeast cell.

Glycerol Re-Uptake

[0387] The recombinant yeast cell may or may not further comprise one or more additional nucleic acid sequences that are part of a glycerol re-uptake pathway. That is, the recombinant yeast cell may or may not further comprise: [0388] one or more heterologous nucleic acid sequences

encoding for a glycerol dehydrogenase; and/or [0389] one or more homologous or heterologous nucleic acid sequences encoding for a dihydroxyacetone kinase; and/or [0390] one or more heterologous nucleic acid sequences encoding for a glycerol transporter.

[0391] Thus, in one preferred embodiment the recombinant yeast cell is a recombinant yeast cell functionally expressing: [0392] one or more heterologous nucleic acid sequences encoding for a ribulose-1,5-phosphate carboxylase/oxygenase (EC4.1.1.39; Rubisco), and optionally one or more nucleic acid sequences encoding for molecular chaperones for Rubisco; [0393] one or more heterologous nucleic acid sequences encoding for phosphoribulokinase (EC2.7.1.19; PRK); [0394] one or more nucleic acid sequences encoding for a transketolase (EC 2.2.1.1), wherein the transketolase is under control of a promoter (the “TKL promoter”) which has a TKL expression ratio .sub.anaerobic/aerobic of 2 or more; [0395] one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase; [0396] one or more homologous or heterologous nucleic acid sequences encoding for a dihydroxyacetone kinase; and [0397] optionally one or more heterologous nucleic acid sequences encoding for a glycerol transporter.

[0398] Without wishing to be bound by any kind of theory it is believed that a recombinant yeast cell that further comprises a combination of glycerol dehydrogenase, dihydroxyacetone kinase and optionally a glycerol transporter has an improved overall performance in the form of higher ethanol yields.

[0399] In an alternative preferred embodiment the recombinant yeast cell is a recombinant yeast cell that does not functionally express: [0400] one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase; and/or [0401] one or more heterologous nucleic acid sequences encoding for a dihydroxyacetone kinase; and/or [0402] one or more heterologous nucleic acid sequences encoding for a glycerol transporter.

[0403] Without wishing to be bound by any kind of theory it is believed that in the absence of one or more of these features of such a glycerol re-uptake pathway, a recombinant yeast cell is obtained that has a very low accumulation of glucose and/or other sugars and has an improved robustness when applied in a medium comprising a high amount of sugars. The application of a recombinant yeast cell that does not comprise one or more of a, heterologous and/or homologous, glycerol dehydrogenase; heterologous and/or homologous dihydroxyacetone kinase and/or heterologous and/or homologous glycerol transporter can therefore be advantageous when applied in a fermentation process where the glucose at the start of or during the fermentation, is preferably equal to or more than 80 g/L, more preferably equal to or more than 90 g/L, even more preferably equal to or more than 100 g/L, still more preferably equal to or more than 110 g/L, yet even more preferably equal to or more than 120 g/L, equal to or more than 130 g/L, equal to or more than 140 g/L, equal to or more than 150 g/L, equal to or more than 160 g/L, equal to or more than 170 g/L, or equal to or more than 180 g/L.

[0404] Most preferably, the recombinant yeast is therefore a recombinant yeast that is functionally expressing: [0405] one or more heterologous nucleic acid sequences encoding for a ribulose-1,5-phosphate carboxylase/oxygenase (EC4.1.1.39; Rubisco), and optionally one or more nucleic acid sequences encoding for molecular chaperones for Rubisco; [0406] one or more heterologous nucleic acid sequences encoding for phosphoribulokinase (EC2.7.1.19; PRK); [0407] one or more nucleic acid sequences encoding for a transketolase (EC 2.2.1.1), wherein the transketolase is under control of a promoter (the “TKL promoter”) which has a TKL expression ratio .sub.anaerobic/aerobic of 2 or more;

wherein the recombinant yeast cell does not functionally express [0408] one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase; and/or [0409] one or more heterologous nucleic acid sequences encoding for a dihydroxyacetone kinase; and/or [0410] one or more heterologous nucleic acid sequences encoding for a glycerol transporter.

Glycerol Dehydrogenase

[0411] As indicated above, the recombinant yeast cell may or may not functionally express [0412]

a nucleic acid sequence encoding for a protein having glycerol dehydrogenase activity (E.C. 1.1.1.6); [0413] a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (E.C. 2.7.1.28 or E.C. 2.7.1.29); and [0414] optionally a nucleic acid sequence encoding a protein having glycerol transporter activity.

[0415] Thus the recombinant yeast cell may or may not functionally express one or more, preferably heterologous, nucleic acid sequences encoding for a glycerol dehydrogenase.

[0416] If a glycerol dehydrogenase is present, the recombinant yeast cell may comprise a NAD^{sup.}+ linked glycerol dehydrogenase (EC 1.1.1.6) and/or a NADP^{sup.}+ linked glycerol dehydrogenase (EC 1.1.1.72). That is, the recombinant yeast cell may or may not comprise a nucleic acid sequence encoding a protein having NAD^{sup.}+ dependent glycerol dehydrogenase activity (EC 1.1.1.6) and/or a nucleic acid sequence encoding a protein having NADP^{sup.}+ dependent glycerol dehydrogenase activity (EC 1.1.1.72).

[0417] In one embodiment the protein having glycerol dehydrogenase activity is preferably a protein having NAD⁺ dependent glycerol dehydrogenase activity (EC 1.1.1.6) and preferably the recombinant yeast cell functionally expresses a nucleic acid sequence encoding a protein having NAD⁺ dependent glycerol dehydrogenase activity (EC 1.1.1.6). Such protein may be from bacterial origin or for instance from fungal origin. An example is *gldA* from *E. coli*.

[0418] In an alternative or additional embodiment, a NADP⁺ dependent glycerol dehydrogenase can be present (EC 1.1.1.72).

[0419] If a glycerol dehydrogenase is present, a NAD⁺ linked glycerol dehydrogenase is preferred.

[0420] A protein having glycerol dehydrogenase activity is herein also referred to as “glycerol dehydrogenase protein”, “glycerol dehydrogenase enzyme” or simply as “glycerol dehydrogenase”. In analogy thereto a protein having NAD⁺ dependent glycerol dehydrogenase activity is herein also referred to as “NAD⁺ dependent glycerol dehydrogenase protein”, “NAD⁺ dependent glycerol dehydrogenase enzyme” or simply as “NAD⁺ dependent glycerol dehydrogenase”. The glycerol dehydrogenase is abbreviated as GLD.

[0421] Preferences for a glycerol dehydrogenase and the nucleic sequences encoding for such are as described in WO2015028582, incorporated herein by reference.

[0422] NAD⁺ dependent glycerol dehydrogenase (EC 1.1.1.6) is an enzyme that catalyzes the chemical reaction:

glycerol+NAD^{sup.}+custom-characterglycerone+NADH+H^{sup.}+

[0423] Thus, the two substrates of this enzyme are glycerol and NAD⁺, whereas its three products are glycerone, NADH, and H^{sup.}+. Glyceron and dihydroxyacetone are herein synonyms.

[0424] The glycerol dehydrogenase enzyme belongs to the family of oxidoreductases, specifically those acting on the CH—OH group of donor with NAD^{sup.}+ or NADP^{sup.}+ as acceptor. The systematic name of this enzyme class is glycerol:NAD^{sup.}+2-oxidoreductase. Other names in common use include glycerin dehydrogenase, and NAD^{sup.}+linked glycerol dehydrogenase. This enzyme participates in glycerolipid metabolism. A glycerol dehydrogenase protein may be further defined by its amino acid sequence. Likewise a glycerol dehydrogenase protein may be further defined by a nucleotide sequence encoding the glycerol dehydrogenase protein. As explained in detail above under definitions, a certain glycerol dehydrogenase protein that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the glycerol dehydrogenase protein.

[0425] The nucleic acid sequence encoding the protein having glycerol dehydrogenase activity can be a heterologous nucleic acid sequence. The protein having glycerol dehydrogenase activity can be a heterologous protein having NAD⁺ dependent glycerol dehydrogenase activity.

[0426] If the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase, the recombinant yeast cell preferably further comprises suitable co-factors to enhance the activity of the glycerol dehydrogenase. For example, the

recombinant yeast cell may comprise zinc, zinc ions or zinc salts and/or one or more pathways to include such in the cell.

[0427] Suitable examples of heterologous proteins having glycerol dehydrogenase activity include the glycerol dehydrogenase proteins of respectively *Klebsiella pneumoniae*, *Enterococcus aerogenes*, *Yersinia aldovae*, and *Escherichia coli*. Their amino acid sequences of such proteins have been illustrated respectively by SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 61 and SEQ ID NO: 62.

[0428] The recombinant yeast cell therefore may or may not include one or more, suitably heterologous, glycerol dehydrogenase proteins having an amino acid sequence of SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 61 and/or SEQ ID NO: 62; and/or functional homologues thereof comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 61 and/or SEQ ID NO: 62; and/or functional homologues thereof comprising an amino acid sequence having one or more mutations, substitutions, insertions and/or deletions as compared to the amino acid sequence of SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 61 and/or SEQ ID NO: 62, wherein more preferably the amino acid sequence of such functional homologues has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions as compared to the amino acid sequence of SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 61 and/or SEQ ID NO: 62.

[0429] A preferred glycerol dehydrogenase protein is the glycerol dehydrogenase protein encoded by the *gldA* gene from *E. coli*. SEQ ID NO: 62 shows the amino acid sequence of this preferred NAD⁺ dependent glycerol dehydrogenase protein, encoded by the *gldA* gene from *E. coli*. The nucleic acid sequence of the *gldA* gene of *E. coli* is illustrated by SEQ ID NO: 63.

[0430] If the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase, the recombinant yeast cell therefore most preferably comprises a heterologous nucleotide sequence encoding a protein having NAD⁺ dependent glycerol dehydrogenase activity (E.C. 1.1.1.6) derived from *E. coli*, optionally codon-optimized for the host cell, as exemplified by the nucleic acid sequence shown in SEQ ID NO:63.

[0431] Preferable the nucleic acid sequence encoding the protein having glycerol dehydrogenase activity thus comprises or consists of: [0432] a nucleic acid sequence of SEQ ID NO:63; or [0433] a functional homologue of SEQ ID NO:63, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO:63; or [0434] a functional homologue of SEQ ID NO:63, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO:63, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO:63.

[0435] If the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase, the recombinant yeast cell therefore most preferably comprises one or more nucleotide sequence encoding a glycerol dehydrogenase (E.C. 1.1.1.6) derived from *E. coli*, optionally codon-optimized for the host cell. Such heterologous nucleic acid sequence (e.g. the gene) encoding for the glycerol dehydrogenase protein may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2015/028583, herein incorporated by reference.

[0436] Further examples of suitable glycerol dehydrogenases are listed in Table 8(a) to 8(d). At the top of each table the gldA that is BLASTED is mentioned.

TABLE-US-00012 TABLE 8(a) BLAST Query - gldA from *Escherichia coli* Description Identity (%) Accession number glycerol dehydrogenase, NAD [*Escherichia coli* str. K-12 100 NP_418380.4 substr. MG1655] glycerol dehydrogenase [*Escherichia coli* 0127:H6 str. 99 YP_002331714.1 E2348/69] glycerol dehydrogenase [*Citrobacter youngae*] 94 WP_006686227.1 glycerol dehydrogenase [*Citrobacter freundii*] 92 WP_003840533.1

TABLE-US-00013 TABLE 8(b) BLAST Query - gldA from *Klebsiella pneumoniae* Description Identity (%) Accession number glycerol dehydrogenase [*Klebsiella pneumoniae* 342] 100 YP_002236495.1 glycerol dehydrogenase [*Citrobacter freundii*] 93 WP_003024745.1 Glycerol dehydrogenase (EC 1.1.1.6) [*Enterobacter* 92 YP_004590977.1 *aerogenes* EA1509E] glycerol dehydrogenase [*Escherichia coli*] 91 WP_016241524.1 glycerol dehydrogenase [*Enterococcus aerogenes*] 87 See examples herein strains with CAS15 glycerol dehydrogenase [*Yersinia aldovae*] 74 WP_004701845.1 glycerol dehydrogenase [*Enterobacteriaceae bacterium* 61 WP_017375113.1 LSJC7] glycerol dehydrogenase [*Citrobacter youngae*] 60 WP_006686227.1

TABLE-US-00014 TABLE 8(c) BLAST Query - gldA from *Enterococcus aerogenes* Description Identity (%) Accession number glycerol dehydrogenase [*Enterobacter aerogenes* KCTC 100 YP_004591726.1 2190] Glycerol dehydrogenase (EC 1.1.1.6) [*Enterobacter* 99 YP_007390021.1 *aerogenes* EA1509E] glycerol dehydrogenase [*Klebsiella pneumoniae*] 92 WP_004203683.1 glycerol dehydrogenase [*Escherichia coli*] 88 WP_001322519.1 See examples herein strains with CAS13 glycerol dehydrogenase [*Enterobacter cloacae* subsp. 87 YP_003615506.1 *cloacae* ATCC 13047]

TABLE-US-00015 TABLE 8(d) BLAST Query - gldA from *Yersinia aldovae* Description Identity (%) Accession number glycerol dehydrogenase [*Yersinia aldovae*] 100 WP_004701845.1 glycerol dehydrogenase [*Yersinia intermedia*] 95 WP_005189747.1 glycerol dehydrogenase [*Serratia liquefaciens* ATCC 81 YP_008232202.1 27592] glycerol dehydrogenase [*Escherichia coli*] 76 WP_016241524.1 See examples herein strains with CAS13. hypothetical protein EAE_03845 [*Enterobacter* 75 YP_004590977.1 *aerogenes* KCTC 2190] glycerol dehydrogenase [*Aeromonas hydrophila*] 65 WP_017410769.1

Dihydroxyacetone Kinase

[0437] As indicated above, the recombinant yeast cell may or may not functionally express [0438] a nucleic acid sequence encoding for a protein having glycerol dehydrogenase activity (E.C. 1.1.1.6); [0439] a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (E.C. 2.7.1.28 or E.C. 2.7.1.29); and [0440] optionally a nucleic acid sequence encoding a protein having glycerol transporter activity.

[0441] That is, the recombinant yeast cell may or may not functionally express one or more, homologous or heterologous, nucleic acid sequences encoding for dihydroxyacetone kinase (E.C. 2.7.1.28 or E.C. 2.7.1.29),

[0442] A protein having dihydroxyacetone kinase activity is herein also referred to as “dihydroxyacetone kinase protein”, “dihydroxyacetone kinase enzyme” or simply as “dihydroxyacetone kinase”. The dihydroxyacetone kinase is abbreviated herein as DAK.

[0443] Preferences for a dihydroxyacetone kinase and the nucleic sequences encoding for such are as described in WO2015028582, incorporated herein by reference.

[0444] The protein having dihydroxy kinase activity may suitably belong to the enzyme categories of E.C. 2.7.1.28 and/or E.C. 2.7.1.29. The recombinant yeast cell thus suitably functionally expresses a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (E.C. 2.7.1.28 and/or E.C. 2.7.1.29).

[0445] A dihydroxyacetone kinase is preferably herein understood as an enzyme that catalyzes the chemical reaction (EC 2.7.1.29):

ATP+glycerone.Math.ADP+glycerone phosphate
and/or the chemical reaction (EC 2.7.1.28):

ATP+D-glyceraldehyde.Math.ADP+D-glyceraldehyde 3-phosphate.

[0446] Other names in common use for a dihydroxyacetone kinase include glycerone kinase, ATP:glycerone phosphotransferase and (phosphorylating) acetol kinase. It is further understood that glycerone and dihydroxyacetone are the same molecule. A dihydroxyacetone kinase protein may be further defined by its amino acid sequence. Likewise a dihydroxyacetone kinase protein may be further defined by a nucleotide sequence encoding the dihydroxyacetone kinase protein. As explained in detail above under definitions, a certain dihydroxyacetone kinase protein that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the dihydroxyacetone kinase protein.

[0447] If present, the recombinant yeast cell preferably functionally expresses a nucleic acid sequence encoding a native protein having dihydroxyacetone kinase activity. More preferably, the nucleic acid sequence encoding the protein having dihydroxyacetone kinase activity is a native nucleic acid sequence.

[0448] Yeast comprises two native isozymes of dihydroxyacetone kinase (DAK1 and DAK2). These native dihydroxyacetone kinase enzymes are preferred according to the invention. Preferably the host cell is a *Saccharomyces cerevisiae* cell and preferably the above native dihydroxyacetone kinase enzymes are the native dihydroxyacetone kinase enzymes of a *Saccharomyces cerevisiae* yeast cell. The amino acid sequences of the native dihydroxyacetone kinase proteins of *Saccharomyces cerevisiae*, DAK1 and DAK2, have been illustrated respectively by SEQ ID NO: 64 and SEQ ID NO: 65. The nucleic acid sequences coding for these native dihydroxyacetone kinase proteins DAK1 and DAK2 have been illustrated respectively by SEQ ID NO: 69 and SEQ ID NO: 70.

[0449] It is also possible for the recombinant yeast cell to functionally express a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity, where the nucleic acid sequence is a heterologous nucleic acid sequence, respectively wherein the protein is a heterologous protein. In an embodiment the recombinant yeast cell comprises a heterologous gene encoding a dihydroxyacetone kinase. Suitable heterologous genes include the genes encoding dihydroxyacetone kinases from *Saccharomyces kudriavzevii*, *Zygosaccharomyces bailii*, *Kluyveromyces lactis*, *Candida glabrata*, *Yarrowia lipolytica*, *Klebsiella pneumoniae*, *Enterobacter aerogenes*, *Escherichia coli*, *Yarrowia lipolytica*, *Schizosaccharomyces pombe*, *Botryotinia fuckeliana*, and *Exophiala dermatitidis*. Preferred heterologous proteins having dihydroxyacetone kinase activity include those derived from respectively *Klebsiella pneumoniae*, *Yarrowia lipolytica* and *Schizosaccharomyces pombe*, as illustrated respectively by SEQ ID NO: 66, SEQ ID NO: 67 and SEQ ID NO: 68.

[0450] The recombinant yeast cell may or may not comprise a genetic modification that causes overexpression of a dihydroxyacetone kinase, for example by overexpression of a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity. The nucleotide sequence encoding the dihydroxyacetone kinase may be native or heterologous to the cell. Nucleic acid sequences that may be used for overexpression of dihydroxyacetone kinase in the cells of the invention are for example the dihydroxyacetone kinase genes from *S. cerevisiae* (DAK1) and (DAK2) as e.g. described by Molin et al., "*Dihydroxy-acetone kinases in Saccharomyces cerevisiae are involved in detoxification of dihydroxyacetone*" (2003), J. Biol. Chem., vol. 278: pages 1415-1423, incorporated herein by reference. In a preferred embodiment a codon-optimised (see above) nucleotide sequence encoding the dihydroxyacetone kinase is overexpressed, such as e.g. a codon optimised nucleotide sequence encoding the dihydroxyacetone kinase of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68.

[0451] As indicated above, the native nucleic acid sequences encoding dihydroxyacetone kinase proteins in *Saccharomyces cerevisiae*, DAK1 and DAK2, have been illustrated respectively by SEQ ID NO: 69 and SEQ ID NO: 70.

[0452] Preferably the recombinant yeast cell does comprise a genetic modification that increases the specific activity of any dihydroxyacetone kinase in the cell. For example, the recombinant yeast cell may comprise one or more native and/or heterologous nucleic acid sequence encoding one or more native and/or heterologous dihydroxyacetone kinase protein(s), such as DAK1 and/or DAK2, that is/are overexpressed. A native dihydroxyacetone kinase, such as DAK1 and/or DAK2, may for example be overexpressed via one or more genetic modifications resulting in more copies of the gene encoding for the dihydroxy acetone kinase than present in the non-genetically modified cell, and/or a non-native promoter may be applied.

[0453] Preferably the recombinant yeast cell is a recombinant yeast cell, wherein the expression of the nucleic acid sequence encoding the protein having dihydroxyacetone kinase activity is under control of a promoter. The promoter can for example be a promoter that is native to another gene in the host cell.

[0454] For overexpression of the nucleotide sequence encoding the dihydroxyacetone kinase, the nucleotide sequence (to be overexpressed) can be placed in an expression construct wherein it is operably linked to suitable expression regulatory regions/sequences to ensure overexpression of the dihydroxyacetone kinase enzyme upon transformation of the expression construct into the host cell of the invention (see above). Suitable promoters for (over)expression of the nucleotide sequence coding for the enzyme having dihydroxyacetone kinase activity include promoters that are preferably insensitive to catabolite (glucose) repression, that are active under anaerobic conditions and/or that preferably do not require xylose or arabinose for induction. Examples of such promoters are given above. A dihydroxyacetone kinase that is overexpressed, is preferably overexpressed by at least a factor 1.1, 1.2, 1.5, 2, 5, 10 or 20 as compared to a strain which is genetically identical except for the genetic modification causing the overexpression. Preferably, the dihydroxyacetone kinase is overexpressed under anaerobic conditions by at least a factor 1.1, 1.2, 1.5, 2, 5, 10 or 20 as compared to a strain which is genetically identical except for the genetic modification causing the overexpression. It is to be understood that these levels of overexpression may apply to the steady state level of the enzyme's activity (specific activity in the cell), the steady state level of the enzyme's protein as well as to the steady state level of the transcript coding for the enzyme in the cell. Overexpression of the nucleotide sequence in the host cell produces a specific dihydroxyacetone kinase activity of at least 0.002, 0.005, 0.01, 0.02 or 0.05 U min⁻¹ (mg protein)⁻¹, determined in cell extracts of the transformed host cells at 30° C. as described e.g. in the Examples of WO2013/081456.

[0455] A most preferred dihydroxyacetone kinase protein is the dihydroxyacetone kinase protein encoded by the *Dak1* gene from *Saccharomyces cerevisiae*. SEQ ID NO: 64 shows the amino acid sequence of a suitable dihydroxyacetone kinase protein, encoded by the *Dak1* gene from *Saccharomyces cerevisiae*. SEQ ID NO: 69 illustrates the nucleic acid sequence of the *Dak1* gene itself.

[0456] If the recombinant yeast cell comprises one or more overexpressed nucleic acid sequences encoding for a dihydroxyacetone kinase, the recombinant yeast cell therefore most preferably comprises one or more overexpressed nucleotide sequence encoding a dihydroxyacetone kinase derived from *Saccharomyces cerevisiae*, as exemplified by the nucleic acid sequence shown in SEQ ID NO: 69.

[0457] Preferably the protein having dihydroxy acetone kinase activity thus comprises or consists of: [0458] an amino acid sequence of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68; or [0459] a functional homologue of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%,

at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68; or [0460] a functional homologue of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68.

The protein having an amino acid sequence of SEQ ID NO: 64 and functional homologues thereof are most preferred.

[0461] Preferable the nucleic acid sequence encoding the protein having dihydroxy acetone kinase activity comprises or consists of: [0462] a nucleic acid sequence of SEQ ID NO: 69 or SEQ ID NO: 70; or [0463] a functional homologue of SEQ ID NO: 69 or SEQ ID NO: 70, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 69 or SEQ ID NO: 70; or [0464] a functional homologue of SEQ ID NO: 69 or SEQ ID NO: 70, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 69 or SEQ ID NO: 70; more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 69 or SEQ ID NO: 70.

[0465] The nucleic acid sequence (e.g. the gene) encoding for the dihydroxy acetone kinase protein may suitably be incorporated in the genome of the recombinant yeast cell.

[0466] Examples of suitable dihydroxyacetone kinases are listed in Table 9(a) to 9(d). At the top of each table the DAK's used in the examples and that is BLASTED is mentioned.

TABLE-US-00016 TABLE 9(a) BLAST Query - DAK1 from *Saccharomyces cerevisiae* Accession Description Identity (%) number Dak1p [*Saccharomyces cerevisiae* S288c] 100 NP_013641.1 dihydroxyacetone kinase [*Saccharomyces cerevisiae* 99 EDN64325.1 YJM789] DAK1-like protein [*Saccharomyces kudriavzevii* IFO 95 EJT44075.1 1802] ZYBAOS11-03576g1_1 [*Zygosaccharomyces bailii* CLIB 77 CDF91470.1 213] hypothetical protein [*Kluyveromyces lactis* NRRL Y-1140] 70 XP_451751.1 hypothetical protein [*Candida glabrata* CBS 138] 63 XP_449263.1 Dak2p [*Saccharomyces cerevisiae* S288c] 44 NP_116602.1 DAK1 [*Yarrowia lipolytica*] 41 See examples herein strains with CAS23

TABLE-US-00017 TABLE 9(b) BLAST Query - dhaK from *Klebsiella pneumoniae* Accession Description Identity (%) Accession number dihydroxyacetone kinase subunit DhaK [*Klebsiella* 100 YP_002236493.1 *pneumoniae* 342] dihydroxyacetone kinase subunit K [*Klebsiella* 99 WP_004149886.1 *pneumoniae*] dihydroxyacetone kinase subunit K [*Enterobacter* 96 WP_020077889.1 *aerogenes*] dihydroxyacetone kinase subunit DhaK [*Escherichia coli* 88 YP_002407536.1 IAI39] dihydroxyacetone kinase, DhaK subunit [*Escherichia* 87 WP_001398949.1 *coli*]

TABLE-US-00018 TABLE 9(c) BLAST Query - DAK1 from *Yarrowia lipolytica* Accession Description Identity (%) number YALIOF09273p [*Yarrowia lipolytica*] 100 XP_505199.1 dihydroxyacetone kinase [*Schizosaccharomyces pombe*] 46 AAC83220.1 dihydroxyacetone kinase Dak1 [*Schizosaccharomyces* 45 NP_593241.1 *pombe* 972h-] dihydroxyacetone kinase [*Saccharomyces cerevisiae* 44 EDV12567.1 RM11-1a] Dak2p [*Saccharomyces cerevisiae* JAY291]

44 EEU04233.1 BN860_19306g1_1 [*Zygosaccharomyces bailii* CLIB 213] 44 CDF87998.1 Dak1p [*Saccharomyces cerevisiae* CEN.PK113-7D] 42 EIW08612.1 See examples herein strains with CAS21

TABLE-US-00019 TABLE 9(d) BLAST Query - DAK1 from *Schizosaccharomyces pombe*
Accession Description Identity (%) number dihydroxyacetone kinase Dak1 [*Schizosaccharomyces*
100 NP 593241.1 *pombe* 972h-] putative dihydroxyacetone kinase protein [*Botryotinia* 48
EMR88164.1 *fuckeliana* BcDW1] Dihydroxyacetone kinase 1 [*Fusarium oxysporum* f. sp. 48
ENH64704.1 *cubense* race 1] Dak1p [*Saccharomyces cerevisiae* CEN.PK113-7D] 46 EIW08612.1
Dak2p [*Saccharomyces cerevisiae* JAY291] 44 EEU04233.1 dihydroxyacetone kinase [*Exophiala*
dermatitidis 42 EHY55064.1 NIH/UT8656]

Glycerol Transporter

[0467] The recombinant yeast cell can optionally, i.e. may or may not, comprise a nucleotide sequence encoding a glycerol transporter. Such a glycerol transporter can allow any glycerol that is externally available in the medium (e.g. from the backset in corn mash) or secreted after internal cellular synthesis to be transported into the cell and converted to ethanol.

[0468] If a glycerol transporter is present, the recombinant yeast preferably comprises one or more nucleic acid sequences encoding a heterologous glycerol transporter represented by amino acid sequence SEQ ID NO: 71, SEQ ID NO: 72 or a functional homologue thereof having an amino acid sequence identity of at least 50%, preferably at least 60%, 70%, 75%, 80%, 85%, 90%, 95%, 98%, or 99% with the amino acid sequence of SEQ ID NO: 71 and/or SEQ ID NO: 72.

[0469] In an embodiment the recombinant yeast can further comprise a deletion or disruption of one or more endogenous nucleotide sequences encoding a glycerol exporter (e.g. FPS1).

Nitrate Reductase

[0470] The recombinant yeast cell may also advantageously comprise, respectively functionally express, a nucleic acid sequences encoding an enzyme having NADH-dependent nitrate reductase activity and/or a nucleic acid sequences encoding an enzyme having NADH-dependent nitrite reductase activity. Details for the expression of such an alternative redox sink have been described in non-pre-published US patent application U.S. 63/087,642 filed with the United States Patent Office on 5 Oct. 2020, the contents of which are herewith incorporated by reference.

[0471] Nitrate reductase (NR) catalyzes the reduction of nitrate (NO.sub.3.sup.-) to nitrite (NO.sub.2.sup.-). Nitrite reductase catalyzes the reduction of nitrite to ammonia (NH.sub.3).

Nitrate reductase and/or nitrite reductase can be part of a so-called nitrogen assimilation pathway in certain cells. Cells comprising nitrate reductase activity and/or nitrite reductase activity include certain plant cells and bacterial cells and a few yeast cells. As indicated by Linder, the ability to assimilate inorganic nitrogen sources other than ammonia is thought to be rare among budding yeasts. Among the few fungi that are naturally capable to assimilate nitrate or nitrite are *Blastobotrys adenivorans* (family Trichomonascaceae) *Candida boidinii* (family Pichiaceae), *Cyberlindnera jadinii* (family Phaffomycetaceae), and *Ogataea polymorpha* (family Pichiaceae).

[0472] Preferably the recombinant yeast cell as described herein comprises at least one or more genes encoding a NADH-dependent nitrate reductase.

[0473] By a NADH-dependent nitrate reductase is herein understood a nitrate reductase that is exclusively depended on NADH as a co-factor or that is predominantly dependent on NADH as a cofactor. Preferably the NADH-dependent nitrate reductase has a ratio of catalytic efficiency for NADPH/NADP⁺ as a cofactor (k.sub.cat/K.sub.m).sup.NADP⁺ to NADH/NAD⁺ as cofactor (k.sub.cat/K.sub.m).sup.NAD⁺ i.e. a catalytic efficiency ratio (k.sub.cat/K.sub.m).sup.NADP⁺: (k.sub.cat/K.sub.m).sup.NAD⁺, of more than 1:1, more preferably of equal to or more than 2:1, still more preferably of equal to or more than 5:1, even more preferably of equal to or more than 10:1, yet even more preferably of equal to or more than 20:1, even still more preferably of equal to or more than 100:1, and most preferably equal to or more than 1000:1. There is no upper limit, but for practical reasons the NADH-dependent nitrate reductase may have a catalytic efficiency ratio

(k.sub.cat/K.sub.m).sup.NADP+:(k.sub.cat/K.sub.m).sup.NAD⁺ of equal to or less than 1.000.000.000:1 (i.e. 1.109). Most preferably the NADH-dependent nitrate reductase is exclusively depended on NADH/NAD⁺ as a co-factor. That is, most preferably the NADH-dependent nitrate reductase has an absolute requirement for NADH/NAD⁺ as a cofactor instead of NADPH/NADP⁺ as a cofactor.

[0474] Preferably the NADH-dependent nitrate reductase is a NADH-dependent nitrate reductase with enzyme classification EC 1.7.1.1. (i.e. with EC number EC 1.7.1.1) or enzyme classification EC.1.6.6.1 (i.e. with EC number 1.6.6.1). Suitably the NADH-dependent nitrate reductase, also referred to as NADH-dependent nitrate oxidoreductase, is an enzyme that catalyzes at least the following chemical reaction:

nitrate+NADH+H⁺→nitrite+NAD⁺+H₂O

[0475] Suitable NADH-dependent nitrate reductases may include one or more NADH-dependent nitrate reductases as obtained or derived from *Agrostemma githago*, *Amaranthus hybridus*, *Amaranthus tricolor*, *Ankistrodesmus braunii*, *Arabidopsis thaliana*, *Aspergillus niger*, *Aspergillus nidulans*, *Auxenochlorella pyrenoidosa*, *Bradyrhizobium* sp., *Bradyrhizobium* sp. 750, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Candida boidinii*, *Candida utilis*, *Capsicum frutescens*, *Chenopodium album*, *Cyberlindnera jadinii*, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Capsicum frutescens*, *Chenopodium album*, *Chlamydomonas reinhardtii*, *Chlorella fusca*, *Chlorella* sp. *Chlorella* sp. Berlin, *Chlorella vulgaris*, *Conticribra weissflogii*, *Cucumis sativus*, *Cucurbita maxima*, *Cucurbita pepo*, *Cucurbita* sp., *Dunaliella tertiolecta*, *Emiliania huxleyi*, *Emericella nidulans*, *Fusarium oxysporum*, *Fusarium oxysporum* JCM 11502, *Glyceria maxima*, *Glycine max*, *Gossypium hirsutum*, *Gracilaria chilensis*, *Gracilaria tenuistipitata*, *Helianthus annuus*, *Hordeum vulgare*, *Lactuca sativa*, *Lemna minor*, *Lupinus albus*, *Mycobacterium tuberculosis*, *Nicotiana plumbaginifolia*, *Nicotiana tabacum*, *Ogataea angusta*, *Ogataea polymorpha*, *Oryza sativa*, *Phaeocystis Antarctica*, *Phragmites australis*, *Physcomitrella patens*, *Pisum arvense*, *Polytrichum commune*, *Pyropia yezoensis*, *Raphanus sativus*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus* E1F1, *Ricinus communis*, *Selaginella kraussiana*, *Sinapis alba*, *Skeletonema costatum*, *Skeletonema tropicum*, *Solanum lycopersicum*, *Spinacia oleracea*, *Suaeda maritima*, *Tetraselmis gracilis*, *Thalassia Testudinum*, *Thalassiosira Antarctica*, *Thalassiosira pseudonana*, *Triticum aestivum*, *Triticum turgidum* subsp durum, *Ulva* sp. And/or *Zea mays*; and/or functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence with at least 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrate reductases; and/or functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrate reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrate reductases.

[0476] Preferred NADH-dependent nitrate reductases include the NADH-dependent nitrate reductases as obtained or derived from *Candida boidinii* (a nitrate reductase capable of utilizing both NADH and NADPH as electron donors), *Candida utilis* (a nitrate reductase capable of utilizing both NADH and NADPH as electron donors), *Fusarium oxysporum* (as described by Fujii et al, in their article titled "Denitrification by the Fungus *Fusarium oxysporum* Involves NADH-Nitrate Reductase" published in Biosci. Biotechnol. Biochem., 72 (2), pages 412-420, 2008, incorporated herein by reference), *Spinacia oleracea* and *Zea mays*.

[0477] Preferred NADH-dependent nitrate reductases hence include: NADH-dependent nitrate reductases comprising a polypeptide having an amino acid sequence of SEQ ID NO:89 and/or SEQ

ID NO:90, as described herein; and/or functional homologues of SEQ ID NO:89 and/or SEQ ID NO:90 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of SEQ ID NO:89 and/or SEQ ID NO:90 respectively; and/or functional homologues of SEQ ID NO:89 and/or SEQ ID NO:90 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of SEQ ID NO:89 and/or SEQ ID NO:90 respectively. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:89 and/or SEQ ID NO:90 respectively.

[0478] Preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrate reductase activity. More preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrate reductase activity selected from the group consisting of NADH-dependent nitrate reductases as obtained or derived from *Agrostemma githago*, *Amaranthus hybridus*, *Amaranthus tricolor*, *Ankistrodesmus braunii*, *Arabidopsis thaliana*, *Aspergillus niger*, *Aspergillus nidulans*, *Auxenochlorella pyrenoidosa*, *Bradyrhizobium* sp., *Bradyrhizobium* sp. 750, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Candida boidinii*, *Candida utilis*, *Capsicum frutescens*, *Chenopodium album*, *Cyberlindnera jadinii*, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Capsicum frutescens*, *Chenopodium album*, *Chlamydomonas reinhardtii*, *Chlorella fusca*, *Chlorella* sp. *Chlorella* sp. Berlin, *Chlorella vulgaris*, *Conticribra weissflogii*, *Cucumis sativus*, *Cucurbita maxima*, *Cucurbita pepo*, *Cucurbita* sp., *Dunaliella tertiolecta*, *Emiliana huxleyi*, *Emericella nidulans*, *Fusarium oxysporum*, *Fusarium oxysporum* JCM 11502, *Glyceria maxima*, *Glycine max*, *Gossypium hirsutum*, *Gracilaria chilensis*, *Gracilaria tenuistipitata*, *Helianthus annuus*, *Hordeum vulgare*, *Lactuca sativa*, *Lemna minor*, *Lupinus albus*, *Mycobacterium tuberculosis*, *Nicotiana plumbaginifolia*, *Nicotiana tabacum*, *Ogataea angusta*, *Ogataea polymorpha*, *Oryza sativa*, *Phaeocystis Antarctica*, *Phragmites australis*, *Physcomitrella patens*, *Pisum arvense*, *Polytrichum commune*, *Pyropia yezoensis*, *Raphanus sativus*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus* E1F1, *Ricinus communis*, *Selaginella kraussiana*, *Sinapis alba*, *Skeletonema costatum*, *Skeletonema tropicum*, *Solanum lycopersicum*, *Spinacia oleracea*, *Suaeda maritima*, *Tetraselmis gracilis*, *Thalassia Testudinum*, *Thalassiosira Antarctica*, *Thalassiosira pseudonana*, *Triticum aestivum*, *Triticum turgidum* subsp durum, *Ulva* sp. and *Zea mays*, and functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence with at least 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrate reductases; and functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrate reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrate reductases.

[0479] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of any of SEQ ID NO:89 and/or SEQ ID NO:90 or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:89 and/or SEQ ID NO:90. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:89 and/or SEQ ID NO:90 respectively.

[0480] The recombinant yeast cell may combine one or more genes encoding the above NADH-dependent nitrate reductase with one or more genes encoding an NADPH-dependent nitrite reductase. Preferably, however, the recombinant yeast cell combines one or more genes encoding the above NADH-dependent nitrate reductase with one or more genes encoding a NADH-

dependent nitrite reductase.

[0481] Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:89, are listed in Table 10 below.

TABLE-US-00020 TABLE 10 Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 89, are listed in Table 10 below. UNIPROT DATABASE ACCESSION IDENTITY TO NUMBER DESCRIPTION ORGANISM SEQ ID 89 [%] K3YG56 Cluster: Nitrate PACMAD clade 94.06 reductase P16081 Cluster: Nitrate cellular organisms 90.54 reductase [NADH] 1 UPI0004DEBCF9 Cluster: nitrate *Zea mays* 88.57 reductase [NADH]-like A0A6G1CIJ1 Cluster: *Oryza meyeriana* 85.7 Uncharacterized var. *granulata* protein P27967 Cluster: Nitrate Pooideae 85.37 reductase [NADH] A0A3B6T8X1 Cluster: Nitrate Triticinae 86.11 reductase A0A4S8IHP6 Cluster: Nitrate Musaceae 78.55 reductase A0A2T7D896 Cluster: Nitrate Panicum 78.22 reductase A0A199UYK8 Cluster: Nitrate Ananas comosus 77.45 reductase A0A4S8J849 Cluster: Nitrate Mesangiospermae 77.67 reductase A0A2H3Y7X8 Cluster: Nitrate Arecaceae 76.68 reductase A0A4S8JVI1 Cluster: Nitrate Musa balbisiana 75.58 reductase A0A200QET5 Cluster: Nitrate Macleaya cordata 72.39 reductase F6H0M5 Cluster: Nitrate Vitis 72.61 reductase A0A4Y7J3A8 Cluster: Nitrate Papaver 71.95 reductase *somniferum* A0A218XU44 Cluster: Nitrate Punica granatum 71.73 reductase A0A2C9PIN3 Cluster: Nitrate Phellodendron 72.06 reductase *amurense* UPI000B8CEBA2 Cluster: nitrate *Carica papaya* 71.51 reductase [NADH]-like A0A2R6RP67 Cluster: Nitrate *Actinidia chinensis* 70.63 reductase var. *chinensis* A0A5J5BUB1 Cluster: Nitrate *Nyssa sinensis* 71.29 reductase V4TCH9 Cluster: Nitrate Citrus 71.51 reductase P36859 Cluster: Nitrate *Petunia hybrida* 71.62 reductase [NADH] A0A0E0GDD5 Cluster: Nitrate Oryzinae 74.76 reductase UPI001263D0A7 Cluster: nitrate *Pistacia vera* 71.4 reductase [NAD(P)H]- like A0A061FFV7 Cluster: Nitrate Byttnerioideae 70.85 reductase K3YPT2 Cluster: Nitrate Panicoideae 75.21 reductase A0A1U7ZDA9 Cluster: Nitrate *Nelumbo nucifera* 72.17 reductase A0A0POVPX4 Cluster: Nitrate *Oryza* 74.17 reductase A0A6A3CP56 Cluster: Nitrate *Hibiscus syriacus* 69.97 reductase A0A067KLT8 Cluster: Nitrate *Jatropha curcas* 70.08 reductase A0A214HOD3 Cluster: Nitrate *Juglans regia* 69.86 reductase A0A5J9UYH6 Cluster: PACMAD clade 74.26 Uncharacterized protein (Fragment) A0A5N6RHU6 Cluster: Nitrate fabids 70.63 reductase A0A1U8KZC0 Cluster: Nitrate Malvaceae 70.74 reductase UPI000C0393D8 Cluster: nitrate *Durio zibethinus* 73.33 reductase [NADH]-like B9H7A1 Cluster: Nitrate Saliceae 70.41 reductase P17569 Cluster: Nitrate Cucurbitaceae 70.74 reductase [NADH] UPI000581598E Cluster: nitrate *Sesamum indicum* 70.41 reductase [NADH] 2- like UPI000F7C3BDF Cluster: nitrate *Abrus precatorius* 74.12 reductase [NADH] 2 isoform X1 P08509 Cluster: Nitrate Solanaceae 70.08 reductase [NADH] 2 P39867 Cluster: Nitrate Brassiceae 71.65 reductase [NADH], clone PBNBR1405 UPI000C1D548F Cluster: nitrate *Olea europaea* var. 71.22 reductase [NADH] 2- *sylvestris* like P39866 Cluster: Nitrate Phaseoleae 73.88 reductase [NADH] 2 P27968 Cluster: Nitrate Triticeae 72.82 reductase [NAD(P)H] A8IJ84 Cluster: Nitrate *Beta vulgaris* 71.14 reductase A0A2U1KIT7 Cluster: Nitrate *Artemisia annua* 69.64 reductase A0A165XNP4 Cluster: Nitrate *Daucus carota* 69.31 reductase subsp. *sativus* B9RJB1 Cluster: Nitrate Mesangiospermae 74 reductase P43101 Cluster: Nitrate Asteraceae 69.86 reductase [NADH] 11LW34 Cluster: Nitrate *Glycine* subgen. 74.49 reductase *Soja* UPI000980E182 Cluster: nitrate *Ananas comosus* 75.54 reductase [NADH]-like A0A4P1R4X1 Cluster: Nitrate *Lupinus* 71.22 reductase *angustifolius* Q8H1T7 Cluster: Nitrate *Tilia platyphyllos* 69.42 reductase 11IER4 Cluster: Nitrate *Brachypodium* 73.47 reductase *distachyon* A0A1U7Z2Y3 Cluster: Nitrate *Nelumbo nucifera* 72.45 reductase A0A068VGB5 Cluster: Nitrate *Coffea* 68.87 reductase A0A2T7F896 Cluster: Nitrate *Panicum* 73.27 reductase A0A2Z7C1Z2 Cluster: Nitrate *Dorcoceras* 69.31 reductase *hygrometricum*

UPI00053C7726 Cluster: nitrate Tarenaya 69.64 reductase [NADH] 2 hassleriana A0A445BT68 Cluster: Nitrate Arachis 72.55 reductase (NADH) P39869 Cluster: Nitrate Lotus japonicus 73.76 reductase [NADH] UPI0010FB6302 Cluster: nitrate Cajanus cajan 73.64 reductase [NADH] 2 Q93XS1 Cluster: Nitrate Prunus 73.76 reductase A0A022Q6Q6 Cluster: Nitrate Erythranthe guttata 69.2 reductase A0A1U8NV83 Cluster: Nitrate Gossypium 73.04 reductase P27783 Cluster: Nitrate fabids 69.53 reductase [NAD(P)H] A0A2T8KWM5 Cluster: Nitrate Panicum hallii 72.93 reductase A0A214GBD5 Cluster: Nitrate Juglans regia 69.42 reductase A0A5N6PSR2 Cluster: Nitrate Mikania micrantha 69.2 reductase A0A2P6PI42 Cluster: Nitrate Mesangiospermae 73.88 reductase E7ELN7 Cluster: Nitrate IRL clade 70.41 reductase A0A4U5QSR8 Cluster: Nitrate Populus 73.64 reductase A0A2K2BFU1 Cluster: Nitrate Saliceae 74.12 reductase A0A1Z5RNB1 Cluster: Nitrate Sorghum bicolor 76.67 reductase A0A0M4P4G3 Cluster: Nitrate rosids 73.28 reductase M4CVT6 Cluster: Nitrate Brassica 70.47 reductase campestris A0A251SAE9 Cluster: Nitrate Helianthus annuus 72.55 reductase A0A2G5DUU4 Cluster: Nitrate Aquilegia coerulea 71.24 reductase P23312 Cluster: Nitrate Chenopodioideae 72.79 reductase [NADH] P11832 Cluster: Nitrate Brassicaceae 71.22 reductase [NADH] 1 P11035 Cluster: Nitrate rosids 68.76 reductase [NADH] 2 W1NJDO Cluster: Nitrate Magnoliopsida 69.31 reductase UPI000C048389 Cluster: nitrate Durio zibethinus 71.36 reductase [NADH]-like A0A6A3C902 Cluster: Nitrate Hibiscus syriacus 73.16 reductase P39870 Cluster: Inducible Glycine subgen. 73.52 nitrate reductase Soja [NADH] 2 UPI000F7C9CC8 Cluster: inducible Abrus precatorius 72.79 nitrate reductase [NADH] 2 A0A6A5MI86 Cluster: Nitrate Lupinus 70.31 reductase

Nitrite Reductase

[0482] As indicated above, nitrite reductase catalyzes the reduction of nitrite to ammonia (NH.sub.3).

[0483] Preferably the recombinant yeast cell as described herein comprises at least one or more genes encoding a NADH-dependent nitrite reductase.

[0484] By a NADH-dependent nitrite reductase is herein understood a nitrite reductase that is exclusively depended on NADH as a co-factor or that is predominantly dependent on NADH as a cofactor. Preferably the NADH-dependent nitrite reductase has a ratio of catalytic efficiency for NADPH/NADP⁺ as a cofactor (k.sub.cat/K.sub.m).sup.NADP⁺ to NADH/NAD⁺ as cofactor (k.sub.cat/K.sub.m).sup.NAD⁺ i.e. a catalytic efficiency ratio (k.sub.cat/K.sub.m).sup.NADP⁺: (k.sub.cat/K.sub.m).sup.NAD⁺, of more than 1:1, more preferably of equal to or more than 2:1, still more preferably of equal to or more than 5:1, even more preferably of equal to or more than 10:1, yet even more preferably of equal to or more than 20:1, even still more preferably of equal to or more than 100:1, and most preferably equal to or more than 1000:1. There is no upper limit, but for practical reasons the NADH-dependent nitrite reductase may have a catalytic efficiency ratio (k.sub.cat/K.sub.m).sup.NADP⁺:(k.sub.cat/K.sub.m).sup.NAD⁺ of equal to or less than 1.000.000.000:1 (i.e. 1.10⁹). Most preferably the NADH-dependent nitrite reductase is exclusively depended on NADH/NAD⁺ as a co-factor. That is, most preferably the NADH-dependent nitrite reductase has an absolute requirement for NADH/NAD⁺ as a cofactor instead of NADPH/NADP⁺ as a cofactor.

[0485] Preferably the NADH-dependent nitrite reductase is a NADH-dependent nitrite reductase with enzyme classification EC 1.7.1.15 (i.e. with EC number EC 1.7.1.15). Suitably the NADH-dependent nitrite reductase, also referred to as NADH-dependent nitrite oxidoreductase, is an enzyme that catalyzes at least the following chemical reaction:

nitrite+3NADH+5H.sub.+.fwdarw.ammonia+3NAD.sup.++2H.sub.2O

The person skilled in the art will understand that the ammonia may also be present and/or referred to as so-called ammonium hydroxide NH.sub.4OH

[0486] Suitable NADH-dependent nitrite reductases may include one or more NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*),

Arcobacter ellisii, *Arcobacter pacificus* *Bacillus subtilis* JH642, *Cupriavidus taiwanensis* *Escherichia coli*, *Ralstonia taiwanensis*, *Ralstonia syzygii*, *Ralstonia solanacearum*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus*, *Paraburkholderia ribeironis*; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrite reductases; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrite reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrite reductases.

[0487] *Escherichia coli* utilizes several distinct enzymes in its nitrite assimilation pathway. The *nirD* gene encodes a NADH-dependent nitrite reductase (NADH) small subunit, whilst the *nirB* gene encodes a NADH-dependent nitrite reductase (NADH) large subunit.

[0488] Preferred NADH-dependent nitrite reductases include the NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*), a nitrite reductase capable of utilizing both NADH and NADPH as electron donors, and/or *Escherichia coli*. At high nitrate and/or nitrite concentrations, the nitrite reductase encoded by the *nirB* gene of *Escherichia coli* is especially preferred.

[0489] Preferred NADH-dependent nitrite reductases hence include: NADH-dependent nitrite reductases comprising a polypeptide having an amino acid sequence of SEQ ID NO:91 (*E. coli* nitrite reductase small subunit encoded by *nirD*) and/or SEQ ID NO:92 (*E. coli* nitrite reductase large subunit encoded by *nirB*) and/or SEQ ID NO:93 (*Emericella nidulans* nitrate reductase encoded by *niiA*), as described herein; and/or functional homologues of SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 respectively; and/or functional homologues of SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 respectively. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 respectively.

[0490] Preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrite reductase activity. More preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrite reductase activity selected from the group consisting of NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*), *Arcobacter ellisii*, *Arcobacter pacificus* *Bacillus subtilis*, *Bacillus subtilis* JH642, *Cupriavidus taiwanensis* *Escherichia coli*, *Ralstonia taiwanensis*, *Ralstonia syzygii*, *Ralstonia solanacearum*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus*, *Paraburkholderia ribeironis*; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrite reductases; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrite reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5

amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrite reductases.

[0491] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of any of SEQ ID NO:91 (*E. coli* nitrate reductase small subunit encoded by nirD) and/or SEQ ID NO:92 (*E. coli* nitrate reductase large subunit encoded by nirB) and/or SEQ ID NO:93 (*Emericella nidulans* nitrate reductase encoded by niiA), or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 respectively.

[0492] The recombinant yeast cell may combine one or more genes encoding one or more of the above NADH-dependent nitrite reductases with one or more genes encoding an NADPH-dependent nitrate reductase. Preferably, however, the recombinant yeast cell combines one or more genes encoding one or more of the above NADH-dependent nitrite reductases with one or more genes encoding a NADH-dependent nitrate reductase.

[0493] Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org) as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:91 (small subunit encoded by nirD), are listed in Table 11 below.

[0494] Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org) as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:92 (large subunit encoded by nirB), are listed in Table 12 below.

TABLE-US-00021 TABLE 11 Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org) as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 91 (small subunit encoded by nirD). UNIPROT DATABASE ACCESSION IDENTITY TO NUMBER DESCRIPTION ORGANISM SEQ ID 91 [%] E2QFM2 Cluster: Nitrite *Gammaproteobacteria* 99.07 reductase A0A078LHD3 Cluster: Nitrite *Enterobacteriaceae* 94.44 reductase small subunit POA230 Cluster: Nitrite Bacteria 93.52 reductase (NADH) small subunit A0A170JME2 Cluster: Nitrite *Enterobacteriaceae* 91.67 reductase [NAD(P)H] small subunit A0A0F6TX22 Cluster: Nitrite *Gammaproteobacteria* 91.67 reductase A0A6DOEPI7 Cluster: Nitrite *Enterobacteriaceae* 87.04 reductase small subunit NirD A0A198H1N5 Cluster: Nitrite Bacteria 86.11 reductase [NAD(P)H] small subunit A0A447MQ21 Cluster: Nitrite *Enterobacteriaceae* 84.26 reductase [NAD(P)H] small subunit UPI00077B73C7 Cluster: nitrite *Enterobacter timonensis* 83.33 reductase small subunit NirD A0A094XNF5 Cluster: Nitrite *Enterobacterales* 82.41 reductase (NADH) small subunit UPI00105F841D Cluster: nitrite *Enterobacter* sp. OV724 81.48 reductase small subunit NirD A0A1X0XB32 Cluster: Nitrite *Enterobacterales* 81.48 reductase small subunit A0A1A9FFN3 Cluster: Nitrite *Lelliottia amnigena* 81.48 reductase small subunit UPI0002D5E0A7 Cluster: nitrite *Metakosakonia* 80.56 reductase small *massiliensis* subunit NirD UPI0004E24438 Cluster: nitrite *Siccibacter colletis* 79.63 reductase small subunit NirD A0A514EKN7 Cluster: Nitrite *Enterobacteriaceae* 77.78 reductase [NAD(P)H] small subunit A0A2P8VHS5 Cluster: Nitrite *Enterobacteriaceae* 80.56 reductase small subunit NirD A0A4V3BP91 Cluster: Nitrite cellular organisms 80.56 reductase (NADH) small subunit A0A3R9F800 Cluster: Nitrite *Enterobacteriaceae* 79.63 reductase small subunit NirD R9VMQ4 Cluster: Nitrite *Enterobacteriaceae* 78.7 reductase small subunit A0A285KH74 Cluster: Nitrite *Enterobacter* sp. 80.37 reductase (NADH) CC120223-11 small subunit A0A658U6S1 Cluster: Assimilatory *Enterobacter* 77.78 nitrite reductase (NAD(P)H) small subunit A0A085A429 Cluster: Nitrite *Enterobacteriaceae* 78.7 reductase [NAD(P)H] small subunit A0A1VOL823

Cluster: Nitrite *Kosakonia* 76.85 reductase small subunit UPI001431C56A Cluster: nitrite *Cedecea*
colo 75 reductase small subunit NirD A0A085G1A1 Cluster: Nitrite *Enterobacteriaceae* 75
 reductase [NAD(P)H] small subunit A0A089PL75 Cluster: Nitrite *Pluralibacter gergoviae* 76.85
 reductase A0A3C0H311 Cluster: Nitrite *Enterobacteriaceae* 77.78 reductase small subunit
 A0A3N2DXC8 Cluster: Assimilatory unclassified 75.93 nitrite reductase *Enterobacter* (NAD(P)H)
 small subunit AOAOL0ATR1 Cluster: Nitrite *Enterobacteriaceae* 78.5 reductase K8AGI3 Cluster:
 Nitrite *Cronobacter condimenti* 75 reductase 1330 [NAD(P)H] small subunit A0A116EAUO
 Cluster: Nitrite *Enterobacter* sp. kpr-6 75 reductase (NADH) small subunit A0A0D5X0Q1 Cluster:
 Nitrite *Enterobacteriaceae* 75 reductase C9Y259 Cluster: Nitrite *Enterobacteriaceae* 74.07
 reductase [NAD(P)H] small subunit A0A4R0GWC0 Cluster: Nitrite *Kosakonia* 75 reductase small
quasisacchari subunit NirD UPI00057BF926 Cluster: nitrite *Enterobacter* sp. Bisph1 73.15
 reductase small subunit NirD A0A495AHMO Cluster: Nitrite *Enterobacter* sp. 72.22 reductase
 small R1(2018) subunit NirD A0A447QKU3 Cluster: Nitrite *Yersiniaceae* 75.47 reductase
 [NAD(P)H] small subunit 12B4C9 Cluster: Nitrite *Shimwellia blattae* 73.83 reductase (strain
 ATCC 29907 / (NAD(P)H) DSM 4481 / JCM 1650 / NBRC 105725 / CDC 9005-74)
 UPI0012B7CAE2 Cluster: nitrite *Erwinia* sp. CPCC 72.22 reductase small 100877 subunit NirD
 UPI00073DB139 Cluster: nitrite [*Erwinia*] *teleogrylli* 73.83 reductase small subunit NirD
 UPI000237D150 Cluster: nitrite *Enterobacter mori* 80.56 reductase small subunit NirD
 A0A4P8YE33 Cluster: Nitrite *Enterobacteriaceae* 72.22 reductase small subunit NirD
 UPI0005DC1E97 Cluster: nitrite *Yersinia rohdei* 71.03 reductase small subunit NirD
 AOAOU1HQI4 Cluster: Nitrite *Yersinia* 71.03 reductase small subunit A0A0KOHM15 Cluster:
 Nitrite *Enterobacterales* 72.9 reductase [NAD(P)H] small subunit H8NP95 Cluster: Nitrite
Yersiniaceae 69.16 reductase small subunit A0A2S4QQ85 Cluster: Nitrite *Enterobacteriaceae*
 69.44 reductase small subunit A0A0A7RXK3 Cluster: Assimilatory *Frischella perrara* 71.3 nitrite
 reductase (NAD(P)H) small subunit D4DYL5 Cluster: Nitrite *Serratia odorifera* DSM 73.58
 reductase 4582 [NAD(P)H], small subunit A0A370QTS9 Cluster: Assimilatory *Enterobacillus*
tribolii 71.7 nitrite reductase (NAD(P)H) small subunit A0A1B9JUBO Cluster: Nitrite *Gilliamella*
apicola 69.16 reductase small subunit A0A085U6Z8 Cluster: Nitrite *Yersinia ruckeri* 72.22
 reductase [NAD(P)H] small subunit UPI00156B8D43 Cluster: unknown 70.75 UPI00156B8D43
 related cluster A0AOT9KRE5 Cluster: Nitrite *Yersinia* 70.37 reductase small subunit A0A066TFS8
 Cluster: Nitrite *Gilliamella* 69.16 reductase [NAD(P)H] small subunit A0A1B9M9X4 Cluster:
 Nitrite *Gilliamella* 68.22 reductase small subunit A0A0E1NM95 Cluster: Nitrite *Yersiniaceae*
 68.52 reductase (NAD(P)H) A0A080KOG0 Cluster: Ferredoxin *Snodgrassella alvi* 69.16 subunits
 of nitrite SCGC AB-598-002 reductase and ring- hydroxylating dioxygenase A0A0Q4N870
 Cluster: Nitrite *Enterobacterales* 67.29 reductase small subunit UPI0004A391D4 Cluster: nitrite
Tatumella saanichensis 69.81 reductase small subunit NirD A0A1B9ZVE6 Cluster: Nitrite
Gilliamella apicola 67.29 reductase small subunit A0AOT9JI92 Cluster: Nitrite *Yersinia* 68.52
 reductase small subunit A0A2N5EJM2 Cluster: Nitrite *Chimaeribacter* 67.59 reductase small
 subunit NirD A0A2V4E983 Cluster: Nitrite *Gilliamella* 65.74 reductase small subunit
 A0A419N5U0 Cluster: Nitrite *Yersiniaceae* 67.29 reductase small subunit NirD A0A0F7H970
 Cluster: Nitrite *Serratia* 70.75 reductase A0A318NZT5 Cluster: Nitrite *Yersiniaceae* 69.81
 reductase small subunit NirD A0A250B1W9 Cluster: Nitrite *Gibbsiella quercinecans* 69.81
 reductase small subunit A0A6G91849 Cluster: Nitrite *Orbus* sp. IPMB12 65.74 reductase small
 subunit NirD A0A419M923 Cluster: Nitrite *Rahnella inusitata* 67.92 reductase small subunit NirD
 A0A389M9P6 Cluster: Nitrite *Enterobacterales* 67.29 reductase (NADH) small subunit
 A0AON8GUB4 Cluster: Nitrite *Vibrionales* 67.96 reductase (NADH) small subunit A0A443IEH9
 Cluster: Nitrite *Pantoea* sp. LMG 27579 68.87 reductase A0A1X1MLP8 Cluster: Nitrite *Vibrio* sp.
 qd031 69.9 reductase A0A542BGK2 Cluster: Assimilatory *Serratia fonticola* 68.87 nitrite
 reductase (NAD(P)H) small subunit A0A2U3B6U3 Cluster: Nitrite *Vibrio* sp. E4404 67.92
 reductase small subunit A0A240CDV1 Cluster: Nitrite *Gammaproteobacteria* 67.92 reductase

[NAD(P)H] small subunit A0A2S918J3 Cluster: Nitrite Bacteria 67.29 reductase small subunit A0A495RE76 Cluster: Assimilatory *Orbus hercynius* 67.59 nitrite reductase (NAD(P)H) small subunit A0A178KL05 Cluster: Nitrite *Photobacterium* 66.04 reductase small subunit A0A198GJ99 Cluster: Nitrite *Cosenzaea myxofaciens* 63.55 reductase ATCC 19692 [NAD(P)H] small subunit A0A2NOWOU3 Cluster: Nitrite *Enterobacterales* 66.36 reductase small subunit A0A1B9JKC9 Cluster: Nitrite *Gilliamella apicola* 66.67 reductase small subunit UPI0011BEE383 Cluster: nitrite *Pantoea* sp. CCBC3-3-1 65.09 reductase small subunit NirD A0A085G554 Cluster: Nitrite *Ewingella americana* 67.29 reductase [NAD(P)H] small subunit A0A1B1NU77 Cluster: Nitrite *Vibrio* 67.96 reductase (NADH) A0A2N7G6C9 Cluster: Nitrite *Vibrio* sp. 67.96 reductase small 10N.261.55.A7 subunit A0A349FFE1 Cluster: Nitrite *Vibrio* sp. 69.52 reductase small subunit A0A1B9JS56 Cluster: Nitrite *Gilliamella apicola* 64.49 reductase small subunit A0A366XH21 Cluster: Nitrite *Vibrionales* 66.67 reductase small subunit UPI00062A46AC Cluster: nitrite *Tatumella morbirosei* 67.59 reductase small subunit NirD UPI00046A4834 Cluster: nitrite *Pantoea* sp. IMH 68.87 reductase small subunit NirD

TABLE-US-00022 TABLE 12 Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 92 (large subunit encoded by nirB). UNIPROT DATABASE ACCESSION IDENTITY TO NUMBER DESCRIPTION ORGANISM SEQ ID 92 [%]

W9B2S1	Cluster: NirB protein Bacteria 93.27	A0A486K457	Cluster: Nitrite Enterobacterales 93.15 reductase [NAD(P)H]	A0A1B7KIV8	Cluster: Nitrite Enterobacterales 85.41 reductase [NAD(P)H] large subunit	A0A087L3N6	Cluster: Nitrite Serratia 84.6 reductase	A0A654DEI2	Cluster: Nitrite Enterobacterales 84.36 reductase, large subunit, nucleotide- and Fe/S-cluster binding	UPI00156BAB07	Cluster: unknown 81.3	UPI00156BAB07 related cluster	A0A2I0FWV2	Cluster: Nitrite Enterobacterales 81.52 reductase (NAD(P)H) bacterium	CwR94 I2B4D0	Cluster: Nitrite Shimwellia blattae 85.24 reductase (NAD(P)H) (strain ATCC 29907 / DSM 4481 / JCM 1650 / NBRC 105725 / CDC 9005-74)	A0A370QTZ7	Cluster: Assimilatory Enterobacillus tribolii 83.39 nitrite reductase (NAD(P)H) large subunit	UPI000DEFBD5D	Cluster: nitrite Edaphovirga creamea 83.53 reductase large subunit	A0A380ASL5	Cluster: Nitric oxide cellular organisms 84.36 reductase FIRD-NAD(+) reductase	A0A443IEI0	Cluster: Nitrite unclassified Pantoea 81.42 reductase	A0A221T4L0	Cluster: Nitrite Enterobacterales 80.45 reductase [NAD(P)H] large subunit	A0A2N5EJN0	Cluster: Nitrite Enterobacterales 82.68 reductase (NAD(P)H)	A0A377N6S4	Cluster: Benzene 1,2- Enterobacterales 82.82 dioxygenase system ferredoxin -- NAD(+) reductase subunit	A0A4P8SFP8	Cluster: Nitrite Nissabacter sp. 82.11 reductase (NAD(P)H)	SGAir0207 A0A0L7TEE8	Cluster: Nitrite Erwinia iniecta 79.62 reductase	A0A1W6B3D3	Cluster: Nitrite Erwinaceae 79.88 reductase large subunit	A0A506V9E3	Cluster: Nitrite Mixta 79.41 reductase large subunit	UPI000468AD1D	Cluster: nitrite Erwinaceae 81.02 reductase large subunit	A0A0N0Z921	Cluster: Nitrite Moellerella 79.34 reductase [NAD(P)H] wisconsensis large subunit	B4EXR2	Cluster: Nitrite Enterobacterales 78.34 reductase [NAD(P)H] large subunit	A0A198GKA3	Cluster: Nitrite Cosenzaea 78.11 reductase [NAD(P)H] myxofaciens ATCC large subunit	19692 A0A0F3LTP0	Cluster: Nitrite Enterobacterales 78.46 reductase	E1SGS9	Cluster: Nitrite Erwinaceae 77.78 reductase (NAD(P)H) subunit	A0A1X1EZA9	Cluster: Nitrite Pantoea cypripedii 78.25 reductase large subunit	A0A3A9I4K8	Cluster: Nitrite Erwinaceae 77.9 reductase large subunit	A0A2S9I8J4	Cluster: Nitrite Erwinaceae 79.48 reductase (NAD(P)H)	A0A0R4FQI6	Cluster: Nitrite Xenorhabdus 76.42 reductase, large nematophila subunit, nucleotide- binding	UPI001265FD68	Cluster: nitrite Erwinia endophytica 77.49 reductase large subunit	UPI00146D663B	Cluster: nitrite Rahnella sp. SAP-1 78.32 reductase large subunit	A0A3S4CWL7	Cluster: Rubredoxin- Enterobacterales 78.65 NAD(+) reductase	A0A1Y2S8S4	Cluster: NADH oxidase Xenorhabdus 76.78 vietnamensis	A0A0A3YW59	Cluster: Nitrite Erwinia typographi 79.31 reductase	A0A506PS78	Cluster: Nitrite Pantoea 76.12 reductase large subunit	D8MY05	Cluster:
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Nitrite Erwinia 79.12 reductase [NAD(P)H] large subunit E6WGY1 Cluster: Nitrite unclassified Pantoea 75.62 reductase (NAD(P)H), large subunit A0A3R8ZPT1 Cluster: Nitrite unclassified Erwinia 75.3 reductase large subunit A0A5M7NEQ9 Cluster: Nitrite Enterobacterales 75.3 reductase large subunit A0A2V2BEC9 Cluster: Assimilatory Pantoea 76.36 nitrite reductase (NAD(P)H) large subunit A0A1X0W567 Cluster: Nitrite Rouxiella 77.88 reductase large subunit A0A6G9I851 Cluster: Nitrite Orbus sp. IPMB12 77.66 reductase large subunit UPI0011BF5221 Cluster: nitrite Pantoea sp. CCBC3- 76.66 reductase large subunit 3-1 A0A556RIT3 Cluster: Nitrite Gilliamella 74.91 reductase large subunit A0A1R4J5I6 Cluster: Nitrite Vibrio 72.82 reductase [NAD(P)H] large subunit A0A1B9MVL8 Cluster: Nitrite Gilliamella 75.03 reductase large subunit UPI0013596BEF Cluster: nitrite unclassified 72.59 reductase large subunit Psychromonas A0A1B9L8J9 Cluster: Nitrite Gilliamella apicola 74.79 reductase large subunit A0A095TED5 Cluster: Nitrite Tatumella 74.67 reductase A0A2S7VXZ8 Cluster: Nitrite Photobacterium 73.6 reductase large subunit A0A495REZ8 Cluster: Assimilatory Orbus hercynius 75.15 nitrite reductase (NAD(P)H) large subunit A0A2T3QB92 Cluster: Nitric oxide Photobacterium 75.18 reductase FIRd-NAD(+) damsela reductase A0A1X0WF55 Cluster: Nitrite Gammaproteobacteria 76.51 reductase large subunit A0A2I8TVB9 Cluster: Nitrite Vibrio 75.33 reductase large subunit A0A2T3NPM2 Cluster: Nitrite Photobacterium 74.85 reductase large subunit sanctipauli R9PFS3 Cluster: Nitrite Agarivorans 73.18 reductase A0A0B7J736 Cluster: Nitrite Photobacterium 73.6 reductase, large subunit, NAD(P)H- binding U4KF90 Cluster: Nitrite Vibrio nigrapulchritudo 73.78 reductase [NAD(P)H] large subunit A0A1Y6IN13 Cluster: Nitrite Vibrio mangrovi 73.54 reductase [NAD(P)H] A0A1L9L1Z8 Cluster: Nitrite Vibrionaceae 73.78 reductase [NAD(P)H] A0A0A7RZE6 Cluster: Assimilatory Frischella perrara 73.24 nitrite reductase (NAD(P)H) large subunit A0A1Q9GSK0 Cluster: Nitrite Photobacterium 74.97 reductase large subunit UPI000200DAF9 Cluster: nitrite Vibrio furnissii 74.97 reductase large subunit A0A378PW16 Cluster: Nitric oxide Vibrionaceae 72.82 reductase FIRd-NAD(+) reductase A0A1B9JKB8 Cluster: Nitrite Gilliamella apicola 72.65 reductase large subunit D4ZBU5 Cluster: Nitrite Shewanella violacea 72.4 reductase [NAD(P)H], (strain JCM 10179 / large subunit CIP 106290 / LMG 19151 / DSS12) UPI00117F52C3 Cluster: nitrite Vibrio furnissii 74.97 reductase large subunit UPI0001B93571 Cluster: nitrite Vibrio furnissii 74.97 reductase large subunit A0A1B1NUU8 Cluster: Nitrite Vibrionaceae 72.94 reductase (NADH) A0A178KL92 Cluster: Nitrite Photobacterium 73.39 reductase large subunit A0A1T4RZM3 Cluster: Nitrite Vibrio cincinnatiensis 74.37 reductase (NADH) large subunit UPI0012ADB8F8 Cluster: nitrite Vibrio furnissii 74.97 reductase large subunit A0A418YIX2 Cluster: Nitrite Motilimonas sp. 73 reductase large subunit PLHSC7-2 Q6LS87 Cluster: Putative nitrite Photobacterium 73.67 reductase (NAD(P)H), large subunit A0A4R3I7X5 Cluster: Nitrite Reinekea 73.72 reductase (NADH) marinisedimentorum large subunit UPI0003F87F52 Cluster: nitrite Psychromonas 72.02 reductase large subunit UPI00037B4F3D Cluster: nitrite Uliginosibacterium 70.2 reductase large subunit gangwonense A0A2N7DBN5 Cluster: Nitrite Vibrio sp. 73.06 reductase large subunit 10N.286.49.B3 A0A484GE91 Cluster: Nitrite Candidatus 72.16 reductase large subunit Schmidhempelia bombi str. Bimp A0A2M8H0G2 Cluster: Nitrite unclassified Vibrio 73.84 reductase (NAD(P)H) U3CNX4 Cluster: Nitrite Vibrio 73.78 reductase large subunit UPI0010A63CB5 Cluster: nitrite Vibrio sp. H11 73.12 reductase large subunit A0A090IJG5 Cluster: Nitrite Moritella 71.92 reductase (NAD(P)H) large subunit A0A2N0WYN5 Cluster: Nitrite Psychromonas 72.87 reductase (NAD(P)H) UPI000B356675 Cluster: nitrite Thaumasiovibrio 73.12 reductase large subunit subtropicus A0A1R4LRM2 Cluster: Nitrite Vibrio 69.41 reductase [NAD(P)H] UPI00082A845F Cluster: nitrite Erwinia sp. ErVv1 72.27 reductase large subunit B8KAR6 Cluster: Nitrite Vibrio 73.3 reductase (NAD(P)H), large subunit A0A1G8E1S6 Cluster: Assimilatory Vibrio xiamenensis 72.59 nitrite reductase (NAD(P)H) large subunit A0A0C5WZ76 Cluster: Nitrite Photobacterium 72.94 reductase (NAD(P)H), large subunit A0A2G4AWG1 Cluster: Nitrite Vibrionaceae 73.18 reductase [NAD(P)H]

Nitrate/Nitrite Transporter

[0495] Preferably, the recombinant yeast cell further comprises one or more genetic modifications that result in an increased transport of oxidized nitrogen source, such as nitrate or nitrite, into the yeast cell. More preferably the recombinant yeast cell further comprising one or more genes encoding a nitrate and/or nitrite transporter.

[0496] Suitable transporters may include the sulphite transporters Ssu1 and Ssu2 (as described by Cabrera et al in their article titled “Molecular Components of Nitrate and Nitrite Efflux in Yeast”, published February 2014 Volume 13 Number 2 Eukaryotic Cell p. 267-278, herein incorporated by reference); and the nitrate/nitrite transporter YNT1 derived from *Pichia angusta* (also referred to as *Hansenula polymorpha*) and/or a functional homologues of one or more of such nitrate/nitrite transporters comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of the aforementioned nitrate/nitrite transporters; and/or functional homologues of one or more of such nitrate/nitrite transporters comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned nitrate/nitrite transporters, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned nitrate/nitrite transporter YNT1.

[0497] Preferably the recombinant yeast cell comprises a nucleic acid sequence encoding the nitrate/nitrite transporter YNT1 derived from *Pichia angusta* and/or a functional homologues of such nitrate/nitrite transporter YNT1 comprising an amino acid sequence with at least 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with nitrate/nitrite transporter YNT1; and/or functional homologues of such nitrate/nitrite transporter YNT1 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned nitrate/nitrite transporter YNT1, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned nitrate/nitrite transporter YNT1.

[0498] Preferred nitrate/nitrite transporter hence include: nitrate/nitrite transporters comprising a polypeptide having an amino acid sequence of SEQ ID NO:94, as described herein; and/or functional homologues of SEQ ID NO:94 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with SEQ ID NO:94; and/or functional homologues of SEQ ID NO:94 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of SEQ ID NO:94. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:94.

[0499] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of SEQ ID NO:94 or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:94. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:94 respectively.

[0500] Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org) as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:94 are listed in Table 13 below.

TABLE-US-00023 TABLE 13 Examples of suitable nitrite/nitrate transporters, their UniProt

Database Accession number (as can be found on the Uniprot website (www.uniprot.org/) as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 94. UNIPROT DATABASE ACCESSION IDENTITY TO NUMBER DESCRIPTION ORGANISM SEQ ID 94 [%] W1QLT0 Cluster: Nitrate/nitrite Ogataea 95.66 transporter W6MFQ5 Cluster: Nitrate/nitrite Kuraishia capsulata 56.89 transporter CBS 1993 A0A1E4TX12 Cluster: Nitrate/nitrite Pachysolen tannophilus 56.55 transporter NRRL Y-2460 A0A1E3NV96 Cluster: Nitrate/nitrite Wickerhamomyces 56.52 transporter anomalus (strain ATCC 58044 / CBS 1984 / NCYC 433 / NRRL Y- 366-8) K0KNB9 Cluster: Nitrate/nitrite Wickerhamomyces 56.46 transporter ciferrii (strain F-60-10 / ATCC 14091 / CBS 111 / JCM 3599 / NBRC 0793 / NRRL Y- 1031) A0A0H5C0H7 Cluster: Nitrate/nitrite Cyberlindnera jadinii 54.44 transporter (strain ATCC 18201 / CBS 1600 / CCRC 20928 / JCM 3617 / NBRC 0987 / NRRL Y- 1542) A0A5P8N917 Cluster: Nitrate/nitrite Cyberlindnera 54.58 transporter americana A0A061BB65 Cluster: Cyberlindnera fabianii 52.72 CYFAOS24e00188g1_1 A0A060T893 Cluster: Nitrate/nitrite Blastobotrys 47.18 transporter adeninivorans B3WFS7 Cluster: Nitrate/nitrite Blastobotrys 47.79 transporter adeninivorans A0A261XZN8 Cluster: Nitrate/nitrite Bifiguratus adelaidae 47.49 transporter A0A0E9N9H8 Cluster: Nitrate/nitrite Saitoella complicata 46.17 transporter (strain BCRC 22490 / CBS 7301 / JCM 7358 / NBRC 10748 / NRRL Y-17804) A0A060T786 Cluster: Nitrate/nitrite Blastobotrys 43.91 transporter adeninivorans A0A0A1T938 Cluster: Nitrate/nitrite Torrubietta 42.57 transporter hemipterigena A0A3D8R1D7 Cluster: Nitrate/nitrite Coleophoma 42.66 transporter A0A370TMI7 Cluster: Nitrate/nitrite Venustampulla 41.72 transporter echinocandica A0A3N4HFN0 Cluster: Nitrate/nitrite Ascobolus immersus 42.16 transporter RN42 A0A167RLM4 Cluster: Nitrate/nitrite Cordyceps 40.04 transporter fumosorosea ARSEF 2679 A0A179I4H5 Cluster: Nitrate/nitrite Akanthomyces lecanii 40.94 transporter A0A0G2E479 Cluster: Nitrate/nitrite Phaeomoniella 42.33 transporter chlamydospora A0A6A6CV94 Cluster: Nitrate/nitrite Zasmidium cellare 42.91 transporter ATCC 36951 S3CRP8 Cluster: Nitrate/nitrite Glarea lozoyensis 40.4 transporter A0A1B8DXT4 Cluster: Nitrate/nitrite unclassified 42.74 transporter Pseudogymnoascus A0A094B0S5 Cluster: Nitrate/nitrite Pseudogymnoascus 42.94 transporter A0A6A6WTZ3 Cluster: Nitrate/nitrite Melanomma pulvis- 42.51 transporter pyrius CBS 109.77 G3JS45 Cluster: Nitrate/nitrite Cordyceps militaris 40.65 transporter A0A0E9NCH7 Cluster: Nitrate/nitrite Saitoella complicata 44.27 transporter (strain BCRC 22490 / CBS 7301 / JCM 7358 / NBRC 10748 / NRRL Y-17804) A0A0C3CTX3 Cluster: Nitrate/nitrite Oidiodendron maius Zn 43.15 transporter A0A6G1J7V7 Cluster: Nitrate Lentithecium fluviatile 42.63 transporter-like protein CBS 122367 A0A4U0XSZ7 Cluster: Nitrate/nitrite Cryomyces minteri 41.8 transporter UPI000CE1C7E3 Cluster: nitrate Quercus suber 41.33 transporter-like A0A6A5XQJ5 Cluster: Nitrate/nitrite Aaosphaeria arxii CBS 42.24 transporter 175.79 A0A1W5CWS1 Cluster: Nitrate/nitrite Lasallia pustulata 43.11 transporter A0A6A6R7R9 Cluster: Nitrate/nitrite Mytilinidiaceae 43.99 transporter U1GKS4 Cluster: Nitrate/nitrite Endocarpon pusillum 42.43 transporter (strain Z07020 / HMAS- L-300199) D5GLI8 Cluster: Nitrate/nitrite Tuber melanosporum 41.75 transporter (strain Mel28) A0A5J5EIC4 Cluster: Nitrate/nitrite Sphaerospora 41.7 transporter brunnea UPI00144AC6B7 Cluster: nitrate Lindgomyces 42.6 transporter ingoldianus A0A6A6RHP6 Cluster: Nitrate/nitrite Massarina eburnea 42.6 transporter CBS 473.64 A0A6A6JM23 Cluster: Nitrate/nitrite Westerdykella ornata 42.63 transporter A0A317SYP8 Cluster: Nitrate/nitrite Tuber 41.34 transporter A0A232LMW8 Cluster: Nitrate/nitrite Elaphomyces 43.09 transporter granulatus A0A1L7WDP7 Cluster: Nitrate/nitrite Phialocephala 42.04 transporter subalpina A0A6A5UCT1 Cluster: Nitrate/nitrite Byssothecium circinans 43.38 transporter A0A6A6BJV5 Cluster: Nitrate/nitrite Aplosporella prunicola 44.08 transporter CBS 121167 A0A6G1LEQ8 Cluster: Nitrate Teratosphaeria 41.68 transporter nubilosa S3DBU1 Cluster: Nitrate/nitrite Glarea lozoyensis 43.51 transporter A0A4U0TQA4 Cluster: Nitrate/nitrite Hortaea thailandica 40.68 transporter A0A4U0X4U1 Cluster: Nitrate/nitrite Friedmanniomyces 40.28 transporter A0A507R2R1 Cluster: Nitrate/nitrite

Monascus purpureus 41.84 transporter A0A2V1DKP2 Cluster: Nitrate/nitrite *Periconia* 41.46 transporter *macrospinoso* A0A1L9RJV0 Cluster: Nitrate/nitrite *Aspergillus wentii* DTO 42.54 transporter 134E9 A0A167P4D7 Cluster: Nitrate/nitrite *Sporothrix insectorum* 41.12 transporter RCEF 264 UPI000CE20498 Cluster: nitrate *Quercus suber* 42.71 transporter-like A0A4Z1P351 Cluster: Nitrate/nitrite *Venturia* 41.63 transporter A0A6G1K4M7 Cluster: Nitrate *Pleomassaria siparia* 42.16 transporter-like protein CBS 279.74 M7U767 Cluster: Nitrate/nitrite *Sclerotiniaceae* 42.65 transporter A0A0D2AUR8 Cluster: Nitrate/nitrite *Verruconis gallopava* 41.41 transporter A0A6G1H5R5 Cluster: Nitrate *Aulographum hederæ* 41.73 transporter CBS 113979 A0A1Y2TEF0 Cluster: Nitrate/nitrite unclassified *Hypoxylon* 41.77 transporter A0A177BV64 Cluster: Nitrate/nitrite *Paraphaeosphaeria* 41.75 transporter *sporulosa* A0A6A5WR45 Cluster: Nitrate/nitrite *Amniculicola lignicola* 42.63 transporter CBS 123094 A0A6G1KNX3 Cluster: Nitrate *Pleomassaria siparia* 41.41 transporter-like protein CBS 279.74 W9WHT3 Cluster: Nitrate/nitrite *Herpotrichiellaceae* 42.14 transporter A0A2T2P8J8 Cluster: Nitrate/nitrite *Corynespora cassicola* 43.06 transporter *Philippines* A0A6A6T246 Cluster: Nitrate/nitrite *Lophiostoma* 41.41 transporter *macrostomum* CBS 122681 A0A2P8ABJ3 Cluster: Nitrate/nitrite *Elsinoe australis* 42.25 transporter A0A1Y1ZNY9 Cluster: Nitrate/nitrite *Clohesyomyces* 40.77 transporter *aquaticus* W2RSI4 Cluster: Nitrate/nitrite *Cyphellophora* 41.24 transporter *europaea* CBS 101466 A0A2V1E4S3 Cluster: Nitrate/nitrite *Periconia* 41.41 transporter *macrospinoso* A0A6A5V026 Cluster: Nitrate/nitrite *Bimuria novae-* 41.15 transporter *zelandiae* CBS 107.79 A0A2K1QG31 Cluster: Nitrate/nitrite *Elsinoaceae* 42.25 transporter A0A094A2I4 Cluster: Nitrate/nitrite *Pseudogymnoascus* 43.54 transporter sp. VKM F-4281 (FW- 2241) A0A6A6XX03 Cluster: Nitrate/nitrite *Melanomma pulvis-* 42.07 transporter *pyrius* CBS 109.77 A0A2J5HKF9 Cluster: Nitrate/nitrite *Aspergillus* 40.56 transporter A0A6A6IDB6 Cluster: Nitrate/nitrite *Trematosphaeria* 42.45 transporter *pertusa* A0A6A5ST79 Cluster: Nitrate/nitrite *Clathrospora elyinae* 41.85 transporter A0A3N4JV10 Cluster: Nitrate/nitrite *Choiromyces venosus* 42.06 transporter 120613-1 A0A6A5XFW9 Cluster: Nitrate/nitrite *Aaosphaeria arxii* CBS 41.43 transporter 175.79 A0A553IBL5 Cluster: Nitrate/nitrite *Xylaria flabelliformis* 40.24 transporter R7Z451 Cluster: Nitrate/nitrite *Coniosporium apollinis* 40.61 transporter (strain CBS 100218) A0A139I046 Cluster: Nitrate/nitrite *Pseudocercospora* 40.24 transporter *musae* A0A2T2NBP3 Cluster: Nitrate/nitrite *Corynespora cassicola* 41.84 transporter *Philippines* A0A177E3H1 Cluster: Nitrate/nitrite *Alternaria* 42.21 transporter B0XPC3 Cluster: Nitrate/nitrite *Neosartorya fumigata* 42.89 transporter A0A150V198 Cluster: Nitrate/nitrite *Acidomyces* 42.08 transporter *richmondensis* BFW A0A3N4KQT8 Cluster: Nitrate/nitrite *Morchella conica* 40.44 transporter CCBAS932 A0A1Y2DSQ4 Cluster: Nitrate/nitrite *Pseudomassariella* 41.05 transporter *vexata* A0A2V1BSJ8 Cluster: Nitrate/nitrite *Helotiales incertae* 42.04 transporter *sedis* A0A1V1TKZ5 Cluster: Nitrate/nitrite *Fungi* 40.24 transporter K1WJ63 Cluster: Nitrate/nitrite *Marssonina brunnea* f. 42.06 transporter sp. *multigermtubi* (strain MB_m1) A0A6A6E950 Cluster: Nitrate/nitrite *Zopfia rhizophila* CBS 41.92 transporter 207.26 A0A2K1R263 Cluster: Nitrate/nitrite *Sphaceloma murrayae* 41.83 transporter A0A2S7Q2X6 Cluster: Nitrate/nitrite unclassified 42.32 transporter *Rutstroemia*

Co-Factors

[0501] Preferably the recombinant yeast cell further comprises suitable co-factors to enhance the activity of the above mentioned NADH-dependent nitrate reductase and/or NADH-dependent nitrite reductase. Preferred cofactors include flavin adenine dinucleotide (FAD), heme prosthetic groups, and/or molybdenum cofactor (MoCo). Preferably the recombinant yeast cell may therefore further comprise one or more genes encoding enzymes for the synthesis of one or more of flavin adenine dinucleotide (FAD), heme prosthetic groups, and/or molybdenum cofactor (MoCo). For example, the recombinant yeast cell may comprise one or more genes encoding for an enzyme having FAD synthase activity. Preferred co-factors are as exemplified in non-pre-published US patent application U.S. 63/087,642 filed with the United States Patent Office on 5 Oct. 2020, the

contents of which are herewith incorporated by reference.

Recombinant Expression

[0502] The recombinant yeast cell is a recombinant cell. That is to say, a recombinant yeast cell comprises, or is transformed with or is genetically modified with a nucleotide sequence that does not naturally occur in the cell in question. Techniques for the recombinant expression of enzymes in a cell, as well as for the additional genetic modifications of a recombinant yeast cell are well known to those skilled in the art. Typically such techniques involve transformation of a cell with nucleic acid construct comprising the relevant sequence. Such methods are, for example, known from standard handbooks, such as Sambrook and Russel (2001) "*Molecular Cloning: A Laboratory Manual*", (3rd edition), published by Cold Spring Harbor Laboratory Press, or F. Ausubel et al., eds., "*Current protocols in molecular biology*", Green Publishing and Wiley Interscience, New York (1987). Methods for transformation and genetic modification of fungal host cells are known from e.g. EP-A-0635574, WO98/46772, WO 99/60102, WO00/37671, WO90/14423, EP-A-0481008, EP-A-0635574 and U.S. Pat. No. 6,265,186.

Fermentation Process

[0503] The invention further provides a process for the production of ethanol, comprising converting a carbon source, preferably a carbohydrate or another organic carbon source, using a recombinant yeast cell as described in this specification, thereby forming ethanol.

[0504] The feed for this fermentation process suitably comprises one or more fermentable carbon sources. The fermentable carbon source preferably comprises or is consisting of one or more fermentable carbohydrates. More preferably, the fermentable carbon source comprises one or more mono-saccharides, disaccharides and/or polysaccharides. For example, the fermentable carbon source may comprise one or more carbohydrates selected from the group consisting of glucose, fructose, sucrose, maltose, xylose, arabinose, galactose, mannose and trehalose. The fermentable carbon source, preferably comprising or consisting of one or more carbohydrates, may suitably be obtained from starch, cellulose, hemicellulose lignocellulose, and/or pectin. Suitably the fermentable carbon source may be in the form of a, preferably aqueous, slurry, suspension, or a liquid.

[0505] The concentration of fermentable carbohydrate, such as for example glucose, during fermentation is preferably equal to or more than 80 g/L. That is, the initial concentration of glucose at the start of the fermentation, is preferably equal to or more than 80 g/L, more preferably equal to or more than 90 g/L, even more preferably equal to or more than 100 g/L, still more preferably equal to or more than 110 g/L, yet even more preferably equal to or more than 120 g/L, equal to or more than 130 g/L, equal to or more than 140 g/L, equal to or more than 150 g/L, equal to or more than 160 g/L, equal to or more than 170 g/L, or equal to or more than 180 g/L. The start of the fermentation may be the moment when the fermentable fermentable carbohydrate is brought into contact with the recombinant cell of the invention.

[0506] The fermentable carbon source may be prepared by contacting starch, lignocellulose, and/or pectin with an enzyme composition, wherein one or more mono-saccharides, disaccharides and/or polysaccharides are produced, and wherein the produced mono-saccharides, disaccharides and/or polysaccharides are subsequently fermented to give a fermentation product.

[0507] Before enzymatic treatment, the lignocellulosic material may be pretreated. The pretreatment may comprise exposing the lignocellulosic material to an acid, a base, a solvent, heat, a peroxide, ozone, mechanical shredding, grinding, milling or rapid depressurization, or a combination of any two or more thereof. This chemical pretreatment is often combined with heat-pretreatment, e.g. between 150-220° C. for 1 to 30 minutes. Subsequently the pretreated material can be subjected to enzymatic hydrolysis to release sugars that may be fermented according to the invention. This may be executed with conventional methods, e.g. contacting with cellulases, for instance cellobiohydrolase(s), endoglucanase(s), beta-glucosidase(s) and optionally other enzymes, The conversion with the cellulases may be executed at ambient temperatures or at higher

temperatures, at a reaction time to release sufficient amounts of sugar(s). The result of the enzymatic hydrolysis is hydrolysis product comprising C5/C6 sugars, herein designated as the sugar composition.

[0508] In one embodiment the fermentable carbohydrate is, or is comprised by a biomass hydrolysate, such as a corn stover or corn fiber hydrolysate. Such biomass hydrolysate may in its turn comprise, or be derived from corn stover and/or corn fiber.

[0509] By a “hydrolysate” is herein understood a polysaccharide-comprising material (such as corn stover, corn starch, corn fiber, or lignocellulosic material, which polysaccharides have been depolymerized through the addition of water to form mono and oligosaccharide sugars.

Hydrolysates may be produced by enzymatic or acid hydrolysis of the polysaccharide-containing material.

[0510] A biomass hydrolysate may be a lignocellulosic biomass hydrolysate. Lignocellulose herein includes hemicellulose and hemicellulose parts of biomass. Also lignocellulose includes lignocellulosic fractions of biomass. Suitable lignocellulosic materials may be found in the following list: orchard primings, chaparral, mill waste, urban wood waste, municipal waste, logging waste, forest thinnings, short-rotation woody crops, industrial waste, wheat straw, oat straw, rice straw, barley straw, rye straw, flax straw, soy hulls, rice hulls, rice straw, corn gluten feed, oat hulls, sugar cane, corn stover, corn stalks, corn cobs, corn husks, switch grass, miscanthus, sweet sorghum, canola stems, soybean stems, prairie grass, gamagrass, foxtail; sugar beet pulp, citrus fruit pulp, seed hulls, cellulosic animal wastes, lawn clippings, cotton, seaweed, algae (including macroalgae and microalgae), trees, softwood, hardwood, poplar, pine, shrubs, grasses, wheat, wheat straw, sugar cane bagasse, corn, corn husks, corn hobs, corn kernel, fiber from kernels, products and by-products from wet or dry milling of grains, municipal solid waste, waste paper, yard waste, herbaceous material, agricultural residues, forestry residues, municipal solid waste, waste paper, pulp, paper mill residues, branches, bushes, canes, corn, corn husks, an energy crop, forest, a fruit, a flower, a grain, a grass, a herbaceous crop, a leaf, bark, a needle, a log, a root, a sapling, a shrub, switch grass, a tree, a vegetable, fruit peel, a vine, sugar beet pulp, wheat midlings, oat hulls, hard or soft wood, organic waste material generated from an agricultural process, forestry wood waste, or a combination of any two or more thereof. Algae, such as macroalgae and microalgae have the advantage that they may comprise considerable amounts of sugar alcohols such as sorbitol and/or mannitol. Lignocellulose, which may be considered as a potential renewable feedstock, generally comprises the polysaccharides cellulose (glucans) and hemicelluloses (xylans, heteroxylans and xyloglucans). In addition, some hemicellulose may be present as glucomannans, for example in wood-derived feedstocks. The enzymatic hydrolysis of these polysaccharides to soluble sugars, including both monomers and multimers, for example glucose, cellobiose, xylose, arabinose, galactose, fructose, mannose, rhamnose, ribose, galacturonic acid, glucuronic acid and other hexoses and pentoses occurs under the action of different enzymes acting in concert. In addition, pectins and other pectic substances such as arabinans may make up considerably proportion of the dry mass of typically cell walls from non-woody plant tissues (about a quarter to half of dry mass may be pectins). Lignocellulosic material may be pretreated. The pretreatment may comprise exposing the lignocellulosic material to an acid, a base, a solvent, heat, a peroxide, ozone, mechanical shredding, grinding, milling or rapid depressurization, or a combination of any two or more thereof. This chemical pretreatment is often combined with heat-pretreatment, e.g. between 150-220° C. for 1 to 30 minutes.

[0511] The process for the production of ethanol may comprise an aerobic propagation step and an anaerobic fermentation step. More preferably the process according to the invention is a process comprising an aerobic propagation step wherein the population of the recombinant yeast cell is increased; and an anaerobic fermentation step wherein the carbon source is converted to ethanol by using the recombinant yeast cell population.

[0512] By propagation is herein understood a process of recombinant yeast cell growth that leads to

increase of an initial recombinant yeast cell population. Main purpose of propagation is to increase the population of the recombinant yeast cell using the recombinant yeast cell's natural reproduction capabilities as living organisms. That is, propagation is directed to the production of biomass and is not directed to the production of ethanol. The conditions of propagation may include adequate carbon source, aeration, temperature and nutrient additions. Propagation is an aerobic process, thus the propagation tank must be properly aerated to maintain a certain level of dissolved oxygen. Adequate aeration is commonly achieved by air inductors installed on the piping going into the propagation tank that pull air into the propagation mix as the tank fills and during recirculation. The capacity for the propagation mix to retain dissolved oxygen is a function of the amount of air added and the consistency of the mix, which is why water is often added at a ratio of between 50:50 to 90:10 mash to water. "Thick" propagation mixes (80:20 mash-to-water ratio and higher) often require the addition of compressed air to make up for the lowered capacity for retaining dissolved oxygen. The amount of dissolved oxygen in the propagation mix is also a function of bubble size, so some ethanol plants add air through spargers that produce smaller bubbles compared to air inductors. Along with lower glucose, adequate aeration is important to promote aerobic respiration during propagation, making the environment during propagation different from the anaerobic environment during fermentation.

[0513] By an anaerobic fermentation process is herein understood a fermentation step run under anaerobic conditions.

[0514] The anaerobic fermentation is preferably run at a temperature that is optimal for the cell. Thus, for most recombinant yeast cells, the fermentation process is performed at a temperature which is less than about 50° C., less than about 42° C., or less than about 38° C. For recombinant yeast cell or filamentous fungal host cells, the fermentation process is preferably performed at a temperature which is lower than about 35, about 33, about 30 or about 28° C. and at a temperature which is higher than about 20, about 22, or about 25° C.

[0515] The ethanol yield, based on xylose and/or glucose, in the process according to the invention is preferably at least about 50, about 60, about 70, about 80, about 90, about 95 or about 98%. The ethanol yield is herein defined as a percentage of the theoretical maximum yield.

[0516] The process according to the invention, and the propagation step and/or fermentation step suitably comprised therein can be carried out in batch, fed-batch or continuous mode. A separate hydrolysis and fermentation (SHF) process or a simultaneous saccharification and fermentation (SSF) process may also be applied.

[0517] The recombinant yeast and process according to the invention advantageously allow for a more robust process. Advantageously the process, or any anaerobic fermentation during the process can be carried out in the presence of high concentrations of carbon source. The process, respectively any anaerobic fermentation step therein, is therefore preferably carried out in the presence of a glucose concentration of 25 g/L or more, 30 g/L or more, 35 g/L or more, 40 g/L or more, 45 g/L or more, 50 g/L or more, 55 g/L or more, 60 g/L or more, 65 g/L or more, 70 g/L or more, 75 g/L or more, 80 g/L or more, 85 g/L or more, 90 g/L or more, 95 g/L or more, 100 g/L or more, 110 g/L or more, 120 g/L or more or may for example be in the range of 25 g/L-250 g/L, 30 g/L-200 g/L, 40 g/L-200 g/L, 50 g/L-200 g/L, 60 g/L-200 g/L, 70 g/L-200 g/L, 80 g/L-200 g/L, or 90 g/L-200 g/L.

[0518] For the recovery of the fermentation product existing technologies are used. For different fermentation products different recovery processes are appropriate. Existing methods of recovering ethanol from aqueous mixtures commonly use fractionation and adsorption techniques. For example, a beer still can be used to process a fermented product, which contains ethanol in an aqueous mixture, to produce an enriched ethanol-containing mixture that is then subjected to fractionation (e.g., fractional distillation or other like techniques). Next, the fractions containing the highest concentrations of ethanol can be passed through an adsorber to remove most, if not all, of the remaining water from the ethanol. In an embodiment in addition to the recovery of fermentation

product, the yeast may be recycled.

[0519] All patent and literature references cited in the present specification are hereby incorporated by reference in their entirety.

[0520] The following examples are offered for illustrative purposes only, and are not intended to limit the scope of the present invention in any way

EXAMPLES

General Molecular Biology Techniques

[0521] Unless indicated otherwise, the methods used are standard biochemical techniques.

Examples of suitable general methodology textbooks include Sambrook et al., *Molecular Cloning, a Laboratory Manual* (1989) and Ausubel et al., *Current Protocols in Molecular Biology* (1995), John Wiley & Sons, Inc.

Hplc Analysis

[0522] HPLC analysis is typically conducted as described in “*Determination of sugars, byproducts and degradation products in liquid fraction in process sample*”; Laboratory Analytical Procedure (LAP, Issue date: Dec. 8, 2006; by A. Sluiter, B. Hames, R. Ruiz, C. Scarlata, J. Sluiter, and D. Templeton; Technical Report (NREL/TP-51042623); January 2008; National Renewable Energy Laboratory.

[0523] After fermentation, samples for HPLC analysis were separated from yeast biomass and insoluble components (corn mash) by passing the clear supernatant after centrifugation through a 0.2 µm pore size filter

Starter Strains

[0524] Strains were prepared using Ethanol Red® as starting strain. Ethanol Red® is a commercial *Saccharomyces cerevisiae* strain, available from Lesaffre.

[0525] A strain construction approach that can be followed is described in WO2013/144257A1 and WO2015/028582, incorporated herein by reference.

[0526] Expression cassettes from various genes of interest can be recombined in vivo into a pathway at a specific locus upon transformation of this yeast (U.S. Pat. No. 9,738,890 B2). The promoter, ORF and terminator sequences are assembled into expression cassettes with Golden Gate technology, as described by Engler et al (2011) and ligated into Bsal-digested backbone vectors that decorated the expression cassettes with the connectors for the in vivo recombination step. The expression cassettes including connectors are amplified by PCR. In addition, a 5'- and a 3'-DNA fragment of the up- and downstream part of the integration locus was amplified using PCR and decorated by a connector sequence. Upon transformation of yeast cells with these DNA fragments, in vivo recombination and integration into the genome takes place at the desired location. CRISPR-Cas9 technology is used to make a unique double stranded break at the integration locus to target the pathway to this specific locus (DiCarlo et al., 2013, *Nucleic Acids Res* 41:4336-4343) and WO16110512 and US2019309268. The gRNA was expressed from a multi-copy yeast shuttling vector that contains a natMX marker which confers resistance to the yeast cells against the antibiotic substance nourseothricin (NTC). The backbone of this plasmid is based on pRS305 (Sikorski and Hieter, *Genetics* 1989, vol. 122, pp. 19-27), including a functional 2 micron ORI sequence. The *Streptococcus pyogenes* CRISPR-associated protein 9 (Cas9) was expressed from a pRS414 plasmid (Sikorski and Hieter, 1989) with kanMX marker which confers resistance to the yeast cells against the antibiotic substance geneticin (G418). The guide RNA and protospacer sequences were designed with a gRNA designer tool (<https://www.atum.bio/eCommerce/cas9/input>).

Example 1: Construction of “Rubisco” Strain (Intermediate Strain IX1)

[0527] In the current example the starter strain was transformed with the cbbM gene encoding the single subunit of ribulose-1,5-biphosphate-carboxylase (RuBisCO) from *Thiobacillus denitrificans* and the genes encoding chaperonins GroEL and GroES from *E. coli* to aid in the proper folding of the RuBisCO protein in the cytosol of *S. cerevisiae*.

[0528] An expression cassette with *S. cerevisiae* TDH3 promoter, cbbM gene and *S. cerevisiae* CYC1 terminators was obtained from plasmid pBTWW002, the expression cassettes with *S. cerevisiae* TEF1 promoter, groEL and *S. cerevisiae* ACT1 terminator was obtained from plasmid pUD232, and the expression cassettes with *S. cerevisiae* TPI1 promoter, groES gene and *S. cerevisiae* PGI1 terminator was obtained from plasmid pUD233, as described by Guadalupe-Medina et al., Biotechnol, Biofuels, 2013, vol. 6, p. 125) and US2019309268.

[0529] These three cassettes were integrated in the INT1 locus between NTR1 (YOR071c) and GYP1 (YOR070c) located on chromosome XV of *S. cerevisiae* with CRISPR-Cas9 using the gRNA expression cassette (as described by Di Carlo et al., Nucleic Acids Res. 2013; pp. 1-8). The cassette was ordered as synthetic DNA cassette (gBLOCK) at Integrated DNA Technologies (Leuven, Belgium) as INT1 gBLOCK and homologous recombination with PCR fragment (5'-INT1) generated with primers BoZ-783 and DBC-18463 genomic DNA of strain CEN.PK113-7D as template and a PCR fragment (3'-INT1) generated with primers DBC-18464 and BoZ-788 using genomic DNA of strain CEN.PK113-7D as template as described in WO2018/114762.

[0530] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT1 locus of the GroES-cbbM-GroEL cassettes. Cas9 and gRNA plasmids were removed with twice a o/n liquid culture with YEPhD, dilution plated on YEPhD plate afterwards for single colonies. Streaks of cells made of the single colonies of the YEPhD plate and re-streaked on YEPhD and YEPhD+G418 (400 mg/ml) and YEPhD+NTC (200 mg/ml). Plasmid free colonies selected which have no growth on the antibiotic plates.

[0531] The above example resulted in intermediate strain IX1 which contained two copies of the GroES, cbbM, and GroEL (see Table 14 for the detailed genotype).

Example 2: Construction of Reference “PRK-Rubisco” Strain (Reference Strain RX2)

[0532] The intermediate strain IX1, as obtained in example 1, was transformed with phosphoribulokinase (prk) from *S. oleacera* (as described by Guadalupe-Medina et al., Biotechnol, Biofuels, 2013, vol. 6, p. 125) and ribulose-1,5-biphosphate-carboxylase (RuBisCO) from *Thiobacfflus denitrificans* encoded by the cbbM gene. The prk gene was expressed by the *S. cerevisiae* ANB1 promoter as described in U.S. Ser. No. 10/689,670 (illustrated by SEQ ID NO: 26 (cbbM), SEQ ID NO: 28 (prk) and SEQ ID NO: 23 (ANB1)) and the *S. cerevisiae* EBO1 terminator (Sc_EBO1t) (illustrated in SEQ ID NO: 79).

[0533] The cbbM gene was cloned between the *S. cerevisiae* ENO2 promoter (illustrated by SEQ ID NO: 80) and *S. cerevisiae* YOX1 terminator (illustrated by SEQ ID NO: 81).

[0534] Cassettes were assembled into a pathway “prk-cbbM” and integrated in the INT14.02 locus, a non-coding region between ORF YNL179c and RPS3 (YNL178W) on chromosome XIV from *S. cerevisiae* using CRISPR-Cas9 with the INT14.02 protospacer (illustrated by SEQ ID NO: 82) and the following two sequences for homologous integration: Sc_INT14.02_FLANK5 (illustrated by SEQ ID NO: 83) and Sc_INT14.02_FLANK3 (illustrated by SEQ ID NO: 84).

[0535] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT14.02 locus of the prk and-cbbM cassettes, and the gRNA and Cas9 plasmids were removed as described above. This resulted in reference strain RX2 which contained two copies of prk, groEL and groES and 4 copies of cbbM (see Table 14 for detailed genotypes).

Example 3: Construction of New Strains NX3 and NX4

[0536] New strains NX3 and NX4 were constructed by transforming the reference strain RX2 obtained in example 2 as follows:

[0537] A DNA fragment was compiled using Golden Gate Cloning (as described for example by Engler et al., “Generation of Families of Construct Variants Using Golden Gate Shuffling”, (2011), published in chapter 11 of Chaofu Lu et al. (eds.), *cDNA Libraries: Methods and Applications, Methods in Molecular Biology*, vol. 729, pages 167-180, incorporated herein by reference) and comprised the *S. cerevisiae* ANB1 promoter (illustrated by SEQ ID NO: 23), *Pichia pastoris* TKL1 gene (illustrated by SEQ ID NO: 18) and the *S. cerevisiae* TDH1 terminator. The DNA fragment

was named “fragmentA” (illustrated by SEQ ID NO: 73). Fragment A was integrated in the INT95 locus between SOD1 (YJR104C) and ADO1 (YJR105W) located on chromosome X of *S cerevisiae* reference strain RX2 using CRISPR-Cas9 and INT95 protospacer (illustrated by SEQ ID NO: 85) and the following two sequences for homologous integration: Sc_INT95B_FLANK5 (illustrated by SEQ ID NO: 86) and Sc_INT95B_FLANK3 (illustrated by SEQ ID NO: 87).

[0538] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT95 locus of the promoted *Pichia* TKL1 expression cassette. Plasmid free colonies were selected and this resulted in new strains NX3 and NX4 which contains two copies of the promoted *Pichia* TKL1 expression cassette (see Table 14 for detailed genotypes).

Example 4: Construction of New Strains NX5 and NX6

[0539] New strains NX5 and NX6 were constructed by transforming reference strain RX2 obtained in example 2 with four expression cassettes:

[0540] The first the cassette contains the DNA fragment named “fragmentA”, comprising the *S. cerevisiae* ANB1 promoter, *Pichia pastoris* TKL1 orf and the *S. cerevisiae* TDH1 terminator as referred to in example 3. (illustrated by SEQ ID NO: 73)

[0541] The second cassette contains a DNA fragment named “fragmentB”, comprising the *S. cerevisiae* MYO4 promoter (Sc_MYO4.pro), *S. cerevisiae* DAK1 orf (Sc_DAK1.orf) and *S. cerevisiae* GPM1 terminator (Sc_GPM1.ter). The nucleic acid sequence of the DNA fragment is illustrated in SEQ ID NO: 74.

[0542] The third cassette contains a DNA fragment named “fragmentC”, comprising the *S. cerevisiae* HHF2 promoter (Sc_HHF2.pro), *E. coli* gldA orf (Ec_gldA.orf) and *S. cerevisiae* EFM1 terminator (Sc_EFM1.ter). The nucleic acid sequence of the DNA fragment is illustrated in SEQ ID NO: 75.

[0543] The fourth cassette contains a DNA fragment named “fragmentD”, comprising the *S. cerevisiae* ANB1 promoter (Sc_ANB1.pro_0001), *Zygosaccharomyces rouxii* orf encoding glycerol transporter GLYT (ZYRO0E01210) (Zrou_T5.orf) and *S. cerevisiae* terminator (Sc_TEF1.ter_0001). The nucleic acid sequence of the DNA fragment is illustrated in SEQ ID NO: 76.

[0544] These four cassettes were integrated in the INT95 locus between SOD1 (YJR104C) and ADO1 (YJR105W) located on chromosome X of *S cerevisiae* reference strain RX2 using CRISPR-Cas9 using the following sequences for homologous integration as described above with Sc_INT95B_FLANK5 (illustrated by SEQ ID NO: 86) and Sc_INT95B_FLANK3 (illustrated by SEQ ID NO: 87).

[0545] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT95 locus of the four expression cassettes. Plasmid free colonies were selected which resulted in new strains NX5 and NX6 (see Table 14 for detailed genotypes).

Example 5: Construction of New Strains NX7 and NX8

[0546] New strains NX7 and NX8 were constructed by transforming reference strain RX2 obtained in example 2 with the cassette containing the DNA fragment comprising *S. cerevisiae* ANB1 promoter, *Pichia pastoris* TKL1 orf and the *S. cerevisiae* TDH1 terminator, i.e. “fragmentA” (illustrated by SEQ ID NO: 73) as referred to in example 3, in the INT95 locus using CRISPR-Cas9 using Sc_INT95B_FLANK5.acc (SEQ ID NO: 86) and Sc_INT95B_FLANK3.acc (SEQ ID NO: 87) for homologous integration as also illustrated in example 3.

[0547] The GPD2 was deleted using CRISPR-Cas9 and GPD2 protospacer (illustrated by SEQ ID NO: 88) and for and a repair DNA fragment (illustrated by SEQ ID NO: 78) for homologous recombination and deleting GPD2. The INT95 integration and GPD2 deletion was checked by diagnostic PCR. Plasmid free colonies were selected which resulted in new strains NX7 and NX8 (see Table 14 for detailed genotypes).

Example 6: Construction of New Strains NX9 and NX10

[0548] New strains NX9 and NX10 were constructed by transforming reference strain RX2 with

three expression cassettes:

[0549] The first the cassette contains the DNA fragment named “fragmentA”, comprising the *S. cerevisiae* ANB1 promoter, *Pichia pastoris* TKL1 orf and the *S. cerevisiae* TDH1 terminator as referred to in example 3. (illustrated by SEQ ID NO: 73)

[0550] The second cassette contains a DNA fragment named “fragmentE” comprising *S. cerevisiae* PFY1promoter (Sc_PFY1.pro), *E. coli* gldA orf (Ec_gldA.orf) and *S. cerevisiae* EFM1 terminator (Sc_EFM1.ter). The nucleic acid sequence of the DNA fragment is illustrated in SEQ ID NO: 77.

[0551] The third cassette contains the DNA fragment named “fragmentD”, comprising the *S. cerevisiae* ANB1 promoter (Sc_ANB1.pro_0001), *Zygosaccharomyces rouxii* orf encoding glycerol transporter GLYT (ZYRO0E01210) (Zrou_T5.orf) and *S. cerevisiae* terminator (Sc_TEF1.ter_0001). The nucleic acid sequence of the DNA fragment is illustrated in SEQ ID NO: 76.

[0552] These three cassettes were integrated in the INT95 locus using CRISPR-Cas9 using INT95B_FLANK5 (SEQ ID NO: 86) and INT95B_FLANK3 (SEQ ID NO: 87) for homologous integration. Diagnostic PCR was performed to confirm the correct assembly and integration at the INT95 locus of the three expression cassettes. Plasmid free colonies were selected which resulted in strain NX9 and NX10 (see Table 14 for detailed genotypes).

TABLE-US-00024 TABLE 14 *S. cerevisiae* strains used in this study

Strain name	Genotype
Parental strain	Wildtype
Intermediate strain	IX1
INT1::GroES-cbbM-GroEL reference strain	RX2
INT1::GroES-cbbM-GroFL	INT70::PRK-cbbM
NX3	INT1::GroES-cobM-GroEL
INT70::PRK-cbbM	INT95::pp_TKL1
NX4	INT1::GroES-cbbM-GroEL
INT70::PRK-cbbM	INT95::pp_TKL1
NX5	INT1::GroES-cbbM-GroEL
INT70::PRK-cbbM	INT95::ppTKL1-DAK1-gldA-T5
NX6	INT1::GroES-cbbM-GroEL
INT70:PRK-cbbM	INT95::ppTKL1-DAK1-gldA-T5
NX7	INT1::GroES-cbbM-GroEL
INT70::PRK-cbbM	INT95::ppTKL1-DAK1-gldA-T5 medium Δ GPD2
NX8	INT1::GroES-cbbM-GroEL
INT70::PRK-cbbM	INT95::ppTKL1-DAK1-gldA-T5 medium Δ GPD2
NX9	INT1::GroES-cbbM-GroEL
INT70::PRK-cbbM	INT95::ppTKL1-gldA-T5
NX10	INT1::GroES-cobM-GroEL
INT70::PRK-cbbM	INT95::ppTKL1-gldA-T5

Example 7: Fermentations

[0553] Precultures of the above new “NX3-NX10” strains were made as follows: Glycerol stocks (-80° C.) were thawed at room temperature and used to inoculate 0.2 L mineral medium [as described by Luttik, M L H. et al (2000) “*The Saccharomyces cerevisiae ICL2 Gene Encodes a Mitochondrial 2-Methylisocitrate Lyase Involved in Propionyl-Coenzyme A Metabolism*”. J. Bacteriol. 182:7007-13] supplemented with 2% (w/v) glucose, at pH 6.0 (adjusted with 2M H₂SO₄/4N KOH), in an unbaffled 0.5 L shake-flask. The precultures were incubated for 18 hours at 32° C. and shaken at 200 RPM. After estimating of the yeast cell dry weight (CDW) through OD600 measurement (using an existing CDW vs OD600 calibration line), a quantity of preculture corresponding to the required 0.5 gCDW/L inoculum concentration for the propagation was centrifuged (3 min, 5300×g), washed once with one sample volume sterile demineralized water, centrifuged once more, and resuspended in propagation medium.

[0554] Propagation of the above “NX3-NX10” strains was carried out as follows: A propagation step was performed in 500 mL shake flasks using 100 mL of filtered and diluted corn mash (70% v/v Corn mash: 30% v/v water) supplemented with 1.25 g/L urea and the antibiotics: neomycin and penicillin G with a final concentration of 50 μ g/mL and 100 μ g/mL respectively. After all additions, the pH was adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes, was dosed at the start of the propagation at a concentration of 0.1 mL/L. All strains were propagated for 6 h at 32° C. and shaken at 200 RPM.

[0555] Main fermentations of the above “NX3-NX10” strains was carried out as follows: A main fermentation step was performed using 200 ml medium in 500 ml Schott bottles equipped with pressure recording/releasing caps (Ankom Technology, Macedon NY, USA), while shaking at 140 rpm and 32° C. pH was not controlled during fermentation. Fermentations were executed with corn

mash having increased dry solids content of 36% w/w DS. Subsequently, the corn mash was supplemented with 1.0 g/L urea, and the antibiotics: neomycin and penicillin G with a final concentration of 50 µg/mL and 100 µg/mL respectively; antifoam (Basildon, approximately 0.5 mL/L). After all additions, the pH was adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes) was dosed at the start of the fermentation at a concentration of 0.24 mL/L. The required yeast pitch from propagation to fermentation was 1.5% on fermentation volume. All strains were tested under a condition of high solids, ie. 36% w/w DS).

[0556] Sampling of the fermentation was carried out as follows: Samples were taken from the main fermentations only. Samples for HPLC analysis were taken at 18, 24, 42, 48, and 66 hours. Table 15 summarizes the ethanol production (g/l) at each point in time. Table 16 summarizes the remaining glucose concentration (g/l) at each point in time.

[0557] Conclusions were as follows: The remaining glucose concentration was an indicator for the robustness of the yeast strain. Due to the presence of glucoamylase, glucose was continuously produced. Without wishing to be limited by any kind of theory it is believed that less robust strains will become more inhibited towards the end of the fermentation and as a result a higher concentration of unconverted glucose will be identified in the sample. A more robust strain will become less inhibited towards the end of the fermentation and as a result a lower concentration of unconverted glucose will be identified in the sample.

[0558] As illustrated by Table 16, the yeasts according to the invention, NX3, NX4, NX5, NX6, NX7, NX8, NX9 and NX10, resulted in lower concentrations of unconverted glucose after 48 hours and 66 hours of fermentation than the reference strain.

TABLE-US-00025 TABLE 15 Ethanol concentration (g/l) in the fermentation medium Time (h)/Strain RX2 NX3 NX4 NX5 NX6 NX7 NX8 NX9 NX10 0 hours 0.395 0.37 0.363 0.363 0.378 0.372 0.364 0.364 0.342 18 hours 61.80 61.80 61.20 61.80 58.80 61.80 63.00 56.60 59.00 24 hours 86.80 86.00 79.00 82.00 80.20 177.2 56.00 81.00 82.60 42 hours 134.20 137.40 137.60 135.30 132.80 137.20 137.60 133.20 135.70 48 hours 133.30 143.70 145.30 140.20 141.40 141.30 143.00 139.10 138.60 66 hours 141.20 150.10 143.40 143.70 149.60 144.90 143.70 142.80 144.10 Percentage (%) increase 0.0 7.8 9.0 5.2 6.1 6.0 7.3 4.4 4.0 in ethanol yield (g/l) vis-à-vis RX2 at 48 hours. Percentage (%) increase 0.0 6.3 1.6 1.8 5.9 2.6 1.8 1.1 2.1 in ethanol yield (g/l) vis-à-vis RX2 at 66 hours.

TABLE-US-00026 TABLE 16 Remaining glucose (grams/liter) concentrations in the fermentation medium Time/Strain RX2 NX3 NX4 NX5 NX6 NX7 NX8 NX9 NX10 0 hours 40.82 40.80 40.81 40.88 40.86 40.90 40.92 40.89 40.94 18 hours 59.80 51.80 50.80 61.80 60.60 46.20 45.60 64.60 58.00 24 hours 50.00 37.80 37.80 52.60 48.60 70.2 24.50 52.00 44.00 42 hours 15.00 6.44 5.22 13.50 17.90 5.72 5.18 15.50 9.79 48 hours 13.80 6.33 4.55 9.42 10.71 5.42 4.56 10.13 7.10 66 hours 18.80 13.16 10.01 14.30 14.50 11.76 11.25 14.60 12.73 Percentage (%) decrease 0.0 -54.1 -67.0 -31.7 -22.4 -60.7 -67.0 -26.6 -48.6 in remaining glucose (g/l) vis-à-vis RX2 at 48 hours. Percentage (%) decrease 0.0 -30.0 -46.8 -23.9 -22.9 -37.4 -40.2 -22.3 -32.3 in remaining glucose (g/l) vis-à-vis RX2 at 66 hours.

Example 8. Construction of Phosphoketolase Pathway-Expressing Reference Strain FGG1-pPATH1 (i.e. Reference Strain RX11)

[0559] WO2018/172328 describes the construction of several phosphoketolase pathway-expressing *Saccharomyces cerevisiae* strains, including FGG1-pPATH1 strain. Strain FGG1-pPATH1 had a relevant genotype comprising PKL, PTA and AADH. A summary of the relevant strains for the below examples is provided in below Table 17. The strains can be constructed in a manner as described in WO2018/172328, herewith incorporated by reference.

[0560] As explained in WO2018/172328, in an industrial environment strains such as the FGG1-pPATH1 strain can be affected in its osmotolerance and its stress response to the external environment.

TABLE-US-00027 TABLE 17 Phosphoketolase pathway-expressing *Saccharomyces cerevisiae*

strains Reference strain New strain RX11 NX12 (FG-pPATH1 of (including promoted Strain: FG WO2018/172328) *Pichia* TKL1) Relevant Wild type PKL, PTA, AADH PKL, PTA, AADH genotype: INT95::pp_TKL1

Example 9: Construction of New Strain NX12 (Prophetic, According to the Invention)

[0561] New strain NX12 can be constructed by transforming the reference strain RX11 (FGG1-pPATH1 as described in WO2018/172328) as follows:

[0562] A DNA fragment is compiled comprising the *S. cerevisiae* ANB1 promoter (illustrated by SEQ ID NO: 23), *Pichia pastoris* TKL1 gene (illustrated by SEQ ID NO: 18) and the *S. cerevisiae* TDH1 terminator. The DNA fragment is named “fragmentA” (illustrated by SEQ ID NO: 73). The DNA fragmentA is assembled using Golden Gate Cloning (as described for example by Engler et al., “*Generation of Families of Construct Variants Using Golden Gate Shuffling*”, (2011), published in chapter 11 of Chaofu Lu et al. (eds.), *cDNA Libraries: Methods and Applications, Methods in Molecular Biology*, vol. 729, pages 167-180, incorporated herein by reference). This expression cassette can be integrated in the INT95 locus between SOD1 (YJR104C) and ADO1 (YJR105W) located on chromosome X of *S. cerevisiae* reference strain RX11 using CRISPR-Cas9 and INT95 protospacer (illustrated by SEQ ID NO: 85) and two sequences for homologous integration: Sc_INT95B_FLANK5 (illustrated by SEQ ID NO: 86) and Sc_INT95B_FLANK3 (illustrated by SEQ ID NO: 87).

[0563] Diagnostic PCR can be performed to confirm the correct assembly and integration at the INT95 locus of the promoted TKL1 expression cassette. Plasmid free colonies are then selected and this results in new strain NX12 which contains two copies of the promoted TKL1 expression cassette (see Table 17 for detailed genotypes).

Example 10: Fermentations (Prophetic)

[0564] Precultures of the above new “NX12” strain can be made as follows: Glycerol stocks (−80° C.) are thawed at room temperature and used to inoculate 0.2 L mineral medium [as described by Luttkik, M L H. et al (2000) “The *Saccharomyces cerevisiae* ICL2 Gene Encodes a Mitochondrial 2-Methylisocitrate Lyase Involved in Propionyl-Coenzyme A Metabolism”. *J. Bacteriol.* 182:7007-supplemented with 2% (w/v) glucose, at pH 6.0 (adjusted with 2M H₂SO₄/4N KOH), in an unbaffled 0.5 L shake-flask. The precultures are incubated for 18 hours at 32° C. and shaken at 200 RPM. After estimating of the yeast cell dry weight (CDW) through OD600 measurement (using an existing CDW vs OD600 calibration line), a quantity of preculture corresponding to the required 0.5 gCDW/L inoculum concentration for the propagation is centrifuged (3 min, 5300×g), ished once with one sample volume sterile demineralized water, centrifuged once more, and resuspended in propagation medium.

[0565] Propagation of the above “NX12” strain can be carried out as follows: A propagation step is performed in 500 mL shake flasks using 100 mL of filtered and diluted corn mash (70% v/v Corn mash: 30% v/v water) supplemented with 1.25 g/L urea and the antibiotics: neomycin and penicillin G with a final concentration of 50 µg/mL and 100 µg/mL respectively. After all additions, the pH is adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes, is dosed at the start of the propagation at a concentration of 0.1 mL/L. All strains are propagated for 6 hours at 32° C. and shaken at 200 RPM.

[0566] Main fermentations of the above “NX12” strain can be carried out as follows: A main fermentation step is performed using 200 ml medium in 500 ml Schott bottles equipped with pressure recording/releasing caps (Ankom Technology, Macedon NY, USA), while shaking at 140 rpm and 32° C. pH is not controlled during fermentation. Fermentations are executed with corn mash having increased dry solids content of 36% w/w DS. Subsequently, the corn mash is supplemented with 1.0 g/L urea, and the antibiotics: neomycin and penicillin G with a final concentration of 50 µg/mL and 100 µg/mL respectively; antifoam (Basildon, approximately 0.5 mL/L). After all additions, the pH is adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes) is dosed at the start of the fermentation at a concentration of 0.24 mL/L.

The required yeast pitch from propagation to fermentation is 1.5% on fermentation volume. All strains are tested under a condition of high solids, ie. 36% w/w DS).

[0567] Sampling of the fermentation can be carried out as follows: Samples are taken from the main fermentations only. Samples for HPLC analysis are taken at 18, 24, 42, 48, and 66 hours. Ethanol production (g/l) at each point in time and remaining glucose concentration (g/l) at each point in time can be analyzed.

[0568] Conclusions can be as follows: The remaining glucose concentration is an indicator for the robustness of the yeast strain. Due to the presence of glucoamylase, glucose is continuously produced. Without wishing to be limited by any kind of theory it is believed that less robust strains such as reference strain RX11 will become more inhibited towards the end of the fermentation and as a result a higher concentration of unconverted glucose will be identified in the sample. A more robust strain such as NX12 will become less inhibited towards the end of the fermentation and as a result a lower concentration of unconverted glucose will be identified in the sample.

Example 11. Construction of Acetylating Acetaldehyde Dehydrogenase-Expressing Reference Strain IMZ132 (i.e. Reference Strain RX13)

[0569] WO2011/010923 describes an acetylating acetaldehyde dehydrogenase-expressing strain IMZ132, further referred to herein as reference strain RX13. The strain IMZ132 can be constructed in a manner as described in WO2011/010923, herewith incorporated by reference. In addition, the strain IMZ132 was deposited at Centraalbureau voor Schimmelcultures on Jul. 16, 2009 under deposit number CBS125049.

TABLE-US-00028 TABLE 18 acetylating acetaldehyde dehydrogenase -expressing *Saccharomyces cerevisiae* strains

Reference strain	New strain
RX13	NX14
Strain: (IMZ132 of WO2011/010923)	
(including promoted <i>Pichia</i> TKL1)	
Relevant MATa <i>ura3 leu2 gpd1</i> (-1, 1133)::loxP MATa <i>ura3 leu2 gpd1</i> (-1, 1133)::loxP	genotype: <i>gpd2</i> (-2, 1281)::hphMX4 <i>gpd2</i> (-2, 1281)::hphMX4
YEplac181(LEU2) YEplac181(LEU2) pUDE43(URA3 pTHD3::mhpF PUDE43(URA3 pTHD3::mhpF	(<i>E. coli</i>)::CYC1t) (<i>E. coli</i>)::CYC1t) INT95::pp_TKL1

Example 12: Construction of New Strain NX14 (Prophetic, According to the Invention)

[0570] New strain NX14 can be constructed by transforming the reference strain RX13 (IMZ132 as described in WO2011/010923) as follows:

[0571] A DNA fragment is compiled comprising the *S. cerevisiae* ANB1 promoter (illustrated by SEQ ID NO: 23), *Pichia pastoris* TKL1 gene (illustrated by SEQ ID NO: 18) and the *S. cerevisiae* TDH1 terminator. The DNA fragment is named “fragmentA” (illustrated by SEQ ID NO: 73). The DNA fragmentA is assembled using Golden Gate Cloning (as described for example by Engler et al., “Generation of Families of Construct Variants Using Golden Gate Shuffling”, (2011), published in chapter 11 of Chaofu Lu et al. (eds.), *cDNA Libraries: Methods and Applications, Methods in Molecular Biology*, vol. 729, pages 167-180, incorporated herein by reference). This expression cassette can be integrated in the INT95 locus between SOD1 (YJR104C) and ADO1 (YJR105W) located on chromosome X of *S. cerevisiae* reference strain RX13 using CRISPR-Cas9 and INT95 protospacer (illustrated by SEQ ID NO: 85) and two sequences for homologous integration: Sc_INT95B_FLANK5 (illustrated by SEQ ID NO: 86) and Sc_INT95B_FLANK3 (illustrated by SEQ ID NO: 87).

[0572] Diagnostic PCR can be performed to confirm the correct assembly and integration at the INT95 locus of the promoted TKL1 expression cassette. Plasmid free colonies are then selected and this results in new strain NX14 which contains two copies of the promoted TKL1 expression cassette (see Table 18 for detailed genotypes).

Example 13: Fermentations (Prophetic)

[0573] Precultures of the above new “NX14” strain can be made as follows: Glycerol stocks (-80° C.) are thawed at room temperature and used to inoculate 0.2 L mineral medium (as described by Luttkik, M L H. et al (2000) “The *Saccharomyces cerevisiae* ICL2 Gene Encodes a Mitochondrial 2-Methylisocitrate Lyase Involved in Propionyl-Coenzyme A Metabolism”. J. Bacteriol. 182:7007-13)

supplemented with 2% (w/v) glucose, at pH 6.0 (adjusted with 2M H₂SO₄/4N KOH), in an unbaffled 0.5 L shake-flask. The precultures are incubated for 18 hours at 32° C. and shaken at 200 RPM. After estimating of the yeast cell dry weight (CDW) through OD600 measurement (using an existing CDW vs OD600 calibration line), a quantity of preculture corresponding to the required 0.5 gCDW/L inoculum concentration for the propagation is centrifuged (3 min, 5300×g), washed once with one sample volume sterile demineralized water, centrifuged once more, and resuspended in propagation medium.

[0574] Propagation of the above “NX14 strain can be carried out as follows: A propagation step is performed in 500 mL shake flasks using 100 mL of filtered and diluted corn mash (70% v/v Corn mash: 30% v/v water) supplemented with 1.25 g/L urea and the antibiotics: neomycin and penicillin G with a final concentration of 50 µg/mL and 100 µg/mL respectively. After all additions, the pH is adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes, is dosed at the start of the propagation at a concentration of 0.1 mL/L. All strains are propagated for 6 hours at 32° C. and shaken at 200 RPM.

[0575] Main fermentations of the above “NX14 strain can be carried out as follows: A main fermentation step is performed using 200 ml medium in 500 ml Schott bottles equipped with pressure recording/releasing caps (Ankom Technology, Macedon NY, USA), while shaking at 140 rpm and applying a temperature of 32° C. pH is not controlled during fermentation. Fermentations are executed with corn mash having increased dry solids content of 36% w/w DS. Subsequently, the corn mash is supplemented with 1.0 g/L urea, and the antibiotics: neomycin and penicillin G with a final concentration of 50 µg/mL and 100 µg/mL respectively; antifoam (Basildon, approximately 0.5 mL/L). After all additions, the pH is adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes) is dosed at the start of the fermentation at a concentration of 0.24 mL/L. The required yeast pitch from propagation to fermentation is 1.5% on fermentation volume. All strains are tested under a condition of high solids, ie. 36% w/w DS).

[0576] Sampling of the fermentation can be carried out as follows: Samples are taken from the main fermentations only. Samples for HPLC analysis are taken at 18, 24, 42, 48, and 66 hours. Ethanol production (g/l) at each point in time and remaining glucose concentration (g/l) at each point in time can be analyzed.

[0577] Conclusions can be as follows: The remaining glucose concentration is an indicator for the robustness of the yeast strain. Due to the presence of glucoamylase, glucose is continuously produced. Without wishing to be limited by any kind of theory it is believed that less robust strains such as reference strain RX13 will become more inhibited towards the end of the fermentation and as a result a higher concentration of unconverted glucose will be identified in the sample. A more robust strain such as NX14 will become less inhibited towards the end of the fermentation and as a result a lower concentration of unconverted glucose will be identified in the sample.

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Claims

1. A recombinant yeast cell functionally expressing: a nucleic acid sequence encoding a native protein having transketolase activity (EC 2.2.1.1); and a nucleic acid sequence encoding a heterologous protein having transketolase activity (EC 2.2.1.1).
2. The recombinant yeast cell according to claim 1, wherein the heterologous protein having transketolase activity comprises or consists of an amino acid sequence having a percentage identity with the amino acid sequence of the native protein having transketolase activity in the range of equal to or more than 30% to equal to or less than 80%.
3. The recombinant yeast cell according to claim 1, wherein the heterologous protein having transketolase activity comprises or consists of: an amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19; or a functional homologue of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19, having at least 70% sequence identity with the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19; or a functional homologue of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19.
4. The recombinant yeast cell according to claim 1, wherein the native protein having transketolase activity comprises or consists of: an amino acid sequence of SEQ ID NO: 1; or a functional homologue of SEQ ID NO: 1, having sequence identity with the amino acid sequence of SEQ ID NO: 1; or a functional homologue of SEQ ID NO: 1, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 1.
5. The recombinant yeast cell according to claim 1, wherein the expression of the nucleic acid sequence encoding the heterologous protein having transketolase activity is under control of a promoter (the "TKL promoter") which TKL promoter has an anaerobic/aerobic expression ratio for the transketolase of 2 or more.
6. The recombinant yeast cell according to claim 1, wherein the expression of the nucleic acid sequence encoding the native protein having transketolase activity is under control of a promoter (the "TKL promoter") which TKL promoter has an anaerobic/aerobic expression ratio for the transketolase of 2 or more.
7. The recombinant yeast cell according to claim 5, wherein the TKL promoter is the promoter of ANB1 and/or DAN1.
8. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell comprises one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink.
9. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses: a nucleic acid sequence encoding a protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or a nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity; and/or a nucleic acid sequence encoding one or more

molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

10. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses: a nucleic acid sequence encoding a protein comprising phosphoketolase activity (EC 4.1.2.9 or EC 4.1.2.22, PKL); and/or a nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8); and/or a nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

11. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses a nucleic acid sequence encoding a protein comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10).

12. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses a nucleic acid sequence encoding an enzyme having NADH-dependent nitrate reductase activity and/or a nucleic acid sequence encoding an enzyme having NADH-dependent nitrite reductase activity.

13. The recombinant yeast cell according to claim 12, wherein the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding an enzyme having nitrate and/or nitrite transporter activity.

14. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell further comprises a deletion or disruption of a nucleic acid sequence encoding a protein having glycerol-3-phosphate dehydrogenase (GPD) activity and/or a nucleic acid sequence encoding a protein having glycerol phosphate phosphatase (GPP) activity.

15. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell further functionally expresses: a nucleic acid sequence encoding for a protein having glycerol dehydrogenase activity (E.C. 1.1.1.6); a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (E.C. 2.7.1.28 or E.C. 2.7.1.29); and/or a nucleic acid sequence encoding a protein having glycerol transporter activity.

16. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding a protein having glucoamylase activity (EC 3.2.1.20 or 3.2.1.3).

17. A process for the production of ethanol, comprising converting a carbon source using a recombinant yeast cell according to claim 1.

18. The process according to claim 17, wherein the process is at least partly carried out in a medium comprising glucose in a glucose concentration of 25 g/L or more.

19. The process according to claim 17, wherein the process is at least partly carried out in the presence of a saccharolytic enzyme.

20. The recombinant yeast cell according to claim 6, wherein the TKL promoter is the promoter of ANB1 and/or DAN1
